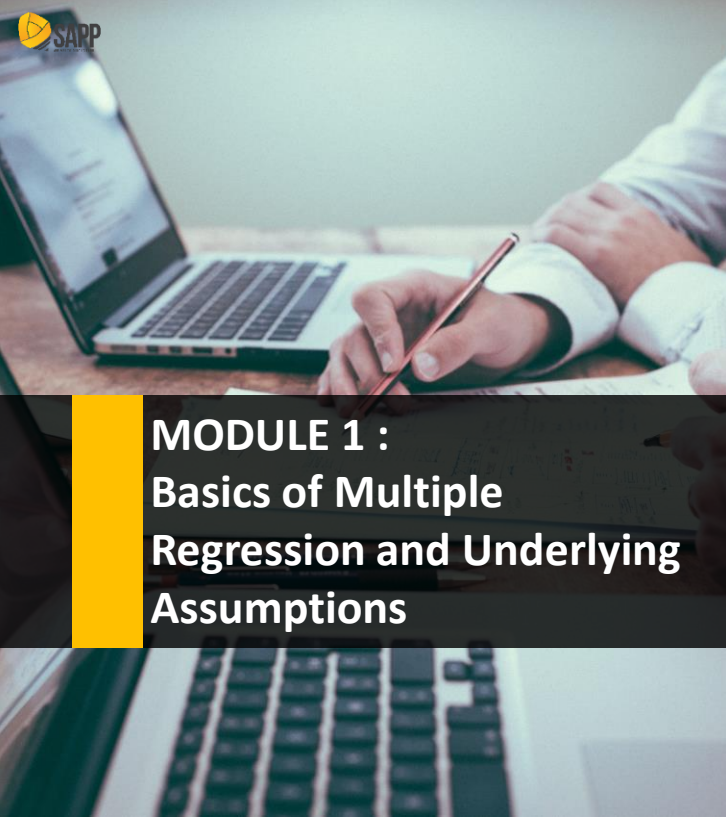




MODULE 1 : **Basics of Multiple** **Regression and Underlying** **Assumptions**

LEARNING OUTCOMES

- 1.b.** Formulate a multiple linear regression model, describe the relation between the dependent variable and several independent variables, and interpret estimated regression coefficients
- 1.c.** Explain the assumptions underlying a multiple linear regression model and interpret residual plots indicating potential violations of these assumptions
- 1.a.** Describe the types of investment problems addressed by multiple linear regression and the regression process

[LOS 1.b] Formulate a multiple linear regression model, describe the relation between the dependent variable and several independent variables, and interpret estimated regression coefficients

1.

General multiple linear regression model

Multiple regression model is the most widely used technique to model the linear relationship between one dependent variable and two or more independent variables.

Multiple linear regression equation:

$$Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + \dots + b_kX_{ki} + \epsilon_i$$

where:

Y_i = i th observation of the dependent variable Y , $i = 1, 2, \dots, n$

X_j = independent variables, $j = 1, 2, \dots, k$

X_{ji} = i th observation of the j th independent variable

b_0 = intercept term (*see next slide*)

b_j = slope coefficient for each of the independent variables (partial slope coefficient or partial regression coefficient) (*see next slide*)

ϵ_i = error term for the i th observation

n = number of observations

k = number of independent variables

[LOS 1.b] Formulate a multiple linear regression model, describe the relation between the dependent variable and...

2.

Implications of intercept term and slope coefficients

$$Y_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + \dots + b_k X_{ki} + \epsilon_i$$

b_0 = intercept term: the value of dependent variable when the independent variable all equal to 0

b_j = slope coefficient (partial slope coefficient or partial regression coefficient): the estimated change in the dependent variable for a one-unit change in that independent variable, holding the other independent variables constant.

[LOS 1.b] Formulate a multiple linear regression model, describe the relation between the dependent variable and...

3.

The multiple regression methodology

We estimate the intercept and slope coefficients using **ordinary least squares** (OLS), such that the sum of the squared error terms, $\sum_{i=1}^n \epsilon_i^2$, is minimized. The result of this procedure is the following regression equation:

$$\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_{1i} + \hat{b}_2 X_{2i} + \dots + \hat{b}_k X_{ki}$$

where the “^” indicates an estimate for the corresponding regression coefficient.

Example 1: Interpret estimated regression coefficients

Consider an estimated regression equation in which the monthly excess returns of a bond index (RET) are regressed against the change in monthly government bond yields (BY) and the change in the investment-grade credit spreads (CS).

The estimated regression, using 60 monthly observations, is:

$$\text{RET} = 0.0023 - 5.0585\text{BY} - 2.1901\text{CS}.$$

1. Interpret the intercept?
2. Interpret the slope coefficient of BY?

[LOS 1.b] Formulate a multiple linear regression model, describe the relation between the dependent variable and...

3.**The multiple regression methodology**

Example 1: Interpret estimated regression coefficients

RET = 0.0023 – 5.0585BY – 2.1901CS.

- 1. Interpret the intercept?*
- 2. Interpret the slope coefficient of BY?*

Answer:

1. The bond index RET yields, on average, 0.0023% per month, or approximately 0.028% per year, if the changes in the government bond yields and investment-grade credit spreads are zero.
2. The change in the bond index return for a given one-unit change in the monthly government bond yield, BY, is –5.0585%, holding CS constant. This means that the bond index has an empirical duration of 5.0585.

[LOS 1.b] Formulate a multiple linear regression model, describe the relation between the dependent variable and...



Practice Example 1: Understanding the multiple regression model

An institutional salesperson has just read the research report in which you estimated a regression of monthly excess returns on a portfolio, RETRF, against the Fama–French three factors:

- MKTRF, the market excess return;
- SMB, the difference in returns between small- and large-capitalization stocks;
- HML, the difference in returns between value and growth stocks.

All returns are stated in whole percentages (that is, 1 for 1%), and the estimated regression equation is:

$$\text{RETRF} = 1.5324 + 0.5892 \text{ MKTRF} - 0.8719 \text{ SMB} - 0.0560 \text{ HML}$$

1. *Interpret the intercept.*
2. *Interpret the slope coefficient of MKTRF.*
3. *Calculate the predicted value of the portfolio's return if MKTRF = 1, SMB = 4, and HML = -2.*

[LOS 1.b] Formulate a multiple linear regression model, describe the relation between the dependent variable and...



Practice Example 1: Understanding the multiple regression model

Solution:

$$\text{RETRF} = 1.5324 + 0.5892\text{MKTRF} + -0.8719\text{SMB} + -0.0560\text{HML}$$

1. Interpret the intercept.

If the market excess return, SMB, and HML are each zero, then we expect a return on the portfolio of 1.534%.

2. Interpret slope coefficient.

For MKTRF, if the market return increases by 1%, we expect the portfolio's return to increase by 0.5892%, holding other variables constant.

3. Calculate the predicted value of the portfolio's return if MKTRF = 1, SMB = 4, and HML = -2.

Given the expected values of the independent variables, the expected return on the portfolio is:

$$R = 1.534 + 0.5892(1) - 0.8719(4) - 0.0560(-2) = -1.2524.$$

[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and interpret residual plots indicating potential violations of these assumptions

1.

Assumptions underlying a multiple regression model

Linearity: The relationship between the dependent variable and the independent variables is linear.

Homoskedasticity: The variance of the regression residuals is the same for all observations ($E(\epsilon_i^2) = \sigma_\epsilon^2$).

Independence of errors: The regression residuals are uncorrelated across observations ($\rho(\epsilon_i, \epsilon_j) = 0 \forall i \neq j$).

Normality: The regression residuals are normally distributed ($\epsilon \sim N$).

Independence of independent variables: Independent are not random. There is no exact linear relation between two or more of the independent variables ($\rho(X_i, X_j) = 0$).

→ We may make invalid conclusions regarding the regression results unless the assumptions are satisfied.

[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and...

1.

Assumptions underlying a multiple regression model

Illustration: Assumptions of multiple regression model

We estimate a regression to analyze 10 years of monthly total excess returns of ABC stock using the Fama–French three-factor model. This model uses market excess return (MKTRF), size (SMB) and value (HML) as independent variables.

$$ABC_RETRF_t = b_0 + b_1 MKTRF_t + b_2 SMB_t + b_3 HML_t + \varepsilon_t$$

Now we will illustrate each assumption underlying a multiple regression model by using Diagnostic plots, where:

- **Scatterplots** of dependent versus and independent variables are useful for detecting nonlinear relationships
- **Residual plots** are useful for detecting violations of homoskedasticity and independence of errors.

(See next slides)

[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and...

2.

Assumption 1: Linearity

The relationship between the dependent variable and the independent variables is linear.

Illustration: Assumptions of multiple regression model

$$ABC_RETRF_t = b_0 + b_1 MKTRF_t + b_2 SMB_t + b_3 HML_t + \varepsilon_t$$

We start our analysis by generating a **scatterplot** which illustrates the relationship between the dependent variable (ABC_RETRF) and MKTRF.

Let's look at how we will interpret it.

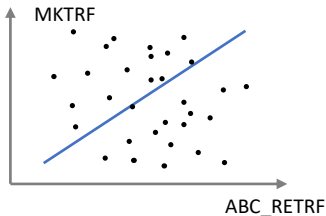
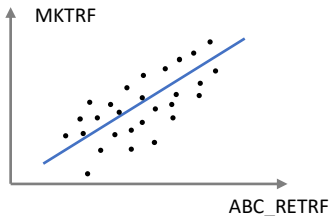
[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and...

2.

Assumption 1: Linearity

The relationship between the dependent variable and the independent variables is linear.

Below are 2 opposite cases of how the scatterplot could show:



✓ There is a **positive relationship** between **ABC_RETF** and the market factor, **MKTRF**.
→ Assumption 1 is supported.

✗ There is **no linear relationship** between **ABC_RETF** and the market factor, **MKTRF**.
→ Assumption 1 is not satisfied.

[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and...

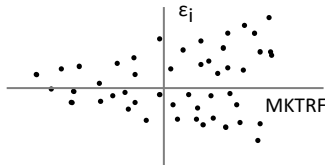
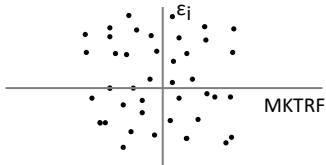
3.

Assumption 2: Homoskedasticity

The variance of the regression residuals is the same for all observations ($E(\epsilon_i^2) = \sigma_\epsilon^2$).

We investigate this assumption by looking at the scatterplot of regression residuals (ϵ_i) and the independent variables.

Below are 2 opposite cases of how the scatterplot of regression residuals (ϵ_i) and MKTRF could show:



✓ The residual plotted on the graph, corresponding to independent variable **MKTRF**, indicates **no directional relationship** between the variance of residuals and the independent variables.
→ Assumption 2 is supported.

✗ The residual plotted on the graph, corresponding to independent variable **MKTRF**, indicates **positive directional relationship** between the variance of residuals and the independent variables.
→ Assumption 2 is not satisfied.

[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and...

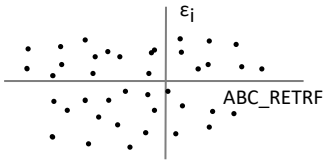
4.

Assumption 3: Independence of errors

The regression residuals are uncorrelated across observations ($\rho(\epsilon_i \epsilon_j) = 0 \forall i \neq j$).

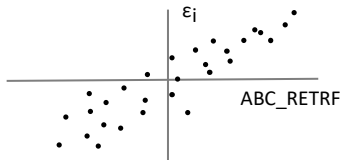
We investigate this assumption by looking at the scatterplot of regression residuals (ϵ_i) and the predicted value of the dependent variable.

Below are 2 opposite cases of how the scatterplot of regression residuals (ϵ_i) and ABC_RETRF could show:



This **residual** plot show that the residuals behave in an **independent manner** among observations.

→ Assumption 3 is supported.



This **residual** plot show that the residuals are **positively correlated** among observations.

→ Assumption 3 is not satisfied.

[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and...

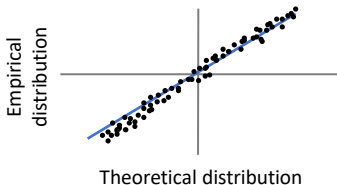
5.

Assumption 4: Normality

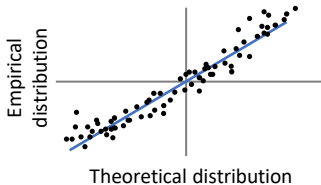
The regression residuals are normally distributed ($\epsilon \sim N$).

To see if this assumption is satisfied, we present a **normal Q-Q plot** to compare the model's standardized residuals to a normal distribution.

Below are 2 opposite cases of how the Q-Q plot could show:



This regression residuals overall are distributed consistently with the **normal distribution**.
→ Assumption 4 is supported.



This regression residuals overall are **not** distributed consistently with the **normal distribution**.
→ Assumption 4 is not satisfied.

[LOS 1.c] Explain the assumptions underlying a multiple linear regression model and...

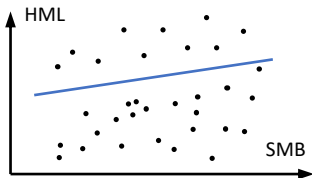
6.

Assumption 5: Independence of independent variables

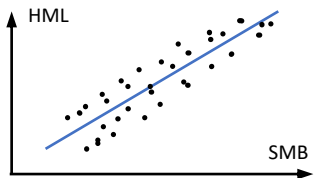
Independent are not random. There is no exact linear relation between two or more of the independent variables ($\rho(X_i, X_j) = 0$).

We investigate this assumption by looking at the scatterplot of between pairs of independent variables.

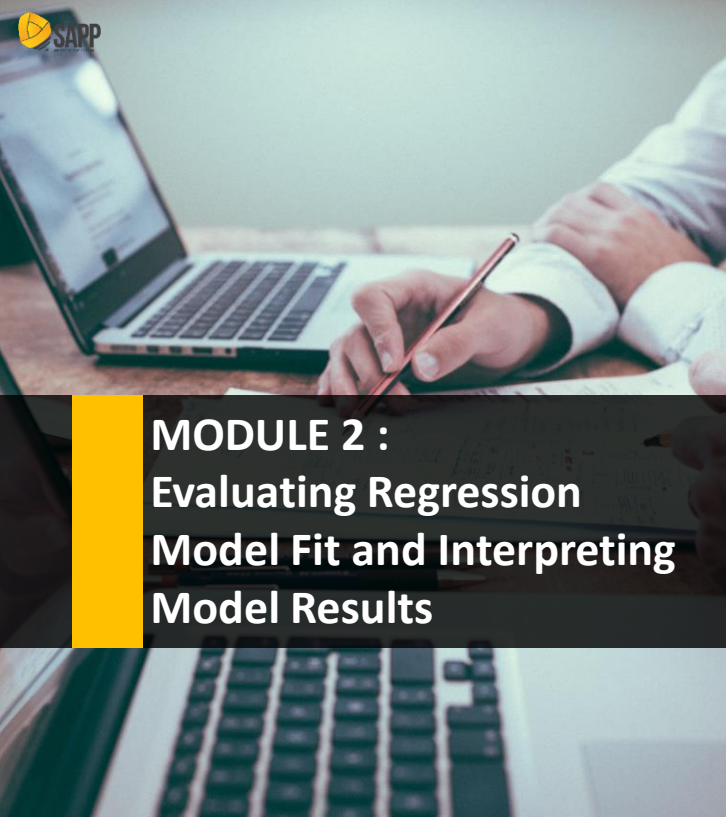
Below are 2 opposite cases of how the scatterplot of SMB and HML could show:



✓ This Scatterplots shows that **SMB** and **HML** have **little or no correlation**. This is a desirable characteristic between independent variables.
→ Assumption 5 is supported.



✗ This Scatterplots shows that **SMB** and **HML** have **positive correlation**. This is not a desirable characteristic between independent variables.
→ Assumption 5 is not satisfied.



MODULE 2 : **Evaluating Regression** **Model Fit and Interpreting** **Model Results**

LEARNING OUTCOMES

- 2.a.** Evaluate how well a multiple regression model explains the dependent variable by analyzing ANOVA table results and measures of goodness of fit
- 2.b.** Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and interpret the results of the joint hypothesis tests
- 2.c.** Calculate and interpret a predicted value for the dependent variable, given the estimated regression model and assumed values for the independent variable

SUMMARY GRAPH

Evaluate the model fit

Measure the goodness of fit [Los 2.a]

Adjusted R^2 [Los 2.a.3]

AIC [Los 2.a.4]

BIC [Los 2.a.5]

Test for the significance of coefficients [Los 2.b]

Single test: T – test [Los 2.b.1]

Group test: F – test [Los 2.b.2-3]

Forecasting the dependent variable [Los 2.c]

Interprete the results

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by analyzing ANOVA table results and measures of goodness of fit

1.

ANOVA Tables

Recall that you have learned about the basics of ANOVA table in the Linear regression module of CFA level I Curriculum.

Analysis of variance (ANOVA) is a statistical procedure that provides information on the explanatory power of a regression.

Source	df (Degrees of Freedom)	SS (Sum of Squares)	MS (Mean Square = SS/df)
Regression	k	SSR	MSR = SSR / k
Error	$n - k - 1$	SSE	MSE = SSE / (n-k-1)
Total	$n - 1$	SST	

The information in an ANOVA table is used to attribute the total variation of the dependent variable to one of two sources: the **regression model** or **the residuals (errors)**.

The information in an ANOVA table can be used to calculate R^2 , $\bar{R}^2$ and other measures that will be discussed later.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

1.

ANOVA Tables

Source	df (Degrees of Freedom)	SS (Sum of Squares)	MS (Mean Square = SS/df)
Regression	k	SSR	MSR = SSR / k
Error	n - k - 1	SSE	MSE = SSE / (n-k-1)
Total	n - 1	SST	

- Regression sum of squares (SSR)** measures the variation in the dependent variable that is explained by the independent variable:

$$SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

- Sum of squared errors (SSE)** measures the unexplained variation in the dependent variable. It's also known as the sum of squared residuals or the residual sum of squares: $SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$.
- Total sum of squares (SST)** measures the total variation in the dependent variable: $SST = SSR + SSE$.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

1.

ANOVA Tables

Example 1: Analyzing ANOVA Table

In an attempt to estimate a regression equation that can be used to forecast BuildCo's future sales, **22** years of Build Co's annual sales were regressed against two independent variables:

GDP = the level of gross domestic product.

ΔI = Changes in 30-year mortgage interest rates (expressed in percentage terms)

Regression Results for BuildCo Sales Data

	Coefficient	Standard Error	t-Statistic	p-Value
Intercept	6.000	4.520	1.327	0.20
Level of gross domestic product (GDP)	0.004	0.003	?	0.20
Changes in 30-year mortgage rates (ΔI)	-20.500	3.560	?	< 0.001

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

1.

ANOVA Tables

Example 1: Analyzing ANOVA Table

In an attempt to estimate a regression equation that can be used to forecast BuildCo's future sales, 22 years of Build Co's annual sales were regressed against two independent variables:

GDP = the level of gross domestic product.

ΔI = Changes in 30-year mortgage interest rates (expressed in percentage terms)

Regression Results for BuildCo Sales Data

Fill in the missing data

ANOVA	<i>df</i>	<i>SS</i>	<i>MS</i>
Regression	?	236.30	?
Error	?	116.11	?
Total	?	?	

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

1.

ANOVA Tables

Example 1: Analyzing ANOVA Table

Fill in the missing data.

Answer:

The computed test statistics for the regression coefficients are:

$$t_{GDP} = \frac{0.004}{0.003} = 1.333$$

$$t_{\Delta I} = \frac{-20.500}{3.560} = -5.758$$

Degrees of freedom are:

$$df_{\text{regression}} = k = 2$$

$$df_{\text{error}} = n - k - 1 = 22 - 2 - 1 = 19$$

$$df_{\text{total}} = n - 1 = 22 - 1 = 21$$

Other calculations:

$$SST = SSR + SSE = 236.30 + 116.11 = 352.41 \quad MSR = \frac{SSR}{k} = \frac{236.30}{2} = 118.15$$

$$MSE = \frac{SSE}{n-k-1} = \frac{116.11}{19} = 6.11$$

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

1.

ANOVA Tables

Example 1: Analyzing ANOVA Table

Fill in the missing data.

	Coefficient	Standard Error	t-Statistic	p-Value
Intercept	6.000	4.520	1.327	0.20
Level of gross domestic product (GDP)	0.004	0.003	1.333	0.20
Changes in 30-year mortgage rates (ΔI)	-20.500	3.560	-5.758	< 0.001
ANOVA	df	SS	MS	
Regression	2	236.30	118.15	
Error	19	116.11	6.11	
Total	21	352.41		

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

2.

Problems with Coefficient of Determination, R^2

- In the simple regression model, the coefficient of determination, also known as R^2 , is a measure of the goodness of fit of an estimated regression to the data.

$$R^2 = \frac{SSR}{SST}$$

- However, it is less appropriate for multiple regression model because R^2 will increase or stay the same, but not decrease as independent variables are added to the model.



Problems with using R^2 in multiple regression include the following:

- The R^2 cannot provide information on whether the coefficients are statistically significant, or whether there are biases in the estimated coefficients and predictions.
- The R^2 cannot tell whether the model fit is good. A good model may have a low R^2 , as in many asset-pricing models, and a bad model may have a high R^2 due to **overfitting** and biases in the model.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

2.

Problems with Coefficient of Determination, R^2

Overfitting of a regression model is a situation in which the model is too complex, meaning there may be too many independent variables relative to the number of observations in the sample.

→ A result of overfitting is that the coefficients on the independent variables may not represent true relationships with the dependent variable.

→ Because an overfitted model can still have a high R^2 , we should consider an alternative measure which can adjust for the problem of R^2 : **adjusted coefficient of determination ($\bar{R}^2$)**

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

3.

Adjusted Coefficient of Determination, $\bar{R}^2$

Adjusted Coefficient of Determination ($\bar{R}^2$) is an measure of model's goodness of fit, which is a modified version of R^2 that takes into account the number of independent variables in the model. It adjusts R^2 value downward to penalize models that have too many variables, which can lead to overfitting.

A benefit of using $\bar{R}^2$ is that it does not automatically increase when another independent variable is added to a regression, because it adjusts for the degrees of freedom as follows:

$$\bar{R}^2 = 1 - \left[\left(\frac{n-1}{n-k-1} \right) (1-R^2) \right]$$

where:

n = number of observations

k = number of independent variables

R^2 = Coefficient of Determination

- $\bar{R}^2 \leq R^2$
- $\bar{R}^2 < 0$ if the R^2 is low enough, n is large enough or k is large enough.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

3.

Adjusted Coefficient of Determination, $\bar{R}^2$

When adding a new variable to a regression:

If the coefficient's t-statistic $> |1.0|$ (the coefficient is likely to be significant).

$\bar{R}^2$ increases.

If the coefficient's t-statistic $< |1.0|$ (the coefficient is not likely to be significant).

$\bar{R}^2$ decreases.

Note that:

- Unlike in simple regression, there is no neat interpretation of $\bar{R}^2$ in a multiple regression setting in terms of percentage of the dependent variable's variation explained.
- $\bar{R}^2$ does not address whether the regression coefficients are significant or the predictions are biased; this requires examining residual plots and other statistics.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

4.

Akaike's information criterion (AIC)

Akaike's information criterion (AIC) is used to evaluate a collection of models that explain the same dependent variable.

$$AIC = n \times \ln\left(\frac{SSE}{n}\right) + 2(k + 1)$$

- The AIC statistic depends on the sample size (n), the number of independent variables in the model (k), and the sum of squares error (SSE) of the model.
- AIC is a measure of model parsimony, so a **lower AIC** indicates a **better-fitting** model.
- The term $2(k + 1)$ is the penalty assessed for adding independent variables to the model.
- AIC is preferred if the model is used for prediction purposes.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

5.

Schwarz's Bayesian information criterion (BIC or SBC)

Schwarz's Bayesian information criterion (BIC or SBC) allows comparison of models with the same dependent variable.

$$\text{BIC} = n \times \ln\left(\frac{\text{SSE}}{n}\right) + \ln(n) \times (k + 1)$$

- Compared to AIC, BIC assesses a greater penalty for having more parameters in a model, so it will tend to prefer small, more parsimonious models. This is because $\ln(n)$ is greater than 2, even for very small sample sizes.
- A **lower BIC** indicates a **better-fitting** model as well.
- BIC is preferred when the best goodness of fit is desired.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

Example 2: Selecting the best model using AIC and BIC

For the regression of portfolio returns on five factors, we present several goodness-of-fit measures here generated for five models out of the 31 possible combinations (C_5^r , $r = 1-5$) of models using these five variables.

Which is the most suitable model we would choose upon AIC and BIC?

	R^2	$\bar{R}^2$	AIC	BIC
Factor 1 only	0.541	0.531	19.079	22.903
Factors 1 and 2	0.541	0.521	21.078	26.814
Factors 1,2 and 3	0.562	0.533	20.743	28.393
Factors 1,2,3 and 4	0.615	0.580	16.331	25.891
Factors 1,2,3,4 and 5	0.615	0.572	18.215	29.687

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

Example 2: Selecting the best model using AIC and BIC

	R^2	$\bar{R}^2$	AIC	BIC
Factor 1 only	0.541	0.531	19.079	22.903
Factors 1 and 2	0.541	0.521	21.078	26.814
Factors 1,2 and 3	0.562	0.533	20.743	28.393
Factors 1,2,3 and 4	0.615	0.580	16.331	25.891
Factors 1,2,3,4 and 5	0.615	0.572	18.215	29.687

- R^2 increases or remains the same as we add variables to the model.
 - $\bar{R}^2$ increases with the addition of some variables (Factors 3 and 4) but decreases with the addition of other variables (Factors 2 and 5).
 - The AIC is minimized with the model using Factors 1, 2, 3, and 4.
 - The BIC is minimized with the model using Factor 1 only.
- If we are developing a model for prediction purposes, then we select the four-factor model that AIC indicates.
- If we are seeking the best-fitting model, we would choose the one-factor model upon BIC's indication.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...



Knowledge check

The research report you are reviewing presents goodness-of-fit statistics for the two models explaining the variation in return on assets (ROA) for the sample of diversified manufacturers, as follows:

You need to interpret the given goodness-of-fit statistics and justify which is the better model using these analysis steps:

- What is the implication of R^2 related to these models?
- What is the implication of $\bar{R}^2$ when ADV is added into model 2?
- Compare AIC and BIC between 2 models.
- Make your conclusion about the better model.

	R^2	$\bar{R}^2$	AIC	BIC
Model 1: CAPEX only	0.880	0.875	23.899	26.523
Model 2: CAPEX and ADV	0.881	0.870	25.804	28.792

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...



Knowledge check

Interpret the given goodness-of-fit statistics and justify which is the better model.

Solution:

The goodness-of-fit results for this sample regression indicate the following:

- 88% of the variation in ROA is explained by CAPEX, and 88.1% of this variation is explained by CAPEX and ADV together according to R^2 .
- $\bar{R}^2$ declines when ADV is added, indicating its coefficient is not significant and has a $|t\text{-statistic}| < 1.0$ (note that for ADV, the p-value = 0.7429 and the t-statistic is -0.3320).
- **AIC** and **BIC** both indicate that Model 1 is a better model as these metrics are lower for Model 1 than for Model 2.

→ To conclude, adding ADV does not improve the explanatory or predicting power of the original model using just CAPEX → model 1 is the better.

[LOS 2.a] Evaluate how well a multiple regression model explains the dependent variable by...

6.

Assessing Model Fit Using Multiple Regression Statistics

Importantly, finding the “best” model is not a straight-line process but is, rather, iterative, because it depends on reviewing the regression results and adjusting the model accordingly.

Statistic	Criterion to use in assessment
Adjusted R^2 ($\bar{R}^2$)	The higher the better
Akaike’s information criterion (AIC)	The lower the better
Schwarz’s Bayesian information criterion (BIC)	The lower the better
t-statistic on a slope coefficient	$ T(\text{test-statistic}) > T_c \text{ (critical value)} = t_{\alpha/2}^{(n-k-1)}$
F-test for joint test of slope coefficients	$F(\text{test-statistic}) > F_c \text{ (critical value)} = F_{\alpha}^{(k, n-k-1)}$

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and interpret the results of the joint hypothesis tests

1.

Using t-test in testing the significance of single coefficient

T-test is a statistical method that can be used to evaluate the means of one or two populations.

Here, we will use a t-test to assess the statistical significance of a coefficient, which means to compare its estimated value to 0.



Hypotheses

If we are testing simply whether a variable is significant in explaining the dependent variable's variation, the hypotheses simply are:

$$H_0: b_j = 0 \text{ versus } H_a: b_j \neq 0$$



Test statistic

$$T = \frac{\widehat{b_1}}{s_{\widehat{b_1}}}$$

where:

$\widehat{b_1}$ is estimated slope coefficient

$s_{\widehat{b_1}}$ is standard error of slope coefficient

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

1.

Using t-test in testing the significance of single coefficient

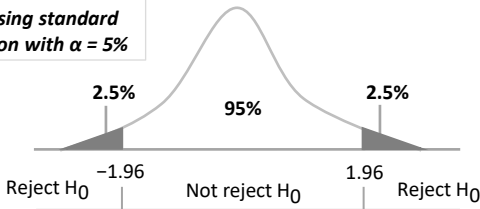


Decision rule

Reject H_0 if: $|T \text{ (test-statistic)}| > T_c \text{ (critical value)}$ with $T_c = t_{\alpha/2}^{(n-k-1)}$

→ The coefficient b_j has the statistical significance in the multiple regression.

Two-tailed test using standard normal distribution with $\alpha = 5\%$



At the 5% level of significance of a two-tailed test, the critical z-values for the confidence intervals are $z_{0.025} = 1.96$ and $-z_{0.025} = -1.96$.

→ Reject H_0 if test statistic > 1.96 or test statistic < -1.96 .

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

1.

Using t-test in testing the significance of single coefficient

An alternative method of testing the significance of the coefficients is to compare the p-value to the significance level (α). **P-value** is the smallest level of significance for which the null hypothesis can be rejected.

Decision rule:

- If the p-value $< \alpha \rightarrow$ reject H_0 .
- If the p-value $> \alpha \rightarrow$ cannot reject H_0 .

❌ **Limitation of t-test:** They provide a less meaningful evaluation of the explanatory power of alternative models, especially when we want to test a subset of variables in a multiple regression jointly.

→ We can use an alternative method to adjust this limitation like F-test that will be described in the next slide.

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

2.

Evaluating overall model fit using F-test

F-test can be used to test whether at least one of the independent variables explains a significant portion of the variation of the dependent variable.



Hypotheses

The F-statistic is always a one-tailed test. The hypotheses are structured as:

$$H_0: b_1^2 + b_2^2 + \dots + b_k^2 = 0 \text{ (all every coefficient is equal to 0).}$$

$$H_a: b_1^2 + b_2^2 + \dots + b_k^2 > 0 \text{ (at least one coefficient is different from 0).}$$



Test statistic

$$F = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n-k-1)}$$

where:

MSR = mean regression sum of squares

MSE = mean squared error

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

2.

Evaluating overall model fit using F-test

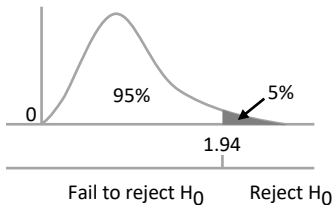


Decision rule

Reject H_0 if F (test-statistic) $> F_c$ (critical value) with $F_c = F_{\alpha}^{(k, n-k-1)}$

→ At least one of the independent variables in the regression model makes a significant contribution to the explanation of the dependent variable.

Rejection regions for F-distribution for two-tailed test at $\alpha = 5\%$, $k = 30$, $n = 71$



At the 5% level of significance of a one-tailed test, the critical z-values for the confidence intervals are $F_{0.05}^{(30, 40)} = 1.94$.

→ Reject H_0 if test statistic > 1.94 .

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

2.

Evaluating overall model fit using F-test

Similar to t-test, we have an alternative method of testing the significance of group of coefficients is to compare the p-value to the significance level (α):

- If the p-value $< \alpha \rightarrow$ reject H_0 .
- If the p-value $> \alpha \rightarrow$ cannot reject H_0 .

Example 4: Calculating and interpreting the F -statistic

An analyst runs a regression of monthly value-stock returns on five independent variables over 60 months.

The total sum of squares is 460, and the sum of squared errors is 170. Test the null hypothesis at the 5% significance level that all five of the independent variables are equal to zero.

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

2.

Evaluating overall model fit using F-test

Example 4: Calculating and interpreting the *F*-statistic

An analyst runs a regression of monthly value-stock returns on five independent variables over 60 months.

The total sum of squares is 460, and the sum of squared errors is 170. Test the null hypothesis at the 5% significance level that all five of the independent variables are equal to zero.

Answer:

- Hypotheses:**

$$H_0 : b_1^2 + b_2^2 + \dots + b_5^2 = 0 \text{ versus } H_a : b_1^2 + b_2^2 + \dots + b_5^2 > 0$$

- Calculate test statistic:**

$$SSR = SST - SSE = 460 - 170 = 290 \quad MSR = \frac{290}{5} = 58.0$$

$$MSE = \frac{170}{60-5-1} = 3.15 \quad \rightarrow F = \frac{58}{3.15} = 18.41$$

- Make the decision:**

$$F = 18.41 > F_{0.05}^{(5, 54)} = 2.4 \rightarrow \text{Reject the null hypothesis}$$

→ At least one of the five independent variables is significantly different than zero.

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

3.

Evaluating a nested model using F-test

Unrestricted model

$$Y_i = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \varepsilon_i$$

We refer to this as **full** model, with all three independent variables, as the **unrestricted** model.

Restricted model

$$Y_i = b_0 + b_1X_1 + \varepsilon_i$$

We refer to this as **restricted** model, in which we restricted the slope coefficients on **X_2** and **X_3** to be equal to zero.

Nested models are models such that **full model** or **unrestricted model** has a higher number of independent variables while the **restricted model** has only a subset of the independent variables.



How to know which model is the better?

The F-test evaluates whether the inclusion of X_2 and X_3 into restricted model can makes the SEE smaller (*which means they are statistically significant*). If it is, the full model is the better, and vice versa.

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

3.

Evaluating a nested model using F-test



Hypotheses

The hypotheses are in this case can be structured as:

$$H_0: b_2^2 + b_3^2 = 0 \text{ (the coefficients of the added variables are equal to 0)}$$

$$H_a: b_2^2 + b_3^2 > 0 \text{ (at least one of the coefficients is different from 0)}$$



Test statistic

$$F = \frac{(SSE_R - SSE_U)/q}{SSE_U/(n - k - 1)}$$

where:

R and U represent the restricted and unrestricted models

q = number of excluded variables in the restricted model.



Decision rule

Reject H_0 if F (test-statistic) $> F_c$ (critical value) with $F_c = F_{\alpha}^{(q, n-k-1)}$

→ The full model is the better.

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

3.

Evaluating a nested model using F-test

Example 3: Calculating and interpreting the F -statistic

An analyst wants to model the determinants of rents for office properties in a large city in the United States. Using a sample of 191 observations, she has estimated the following model:

$$\text{rent}_i = b_0 + b_1 \text{age}_i + b_2 \text{distance}_i + b_3 \text{restaurant}_i + \varepsilon_i$$

Using the information below, conduct a joint test of hypotheses for the slope coefficients 2 and 3 at a 5% level of significance.

Model	k	SSE	n
Unrestricted	3	13,550.64	191
Restricted	1	32,627.29	191

Partial F-table (5% level of significance)

Denominator degrees of freedom	Numerator degrees of freedom		
	1	2	3
1	161	200	216
30	4.17	3.32	2.92
60	4.00	3.15	2.76
120	3.92	3.07	2.68
190	3.89	3.04	2.65
∞	3.84	3.00	2.60

[LOS 2.b] Formulate hypotheses on the significance of two or more coefficients in a multiple regression model and...

3.

Evaluating a nested model using F-test

Example 3: Calculating and interpreting the F -statistic

Answer:

- Hypothesis:

$$H_0: b_2^2 + b_3^2 = 0 \text{ versus } H_a: b_2^2 + b_3^2 > 0$$

- Calculate test statistic:

$$F = \frac{(SSE_R - SSE_U)/q}{SSE_U/(n - k - 1)} = \frac{(32,627.29 - 12,550.64)/2}{13,550.64/(191 - 3 - 1)} = 131.63$$

- Make the decision:

	Numerator degrees of freedom		
Denominator degrees of freedom	1	2	3
120	3.92	3.07	2.68
190	3.89	3.04	2.65
∞	3.84	3.00	2.60

$F \text{ statistic} = 131.63 > F_{0.05}^{(2, 187)} = 3.04 \rightarrow \text{Reject } H_0 \rightarrow \text{At least one of the two slope coefficients of the excluded variables is statistically different from 0.}$

[LOS 2.c] Calculate and interpret a predicted value for the dependent variable, given the estimated regression model and assumed values for the independent variable

1.

Calculating the predicted value

The process of calculating the predicted value is similar to forecasting with simple linear regression, only now we need predicted values for more than one independent variable.

The predicted value of dependent variable Y is:

$$\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_{1i} + \hat{b}_2 X_{2i} + \dots + \hat{b}_k X_{ki}$$

where:

$\hat{Y}_i$ = the predicted value of the dependent variable

$\hat{b}_j$ = the estimated slope coefficient for the j^{th} independent variable

X_{ji} = the forecast of the j^{th} independent variable, $j = 1, 2, \dots, k$

[LOS 2.c] Calculate and interpret a predicted value for the dependent variable, given...

Knowledge check

An analyst would like to use this estimated regression equation to calculate the predicted 10-year real earnings growth for the S&P 500, assuming the payout ratio of the index is 50%. He observes that the slope of the yield curve is currently 4%, and the coefficients estimated are as following:

	Coefficient
Intercept ($\widehat{b_0}$)	-11.6%
PR ($\widehat{b_1}$)	0.25
YCS ($\widehat{b_2}$)	0.14

What is the predicted 10-year real earnings growth for the S&P 500?

[LOS 2.c] Calculate and interpret a predicted value for the dependent variable, given...

Knowledge check

An analyst would like to use this estimated regression equation to calculate the predicted 10-year real earnings growth for the S&P 500, assuming the payout ratio of the index is 50%. He observes that the slope of the yield curve is currently 4%, and the coefficients estimated are as following:

	Coefficient
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PR ($\widehat{b_1}$)	0.25
YCS ($\widehat{b_2}$)	0.14

What is the predicted 10-year real earnings growth for the S&P 500?

Answer:

The predicted 10-year real earnings growth for the S&P 500:

$$\widehat{EG}_{10} = -11.6\% + 0.25(50\%) + 0.14(4\%) = 1.46\%$$

The model predicts a 1.46% real earnings growth rate for the S&P 500, assuming a 50% payout ratio, when the slope of the yield curve is 4%.



MODULE 3 : Model Misspecification

LEARNING OUTCOMES

- 3.a.** Describe how model misspecification affects the results of a regression analysis and how to avoid common forms of misspecification
- 3.b.** Explain the types of heteroskedasticity and how it affects statistical inference
- 3.c.** Explain serial correlation and how it affects statistical inference
- 3.d.** Explain multicollinearity and how it affects regression analysis

[LOS 3.a] Describe how model misspecification affects the results of a regression analysis and how to avoid common forms of misspecification

1.

Model specification

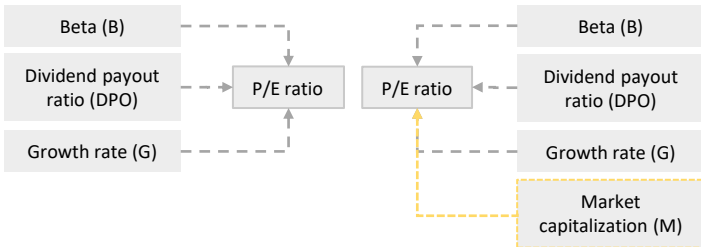
Regression model specification is the selection of the explanatory (independent) variables to be included in the regression and the transformations, if any, of those explanatory variables.

Example: Suppose we're trying to predict a P/E ratio using a cross-sectional regression with fundamental variables that are related to P/E.


Specification 1

Specification 2

$$P/E = b_0 + b_1 DPO + b_2 G + b_3 B + \varepsilon$$



[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

2.

Principles for Proper Regression Model Specification

Principle

Explanation

Model should be grounded in economic reasoning.

Provide economic reasoning behind choice of variables.

Model should be parsimonious.

Each variable included in regression should play an essential role.

Model should perform well out of sample.

Model may explain only the specific dataset on which it was trained, meaning it is overfit.

Model functional form should be appropriate.

If a nonlinear relationship between regressors is expected, model should incorporate the appropriate nonlinear terms.

Model should satisfy regression assumptions.

If heteroskedasticity, serial correlation, or multicollinearity are detected, revise regression variables and/or functional form.

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

3.

Types of Functional Form Model Misspecifications

Misspecification	Description	Effect
Omission of important variables	Based on economic theory, one or more variables that should have been included are omitted.	Biased and inconsistent regression parameters
Inappropriate variable form	The relationship between the dependent and independent variables may be non-linear.	May lead to serial correlation or heteroskedasticity in the residuals
Inappropriate variable scaling	Variables may need to be transformed before estimating the regression.	May lead to heteroskedasticity in the residuals or multicollinearity
Data is improperly pooled	Sample has periods of dissimilar economic environments (that should not be pooled).	May lead to heteroskedasticity or serial correlation in the residuals

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

3.

Types of Functional Form Model Misspecifications

Illustration of Misspecification of functional form

Let's start with a regression that predicts monthly returns on portfolios of Chinese stocks (R) using four independent variables: portfolio beta (B), the natural log of market capitalization ($\ln M$), the natural log of the price-to-book ratio $\ln(PB)$, and free float (FF).

The regression is estimated with 72 monthly observations from July 1996 to June 2002. The **correct** specification of the model is as follows:

$$R = b_0 + b_1 B + b_2 \ln M + b_3 \ln PB + b_4 FF + \epsilon$$

Portfolio beta (B)

Natural log of market capitalization
($\ln M$)

Natural log of P/B ($\ln PB$)

Free float (FF)

Monthly return (R)

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

3.

Types of Functional Form Model Misspecifications

3.1. Misspecification 1: Omitted variables

Suppose we do not include $\ln M$ in the regression model:

$$R = a_0 + a_1 B + a_2 \ln PB + a_3 FF + \epsilon$$

→ This can bias our coefficients if the omitted variable ($\ln M$) is correlated with either:

The dependent variable

One or more other independent variables (B , $\ln PB$, or FF)



The error term is correlated with the dependent variables and the resulting regression coefficients are biased and inconsistent.

The error term is also correlated with the same independent variables and the resulting regression coefficients are biased and inconsistent.



Our hypothesis tests and predictions using the model are unreliable.

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

3.

Types of Functional Form Model Misspecifications

3.2. Misspecification 2: Inappropriate form of variables

Suppose we include market cap in the regression without transforming it by taking the natural log - use M and not $\ln(M)$:

$$\times R = a_0 + a_1B + a_2M + a_3\ln PB + a_4FF + \varepsilon$$

We have misspecified the model because typically, market capitalization is **not linearly** related to portfolio returns, but rather the **natural log** of market cap is linearly related:

$$R = a_0 + a_1B + a_2\ln M + a_3\ln PB + a_4FF + \varepsilon$$



Our hypothesis tests and predictions using the model are unreliable.

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

3.

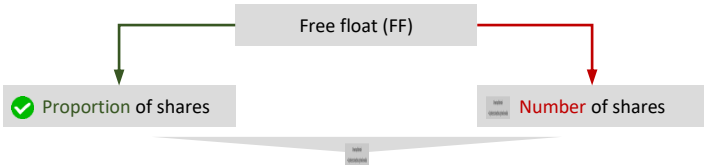
Types of Functional Form Model Misspecifications

3.3. Misspecification 3: Inappropriate scaling of variables

Suppose that rather than defining FF as the **proportion** of shares available to be traded, we instead use the **number** of shares available to be traded.

$$R = a_0 + a_1B + a_2\ln M + a_3\ln PB + a_4FF + \epsilon$$

We could have **similar companies** with very **different values for FF**, an example of inappropriate scaling.



We have misspecified the model

→ Our hypothesis tests and predictions using the model are unreliable.

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

3.

Types of Functional Form Model Misspecifications

3.4. Misspecification 4: Inappropriate Pooling of Data

Suppose that the relationship between returns and the independent variables during the **first three years** is:

$$R = a_0 + a_1B + a_2\ln M + a_3\ln PB + a_4FF + \varepsilon$$

and it during the **second three years** is:

$$R = b_0 + b_1B + b_2\ln M + b_3\ln PB + b_4FF + \varepsilon$$

The regression coefficients are different from one period to the next ($a_i \neq b_i$).

If we pool the data and estimating **one regression over the entire period**, rather than estimating **two separate regressions over each of the sub-periods**.



We have misspecified the model and our hypothesis tests and predictions of portfolio returns will be misleading.

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

4.

Recall: Violation of regression assumptions

Most of the assumptions made with the multiple regression pertain to ϵ , the model's error term:

01

Linearity: The relationship between the dependent variable and the independent variables is linear.

02

Homoskedasticity: The variance of the regression residuals is the same for all observations ($E(\epsilon_i^2) = \sigma_\epsilon^2$).

03

Independence of errors: The regression residuals are uncorrelated across observations ($\rho(\epsilon_i, \epsilon_j) = 0 \forall i \neq j$).

04

Independence of independent variables: There is no exact linear relation between two or more of the independent variables ($\rho(X_i, X_j) = 0$).

05

Normality: The regression residuals are normally distributed ($\epsilon \sim N$).

[LOS 3.a] *Describe* how model misspecification affects the results of a regression analysis and how to avoid misspecification

4.

Recall: Violation of regression assumptions

There are **three** primary assumption violations

02

03

04

Heteroskedasticity

Serial correlation

Multicollinearity

From Los 3.b, 3.c to 3.d, we will discuss these 03 violations by looking into the following aspects:

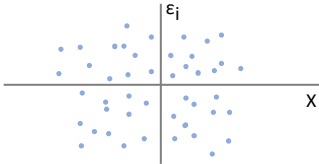
- What are they?
- What are the consequences?
- How to detect them?
- How to correct them?

[LOS 3.b] Explain the types of heteroskedasticity and how it affects statistical inference

1.

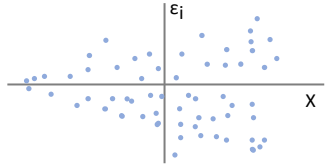
Heteroskedasticity

- **Heteroskedasticity** occurs when the variance of the residuals is not the same across all observations in the sample.
- **Heteroskedasticity** may arise from model misspecification, including omitted variables, incorrect functional form, and incorrect data transformations, as well as from extreme values of independent variables.



No heteroskedasticity

The variance of residuals is unchanged across observations (no relationship with the value of independent variable X).



Heteroskedasticity

The variance of residuals changes across observations (positive relationship with the value of independent variable X).

[LOS 3.b] *Explain* the types of heteroskedasticity and how it affects statistical inference

1.

Heteroskedasticity

Unconditional heteroskedasticity

- The heteroskedasticity is **not** related to the level of the independent variables.
- Usually cause **no major problems** with the regression.

Conditional heteroskedasticity

- The heteroskedasticity is related to the level of the independent variables.
- Create **significant problems** for statistical inference.

We will only consider **Conditional Heteroskedasticity** in the upcoming slides.

[LOS 3.b] *Explain* the types of heteroskedasticity and how it affects statistical inference

2.

Consequences of Heteroskedasticity

Conditional heteroskedasticity

- The heteroskedasticity is related to the level of the independent variables
- Create **significant problems** for statistical inference as the followings:

! The standard errors are usually unreliable estimates

- If standard errors are too **small**, but the coefficient estimates are not affected → the t -statistics will be too large → the null hypothesis (H_0) is rejected too often (**Type I error**).
- Similarly, if standard errors are too **large** → it is hard to reject H_0 (**Type II error**).

! The F -test is unreliable

The MSE is unreliable estimate as SSE is unreliable. If MSE is too **small**, but MSR is not affected
→ the F -statistics will be too large
→ the null hypothesis (H_0) is rejected too often, and vice versa.

[LOS 3.b] *Explain* the types of heteroskedasticity and how it affects statistical inference

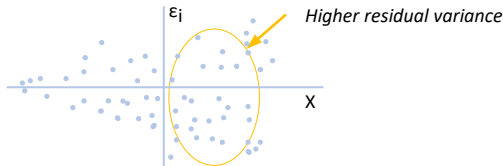
3.

Detecting Heteroskedasticity

Method 1: Examining scatter plots of the residuals

We can use scatter plot of residual to detect Heteroskedasticity. If the variance of residuals changes across observations, we can conclude that there is Conditional Heteroskedasticity.

Let's look at the following scatter plot of the error terms and the independent variables X :



The variation in the regression residuals increases as the independent variable increases.

→ **Conditional Heteroskedasticity.**

[LOS 3.b] Explain the types of heteroskedasticity and how it affects statistical inference

3.

Detecting Heteroskedasticity

Method 2: Using the Breusch-Pagan chi-square (χ^2) test

If conditional heteroskedasticity is present, the independent variables will significantly contribute to the explanation of the squared residuals.

Breusch-Pagan test calls for the regression of the squared residuals on the independent variables as the following:

$$\epsilon_i^2 = a_0 + a_1X_1 + a_2X_2 + \dots + a_kX_k \quad (1)$$



Hypotheses

Breusch-Pagan test is a one-tailed test and the hypotheses are structured as:

$$H_0: a_1^2 + a_2^2 + \dots + a_k^2 = 0 \text{ (no conditional heteroskedasticity)}$$

$$H_a: a_1^2 + a_2^2 + \dots + a_k^2 > 0 \text{ (conditional heteroskedasticity)}$$



Test statistic

$$\text{BP chi-square test} = n \times R_{\text{resid}}^2 = n \times \frac{\text{SSR}(1)}{\text{SST}(1)}$$

where:

n = the number of observations

$R_{\text{resid}}^2 = R^2$ from regression (1), $\text{SSR}(1)$ is regression sum of squares from regression (1), $\text{SST}(1)$ is total sum of squares from regression (1)

[LOS 3.b] *Explain* the types of heteroskedasticity and how it affects statistical inference

3.

Detecting Heteroskedasticity



Decision rule

Reject H_0 if BP chi-square test $> \chi_c$ (critical value) with $\chi_c = \chi_{\alpha}^{2(k)}$

→ The heteroskedasticity is present (*there is a significant relationship between independent variables and residuals*).

Example 1: The Breusch-Pagan test

The residual plot of mutual fund returns over time shows evidence of heteroskedasticity. To confirm your suspicions, you regress the squared residuals from the original regression on the independent variable, S&P 500 index returns. The R^2 from that regression is 8%. Use the Breusch-Pagan test to determine whether heteroskedasticity is present at the 5% significance level, known that $\chi_{0.05}^{2(1)} = 3.841$.

[LOS 3.b] *Explain* the types of heteroskedasticity and how it affects statistical inference

3.

Detecting Heteroskedasticity

Example 1: The Breusch-Pagan test

The residual plot of mutual fund returns over time shows evidence of heteroskedasticity. To confirm your suspicions, you regress the squared residuals from the original regression on the independent variable, S&P 500 index returns. The R^2 from that regression is 8%. Use the Breusch-Pagan test to determine whether heteroskedasticity is present at the 5% significance level, known that $\chi_{0.05}^{2(1)} = 3.841$.

Answer:

Hypotheses:

H_0 : no conditional heteroskedasticity

H_a : conditional heteroskedasticity

Calculate test statistic:

BP chi-square test = $n \times R^2 = 60 \times 0.08 = 4.8$

Make decision:

Reject H_0 because BP chi-square test = $4.8 > \chi_{0.05}^{2(1)} = 3.841$

→ Conditional heteroscedasticity exists.

[LOS 3.b] Explain the types of heteroskedasticity and how it affects statistical inference

4.

Correcting Heteroskedasticity

- The most common remedy for heteroskedasticity is to use **Robust standard errors** as they **adjust the standard errors** of the regression's estimated coefficients to account for the heteroskedasticity.
- This correction is also known as White-corrected standard errors or heteroskedasticity consistent standard errors.
- We then use the Robust standard errors to recalculate the t-statistics using the original regression coefficients.

	Coefficient	Standard Error
Intercept	-0.0026	0.0024
MKTRF	1.2414	0.0601
SMB	1.0953	0.1042
HML	-0.1065	0.0872

Robust SE correction



...	Robust Standard Error
...	0.0021
...	0.0910
...	0.1111
...	0.1015

Note: The regression coefficients did not change under Robust SE correction.

[LOS 3.b] Explain the types of heteroskedasticity and how it affects statistical inference

4.

Correcting Heteroskedasticity

Example 2: Using White-corrected standard errors

An analyst runs a regression of annualized Treasury bill rates (the dependent variable) on annual inflation rates (the independent variable) using monthly data for 10 years. The results of the regression are shown in the following table.

Regression of T-Bill Rates on Inflation Rates

Variable	Coefficient	Standard Error	t-Statistic	p-Value
<i>Intercept</i>	4.82	0.85	5.67	< 0.0001
<i>Inflation</i>	0.60	0.28	2.14	0.0340

He determines using the Breusch-Pagan test that heteroskedasticity is present, so he also estimates the White-corrected standard error for the coefficient on inflation to be 0.31. The critical two-tail 5% t-value for 118 degrees of freedom is 1.98. Is inflation statistically significant at the 5% level?

[LOS 3.b] *Explain* the types of heteroskedasticity and how it affects statistical inference

4.

Correcting Heteroskedasticity

Example 2: Using White-corrected standard errors

Is inflation statistically significant at the 5% level?

Answer:

There is evidence of heteroskedasticity

→ We used White-corrected standard error and the original regression coefficients to recalculate the t -statistics.

$$t = \frac{\text{Coefficient}}{\text{White-corrected standard error}} = \frac{0.6}{0.31} = 1.94$$

Because critical t -value = 1.98

→ t -statistic < critical t -value

→ The null hypothesis that the inflation coefficient is zero cannot be rejected.

→ Inflation is not statistically significant.

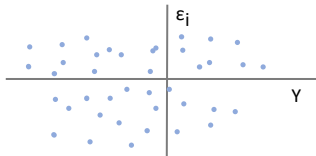
Note: The original standard error of 0.28 was too low (underestimated) → the original t -statistic of 2.14 was too high. After using the higher White-corrected standard error of 0.31, the t -statistic fell to 1.94.

[LOS 3.c] *Explain* serial correlation and how it affects statistical inference

1.

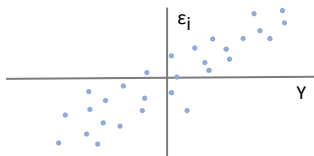
Serial correlation

Serial correlation refers to the situation in which the residual terms are correlated with one another (common problem with time series data, which will be illustrated in Module 5).



No serial correlation

The residuals behave in an independent manner among observations (not correlated with dependent variable Y).



Serial correlation

The residuals are positively correlated among observations (positively correlated with dependent variable Y).

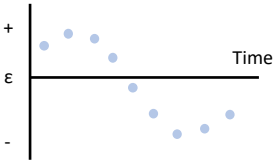
[LOS 3.c] Explain serial correlation and how it affects statistical inference

1.

Serial correlation

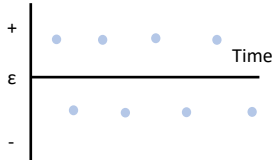
Positive serial correlation

When a *positive regression error* in one time period increases the *probability of observing a **positive regression error*** for the next time period



Negative serial correlation

When a *positive regression error* in one time period *increases the probability of observing a **negative regression error*** for the next time period



We focus solely on **positive serial correlation** and **first-order serial correlation**, which involves the correlation between adjacent observations, as these are the most common types in reality.

[LOS 3.c] Explain serial correlation and how it affects statistical inference

2.

Consequences of Serial correlation

Impact of Serial Correlation on Multiple Regression Model

Independent Variable Is Lagged Value of Dependent Variable	Invalid Coefficient Estimates	Invalid Standard Error Estimates
No	No	Yes
Yes	Yes	Yes

The **main problem** caused by serial correlation in linear regression is an incorrect estimate of the regression coefficients' **standard errors** and **MSE** as the following:



Coefficient standard errors underestimated



Mean squared error (MSE) underestimated

Inflated t -stat

Inflated F -stat

More likely to reject H_0 (Too many Type I errors)

[LOS 3.c] Explain serial correlation and how it affects statistical inference

3. Detecting Serial correlation

Method 1: Breusch-Godfrey (BG) test (more generally used)

We use **Breusch-Godfrey (BG) test** for detecting serial correlation *at multiple lags*, assume m lags.

- Regress the **regression residuals** against the **original set of independent variables**, plus m additional variables representing **lagged residuals**.

$$\hat{\epsilon}_t = a_0 + a_1 X_{1t} + a_2 X_{2t} + \dots + a_k X_{kt} + p_1 \epsilon_{t-1} + p_2 \epsilon_{t-2} + \dots + p_m \epsilon_{t-m}$$

- Test whether slope coefficient for **at least one of the lags** is statistically significantly **different from 0**.



Hypotheses

Breusch-Godfrey (BG) test is a F-test and the hypotheses are structured as:

$$H_0: p_1^2 + p_2^2 + \dots + p_m^2 = 0 \text{ (no serial correlation)}$$

$$H_a: p_1^2 + p_2^2 + \dots + p_m^2 > 0 \text{ (serial correlation)}$$



Test statistic

The BG F-statistic is provided in most statistical software.

[LOS 3.c] *Explain* serial correlation and how it affects statistical inference

3.

Detecting Serial correlation

Method 1: Breusch-Godfrey (BG) test (more general used)

We use **Breusch-Godfrey (BG) test** for detecting serial correlation *at multiple lags*, assume m lags.

$$\hat{\varepsilon}_t = a_0 + a_1 X_{1t} + a_2 X_{2t} + \dots + a_k X_{kt} + p_1 \varepsilon_{t-1} + p_2 \varepsilon_{t-2} + \dots + p_m \varepsilon_{t-m}$$



Decision rule

Reject H_0 if $F\text{-test} > F_C$ (critical value) with $F_C = F_{\alpha}^{(m, n-k-1-m)}$

→ The serial correlation is present (there is a significant relationship between two or more residuals).

[LOS 3.c] Explain serial correlation and how it affects statistical inference

3. Detecting Serial correlation

Method 2: Durbin-Watson test

We can use **Durbin-Watson** test for detecting residual serial correlation at a *single lag*.



Hypotheses

The hypotheses of DW test are simply structured as:

H_0 : no serial correlation

H_a : serial correlation



Test statistic

- $DW = \frac{\sum_{t=2}^T (\hat{\epsilon}_t - \hat{\epsilon}_{t-1})^2}{\sum_{t=2}^T \hat{\epsilon}_t^2}$ (This DW-statistic is provided in most statistical software).
- $DW \approx 2(1 - r)$ if the sample is very large.

[LOS 3.c] Explain serial correlation and how it affects statistical inference

3. Detecting Serial correlation

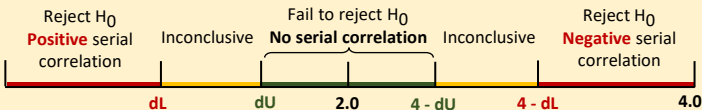
Method 2: Durbin-Watson test

We can use **Durbin-Watson** test for detecting residual serial correlation at a single lag.



Decision rule

Compare DW to the upper and lower critical DW-values (d_U and d_L)



[LOS 3.c] Explain serial correlation and how it affects statistical inference

4. Correcting Serial correlation

- The most common remedy for Serial correlation is to use **Robust standard errors** as they **adjust the standard errors** of the regression's estimated coefficients to account for the serial correlation (*and also correct for Conditional heteroskedasticity*).
- This correction is also known as serial-correlation consistent standard errors, serial correlation and heteroskedasticity adjusted standard errors, Newey–West standard errors.
- We then use the Robust standard errors to recalculate the t-statistics using the original regression coefficients.

	Coefficient	Standard Error
Intercept	0,0991	0.050
CONS	0,8696	0.027

Robust SE
correction



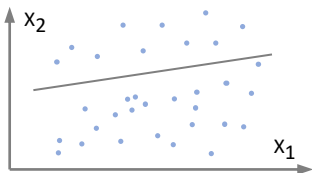
	Robust Standard Error
...	0.058
...	0.036

Note: The regression coefficients did not change under Robust SE correction.

[LOS 3.d] Explain multicollinearity and how it affects regression analysis

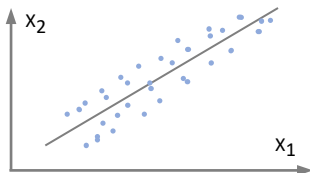
1. Multicollinearity

- **Multicollinearity** refers to the condition when two or more of the independent variables, or linear combinations of the independent variables, in a multiple regression are highly correlated with each other.
- **Multicollinearity** is a serious concern because approximate linear relationships among economic and financial variables are common.



✓ No heteroskedasticity

Two independent variables X_1 and X_2 have **little or no correlation**. This is a desirable characteristic between independent variables.



! Heteroskedasticity

Two independent variables X_1 and X_2 have **positive correlation**. This may be a sign of multicollinearity.

[LOS 3.d] *Explain* multicollinearity and how it affects regression analysis

2.

Consequences of Multicollinearity



The slope coefficients themselves are consistent but tend to be unreliable



The standard errors of the slope coefficients are artificially *inflated*

Standard errors are too **large** because we cannot assign precise credit for explaining the variation in the dependent variable to the independent variables.

→ It is hard to reject the null hypothesis (H_0).

→ Greater probability that we will incorrectly conclude that a variable is not statistically significant (Type II error).

[LOS 3.d] *Explain* multicollinearity and how it affects regression analysis

3.

Detecting Multicollinearity

Method 1: The most common sign

T-tests indicate that none of the individual coefficients is significantly different than zero.



F-test is statistically significant and the R^2 is high.

The individual independent variables do not explain much of the variation.

The variables together explain much of the variation in the dependent variable.

Multicollinearity

Method 2: Using Variance inflation factor (VIF)

More formally, we can quantify multicollinearity using the **variance inflation factor (VIF)** for each of the independent variables.

We start by regressing one of the independent variable j against the remainders :

$$\hat{x}_{jt} = a_0 + a_1x_{1t} + a_2x_{2t} + \dots + a_kx_{kt}$$

[LOS 3.d] *Explain* multicollinearity and how it affects regression analysis

3.

Detecting Multicollinearity



Hypotheses

The hypotheses of VIF test are simply structured as:

H_0 : no multicollinearity

H_a : multicollinearity



Test statistic

$$VIF_j = 1/(1 - R_j^2)$$

where R_j^2 - the variation in X_j explained by the other k independent variables



Decision rule

Reject H_0 if VIF_j is large enough, because VIF increases as the correlation increases:

$VIF_j > 5 \rightarrow$ further investigation of the given independent variable.

$VIF_j > 10 \rightarrow$ serious multicollinearity requiring correction.

For a given independent variable, X_j , the minimum VIF_j is 1, which occurs when R_j^2 is 0.

$\rightarrow$ Do not reject H_0

$\rightarrow$ No multicollinearity (No correlation between X_j and the remaining independent variables).

[LOS 3.d] *Explain* multicollinearity and how it affects regression analysis

3. Detecting Multicollinearity

Example 3: Detecting multicollinearity using common sign

Bob Watson, CFA, runs a regression of mutual fund returns on average P/B, average P/E, and average market capitalization, with the following results:

Variable	Coefficient	p-Value
Average P/B	3.52	0.15
Average P/E	2.78	0.21
Market Cap	4.03	0.11
F-test	34.6	< 0.001
R^2	89.6%	

Determine whether or not multicollinearity is a problem in this regression.

[LOS 3.d] *Explain* multicollinearity and how it affects regression analysis

3.

Detecting Multicollinearity

Example 3: Detecting multicollinearity using common sign

Determine whether or not multicollinearity is a problem in this regression.

Answer:

- The R^2 is high and the F -test is statistically significant → the three variables *as a group* can explain the variation in mutual fund returns quite well.
 - However, none of the independent variables *individually* is statistically significant to any reasonable degree, since the p -values are larger than 10%.
- This is a indication of multicollinearity.

[LOS 3.d] *Explain* multicollinearity and how it affects regression analysis

4.

Correcting Multicollinearity

- 01 Excluding one or more of the regression variables
- 02 Using a different proxy for one of the variables
- 03 Increasing the sample size

There is no easy solution for handling multicollinearity.

→ Experiment with including or excluding different independent variables to determine the source of and best solution to multicollinearity.

If the goal is simply to predict the dependent variable - rather than to understand the roles of the independent variables

→ **Multicollinearity may not be a major issue.**

[LOS 3.d] *Explain multicollinearity and how it affects regression analysis*

Knowledge check

Returning to the example of the three-feature (CAPEX, ADV, R&D) ROA model for diversified manufacturers, for which the $\bar{R}^2$ is 86.50% and the overall F-statistic is 54.4039 (partial ANOVA results shown below), the senior analyst is concerned about potential multicollinearity. So, she asks you, the junior analyst, to estimate VIF for each feature and then do the following:

	Coefficient	Standard Error	t-Statistic	p-Value	VIF
Intercept	4.7022	1.2349	3.8078	0.0010	
CAPEX	1.2302	0.1056	11.6490	0.0000	1.164
ADC	-0.0371	0.1062	-0.3490	0.7304	1.068
R&D	0.1029	0.2837	0.3628	0.72003	1.095

1. Determine whether multicollinearity is a concern using the regression results alone.
2. Determine whether multicollinearity is a concern using the VIF results.

[LOS 3.d] *Explain* multicollinearity and how it affects regression analysis



Knowledge check

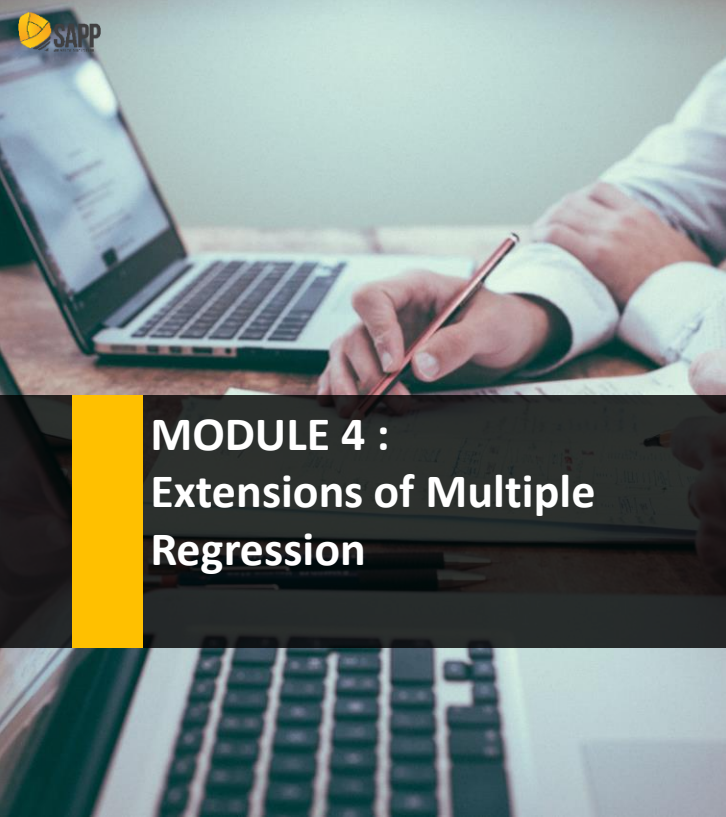
1. *Determine whether multicollinearity is a concern using the regression results alone.*
2. *Determine whether multicollinearity is a concern using the VIF results.*

Solution:

1. The overall F-statistic is significant, indicating at least one slope coefficient is non-zero. This is consistent with CAPEX's highly significant t-statistic, so multicollinearity should not be a concern.
2. VIF for all features is well below 5 (closer to 1), indicating variances are not inflated, t-statistics are reliable, and multicollinearity is not an issue.

Summary of Violations of Assumptions from Model Misspecification

Assumption	Violation	Issue	Detection	Correction
Homoskedastic error terms	Heteroskedastic error terms	Biased estimates of coefficients' standard errors	Visual inspection of residuals; Breusch – Pagan (BP) test	Revise model; use robust standard errors
Independence of observations	Serial correlation	Inconsistent estimates of coefficients and biased standard errors	Breusch – Godfrey (BG) test	Revise model; use serial correlation consistent standard errors
Independence of independent variables	Multicollinearity	Inflated standard errors	Variance inflation factor	Revise model; increase sample size



MODULE 4 : Extensions of Multiple Regression

LEARNING OUTCOMES

- 4.a.** Describe influence analysis and methods of detecting influential data points
- 4.b.** Formulate and interpret a multiple regression model that includes qualitative independent variables
- 4.c.** Formulate and interpret a logistic regression model

[LOS 4.a] *Describe* influence analysis and methods of detecting influential data points

1.

Influential observation

Definition

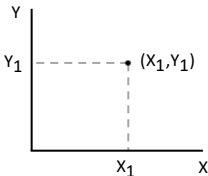
An **influential observation** is an observation whose inclusion may significantly alter regression results.



Recall: Observation

On a scatterplot, an observation has its own x and y values.

As a result, extreme values of x and y may make that observation influential and affect the regression results.



An
influential
observation
may possess

High-leverage point

A data point having an extreme value of an independent variable (X)

Outlier

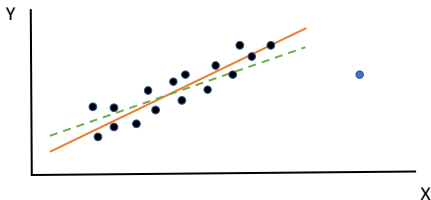
A data point having an extreme value of a dependent variable (Y)

[LOS 4.a] Describe influence analysis and methods of detecting influential data points

1.

Influential observation

Illustration of High-Leverage Point



This plot shows a **high-leverage point** in a sample of observations:

- Its X value does not follow the trend and is extremely high relative to the other observations.

There are two estimated regression lines:

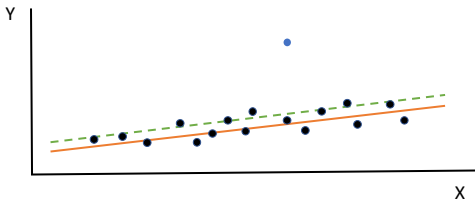
- The **dashed line** includes the **high-leverage point** in the regression sample.
- The **solid line** deletes it from the regression sample.

[LOS 4.a] Describe influence analysis and methods of detecting influential data points

1.

Influential observation

Illustration of Potentially Influential Outlier



This plot shows **an outlier data point** in a sample of observations:

- Its Y value does not follow the trend and is extremely high relative to its predicted value $\hat{Y}$, resulting in a large residual $(Y - \hat{Y})$.

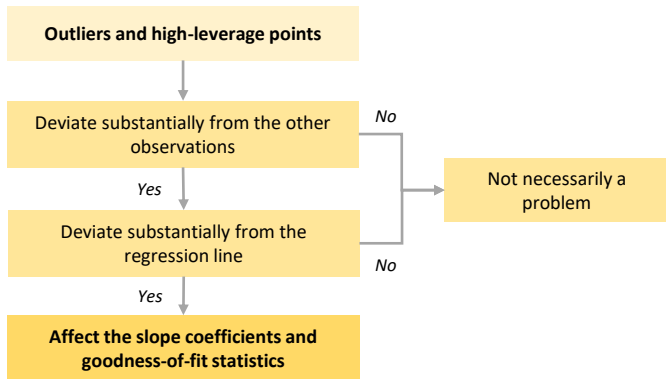
There are two estimated regression lines:

- The **dashed line** includes the **outlier** in the regression sample.
- The **solid line** deletes it from the regression sample.

[LOS 4.a] *Describe* influence analysis and methods of detecting influential data points

2.

Consequences of Influential observation



Both outliers and high-leverage points are substantially different from the majority of sample observations, but each presents itself in different ways.

→ These observations should be investigated to determine whether they are influential.

[LOS 4.a] Describe influence analysis and methods of detecting influential data points

3.

Using Leverage for detecting High-leverage point

Definition

Leverage (h_{ij}) is a measure of the distance between the i th observation of one independent variable relative to its sample mean.

Value of Leverage

$$0 < h_{ij} < 1$$

Higher value of leverage

Greater distance from mean

Higher potential influence

Detecting influential observation

$$h_{ij} > \frac{3(k+1)}{n}$$

→ The observation is potentially influential

where:

k is the number of independent variables

n is the number of observations

[LOS 4.a] Describe influence analysis and methods of detecting influential data points

4.

Using Studentized residuals for detecting Outliers

Definition

Studentized residuals (t_i^*) is a form of a Student's t-statistic resulting from the division of a residual by an estimate of its standard deviation.

Detecting Outliers with Studentized residuals

1

Estimate the initial regression model with n observations, then sequentially delete observations, one at a time, and each time re-estimate the regression model on the remaining $n - 1$ observations.

Illustration: The initial regression model has 10 observations, with 4 independent variables (A, B, C, D)

	Variable A	Variable B	Variable C	Variable D	Predicted value
1 st observation	A_1	B_1	C_1	D_1	$\widehat{Y}_1$
2 nd observation	A_2	B_2	C_2	D_2	$\widehat{Y}_2$
...	...	...	...	...	
10 th observation	A_{10}	B_{10}	C_{10}	D_{10}	$\widehat{Y}_{10}$

[LOS 4.a] Describe influence analysis and methods of detecting influential data points

4.

Using Studentized residuals for detecting Outliers

Detecting Outliers with Studentized residuals

1

Estimate the initial regression model with n observations, then sequentially delete observations, one at a time, and each time re-estimate the regression model on the remaining $n - 1$ observations.

Illustration: Delete the 2nd observation, we get a new regression model with 9 observations, with 4 independent variables (A, B, C, D) and new coefficients.

We want to see if the model can predict the value of the deleted observation, if not, it is an outlier.

	Variable A	Variable B	Variable C	Variable D	Predicted value
1 st observation	A_1	B_1	C_1	D_1	$\widehat{Y}_1$
3 rd observation	A_3	B_3	C_3	D_3	$\widehat{Y}_3$
...	...	...	...	...	
10 th observation	A_{10}	B_{10}	C_{10}	D_{10}	$\widehat{Y}_{10}$

Regression model with 9 observation generates $\widehat{Y}^*$

→ Substitute the independent variables of the deleted observation 2 to get $\widehat{Y}_2^*$

[LOS 4.a] *Describe* influence analysis and methods of detecting influential data points

4.

Using Studentized residuals for detecting Outliers

Detecting Outliers with Studentized residuals

2

For a given observation i , calculate the difference (residual) between actual Y value (Y_i) with the predicted Y -values resulting from the models with the i th observation deleted ($\widehat{Y}_i^*$), to get $e_i^* = Y_i - \widehat{Y}_i^*$.

Illustration: We calculate the difference between actual Y value of the deleted observation 2 with its predicted Y value from the new model generated from *step 1*.

$$e_2^* = Y_2 - \widehat{Y}_2^*$$

Note: This process is repeated $n - 1$ times. Each of the observation will be removed each attempt to generate a new regression model, and we get a residual for that observation.

From the set of residual, we can calculate the standard deviation of the residuals (s_e^*).

[LOS 4.a] *Describe* influence analysis and methods of detecting influential data points

4.

Using Studentized residuals for detecting Outliers

Detecting Outliers with Studentized residuals

3

Divide the residual calculated in step 2 by the estimated standard deviation to calculate studentized residual.

$$t_i^* = \frac{e_i^*}{s_e^*} = e_i \cdot \sqrt{\frac{n - k - 1}{SSE(1 - h_{ii}) - e_i^2}}$$

where:

e_i^* is the residual with the i th observation deleted

s_e^* is the standard deviation of the residuals

k is the number of independent variables

SSE is the sum of squares error of the initial regression model

h_{ii} is the leverage value for the i th observation

4

Compare t_i^* from step 3 with critical values from t -distribution with $n-k-2$ degrees of freedom.

If $|t_i^*| > \text{critical value } t\text{-statistic} \rightarrow \text{The observation is potentially influential.}$

[LOS 4.a] Describe influence analysis and methods of detecting influential data points

5.

Using Cook's D for detecting Influential observations

Definition

Cook's distance (D_j) is a metric for identifying influential data points. It measures *how much* the estimated values of the regression change if observation i is deleted from the sample.

Cook's D value

$$D_j = \frac{e_i^2}{k \times \text{MSE}} \left[\frac{h_{ii}}{(1 - h_{ii}^2)} \right]$$

where:

e_j is the residual for observation i

MSE is the mean square error of the estimated regression model

Guidelines

If D_j is greater than

Conclusion

0.5

The i th observation may be influential and merits further investigation

1.0

The i th observation is highly likely to be an influential data point

 $2\sqrt{k/n}$

The i th observation is highly likely to be an influential data point

[LOS 4.a] *Describe* influence analysis and methods of detecting influential data points

5.

Using Cook's D for detecting Influential observations

Key points about Cook's D

- It depends on both residuals and leverages (or dependent and independent variable information plays a role).
→ *Composite measure* for detecting extreme values of both types of variables.
- In one measure, it summarizes how much all of the regression's estimated values change when the i th observation is deleted from the sample.
- A large D_i indicates that the i th observation strongly influences the regression's estimated value.

[LOS 4.a] Describe influence analysis and methods of detecting influential data points

6.

Summary of Measures of Influential Observations

Measure	Influence of dependent variable	Influence of independent variable	Value	When observation is influential
Leverage		x	$0 < h_{ii} < 1$	$h_{ii} > \frac{3(k+1)}{n}$
Studentized residual	x		$t_i^* = \frac{e_i^*}{s_e^*}$ $= e_i \cdot \sqrt{\frac{n-k-1}{\text{SEE}(1-h_{ii}) - e_i^2}}$	$ t_i^* > t_{\alpha}^{n-k-2}$
Cook's distance	x	x	$D_i = \frac{e_i^2}{k \times \text{MSE}} \left[\frac{h_{ii}}{(1-h_{ii}^2)} \right]$	$D_i > 2\sqrt{k/n}$

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

1.

Defining a Dummy Variable

Definition

Dummy variable (or *indicator variable*) is the independent variable that is binary in nature.



Recall: Observation

Qualitative data represent qualitative outcomes (i.e., quality or characteristic) of a group of observations.

E.g.: Gender representation - 1 to represent “male” and 0 to represent “female”.

Qualitative events

A particular condition
is **true**

A particular condition
is **false**

Dummy variable takes
on value of “1”

Dummy variable takes
on value of “0”

Example: in a time-series regression (discussed later) of monthly stock returns, we could employ a “January” dummy variable that would take on “1”: **stock return occurred in January** and “0”: **it occurred in any other month**.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

1.

Defining a Dummy Variable

Choosing the number of dummy variables: To distinguish among n categories, we need $n - 1$ dummy variables.

Example: The following table illustrates how we use four dummy variables to denote companies belonging to one of five industry sectors: Technology, Financial Services, Health Care, Energy, Food & Beverage.

Industry sector	Dummy variables			
	Technology	Financial Services	Health Care	Energy
Technology	1	0	0	0
Financial Services	0	1	0	0
Health Care	0	0	1	0
Energy	0	0	0	1
Food & Beverage	0	0	0	0

The category not assigned becomes the “base” or “control” group, and the slope of each dummy variable is interpreted relative to the base.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

2.

Visualizing and Interpreting Dummy Variables

2.1. Intercept dummy

Consider a regression with one continuous X variable and one intercept dummy variable (D):

$$Y_i = b_0 + d_0 D_i + b_1 X_i + \varepsilon_i$$

Intercept dummy

$$D = 0$$

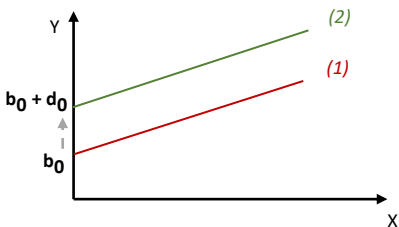
(D belongs in category 1)

$$Y = b_0 + b_1 X + \varepsilon \quad (1)$$

$$D = 1$$

(D belongs in category 2)

$$Y = (b_0 + d_0) + b_1 X + \varepsilon \quad (2)$$



The intercept of the line shifts from b_0 to $b_0 + d_0$.

The value of this shift is the vertical distance d_0 , which can be positive or negative.

Interpretation of d_0 : When X is constant, Y of category 2 is more than that of the category 1 by d_0 unit.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

2.

Visualizing and Interpreting Dummy Variables

2.2. Slope dummy

Consider a regression with one continuous X variable and one slope dummy variable (D):

$$Y_i = b_0 + b_1X_i + d_1 D_i X_i + \varepsilon_i$$

Slope dummy

$$D = 0$$

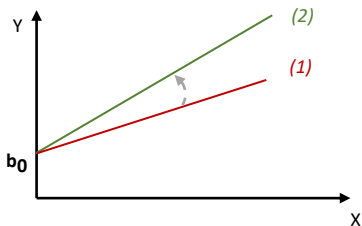
(D belongs in category 1)

$$Y = b_0 + b_1X + \varepsilon \quad (1)$$

$$D = 1$$

(D belongs in category 2)

$$Y = b_0 + (b_1 + d_1)X + \varepsilon \quad (2)$$



The slopes differ between the two categories.

For category 2, the relationship between Y and X is shown by the steeper-sloping line.

Interpretation of d_1 : When X increases by 1 unit, category 2 makes Y increases more than that of category 1 by d_1 unit.

[LOS 4.b] *Formulate* and *interpret* a multiple regression model that includes qualitative independent variables

2.

Visualizing and Interpreting Dummy Variables

2.3. Slope and intercept dummy

Combine the two previous models, we get dummies in both slope and intercept:

$$Y_i = b_0 + d_0 D_i + b_1 X_i + d_1 D_i X_i + \varepsilon_i$$

Intercept
dummy

Slope
dummy

D = 0

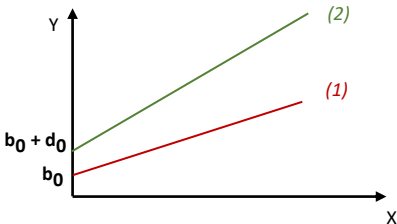
(D belongs in category 1)

$$Y = b_0 + b_1 X + \varepsilon \quad (1)$$

D = 1

(D belongs in category 2)

$$Y = (b_0 + d_0) + (b_1 + d_1) X + \varepsilon \quad (2)$$



The difference between the two categories depends on both an intercept effect (d_0) and a slope effect ($d_1 X$).

In this case, $d_1 > 0$.

Note: We may have more categories with more dummies and more independent variables.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

We will demonstrate the testing based on the following example:

An analyst has been tasked with analyzing how mutual fund characteristics (fund expense ratio, portfolio cash ratio, fund age and the natural logarithm of fund size) affect fund returns. She uses a large database of mutual funds that includes several styles: blend, growth, and value.

What are the dependent variable and independent variables?

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

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What are the dependent variable and independent variables?

Dependent variable: Fund returns

Continuous independent variables:

- Fund expense ratio (EXP)
- Fund portfolio cash ratio (CASH)
- Fund age (AGE)
- Natural logarithm of fund size (SIZE)

There are three possible style categories, the analyst use 2 ($= n - 1$) intercept dummy for these independent variables:

- BLEND, which takes a value of 1 if the fund is a blend fund and 0 otherwise;
- GROWTH, which takes a value of 1 if the fund is a growth fund and 0 otherwise;
- VALUE, the base category without a dummy.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

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What is the regression model?

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

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Testing for Statistical Significance of Dummy Variables

We will demonstrate the testing based on the following example:

An analyst has been tasked with analyzing how mutual fund characteristics (fund expense ratio, portfolio cash ratio, fund age and the natural logarithm of fund size) affect fund returns. She uses a large database of mutual funds that includes several styles: blend, growth, and value.

What is the regression model?

$$\begin{aligned} \text{Returns}_i = & b_0 + b_1 \text{EXP}_i + b_2 \text{CASH}_i + b_3 \text{AGE}_i + b_4 \text{SIZE}_i \\ & + d_1 \text{BLEND}_i + d_2 \text{GROWTH}_i + \varepsilon_i \end{aligned}$$

The regression output is presented in the next slide.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

We will demonstrate the testing based on the following example:

$$\text{Returns}_i = b_0 + b_1 \text{EXP}_i + b_2 \text{CASH}_i + b_3 \text{AGE}_i + b_4 \text{SIZE}_i \\ + d_1 \text{BLEND}_i + d_2 \text{GROWTH}_i + \varepsilon_i$$

Regression Statistics

R^2 0.1230

Adjusted R^2 0.1228

Standard Error 4.224

Observations 23,025

ANOVA

	df	SS	MS	F	Significance F
Regression	6	57,636.46	9,606	538	0
Residual	23,018	410,816.9	17.85		
Total	23,024	468,453.3			

Are the independent variables, as a group, explains the variation in the dependent variable at the significance level of 1%?

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

Are the independent variables, as a group, explains the variation in the dependent variable at the significance level of 1%?

In order to test the ability of the set of independent variables to explain the variation in the dependent variable, we use the F-test.

Regression Statistics	
F	Significance F
538	0

- Hypotheses:**

- H_0 : The set of independent variables, as a group, **can not** explain the variation in the dependent variable at the significant level of 1%.
- H_a : The set of independent variables, as a group, **can** explain the variation in the dependent variable at the significant level of 1%.

- Decision rule:**

- P-value > Significance level $\rightarrow$ Cannot reject H_0
- P-value < Significance level $\rightarrow$ Reject H_0

p-value (Significance F $\approx$ 0) < Significance level ($\alpha = 1\%$) $\rightarrow$ Reject H_0

The set of independent variables, as a group, **can** explain the variation in the dependent variable at the significant level of 1%.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

We will demonstrate the testing based on the following example:

$$\text{Returns}_i = b_0 + b_1 \text{EXP}_i + b_2 \text{CASH}_i + b_3 \text{AGE}_i + b_4 \text{SIZE}_i + d_1 \text{BLEND}_i + d_2 \text{GROWTH}_i + \varepsilon_i$$

	Coefficients	Standard Error	t-Statistic	P-value
Intercept	-2.909	0.299	-9.7376	0.00
Annual expense ratio	-0.586	0.0495	-11.824	0.00
Portfolio cash	-0.032	0.0029	-11.168	0.00
Fund age	0.074	0.0033	22.605	0.00
Log of assets	0.267	0.0141	18.924	0.00
Blend dummy	0.661	0.0678	9.749	0.00
Growth dummy	2.498	0.0748	33.394	0.00

Are the independent variables statistically significant, and how do we interpret them?

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

	Coefficients	Standard Error	t-Statistic	P-value
Intercept	−2.909	0.299	−9.7376	0.00
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Log of assets	0.267	0.0141	18.924	0.00
Blend dummy	0.661	0.0678	9.749	0.00
Growth dummy	2.498	0.0748	33.394	0.00

Are the independent variables statistically significant, and how do we interpret them?

The p-values < 5% and |t-statistics| > 3 suggest that all the slope coefficients, including 02 coefficients of dummy variables are significantly different from zero.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

Are the independent variables statistically significant, and how do we interpret them?

	Coefficient	Interpretation
<i>Intercept</i>	-2.909	Average annual negative return of 2.91% is unexplained by the independent variables in the model.
<i>EXP</i>	-0.586	Fund returns are negatively impacted by the level of expenses.
<i>CASH</i>	-0.032	Fund returns are negatively impacted by the level of cash holding.
<i>AGE</i>	0.074	Older funds perform better.
<i>SIZE</i>	0.267	Fund returns are positively impacted by the level of fund's asset.
<i>BLEND</i>	0.661	Blend funds deliver average returns that exceed those in the Value category by 0.66% per year.
<i>GROWTH</i>	2.498	Growth funds deliver average returns that exceed those in the Value category by 2.50% per year.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

We will demonstrate the testing based on the following example:

The analyst wonders whether the relationship between age and performance differ by fund type (i.e., Does the age factor affect growth or blend funds differently from how it affects value funds?). The analyst extends the study by adding 2 slope dummies: AGE_BLEND and AGE_GROWTH

What are the dependent variable and independent variables? What is the regression model?

Besides the existing dependent and independent variables, two interaction variables are introduced:

- AGE_BLEND: When BLEND = 1, the interaction term AGE_BLEND takes the value of AGE
- AGE_GROWTH: When GROWTH = 1, the interaction term AGE_GROWTH takes the value of AGE

The new regression model is:

$$\begin{aligned} \text{Returns}_i = & b_0 + b_1 \text{EXP}_i + b_2 \text{CASH}_i + b_3 \text{AGE}_i + b_4 \text{SIZE}_i \\ & + d_1 \text{BLEND}_i + d_2 \text{GROWTH}_i + d_3 \text{AGE_BLEND}_i + d_4 \text{AGE_GROWTH}_i + \varepsilon_i \end{aligned}$$

The regression output is presented in the next slide.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

$$\text{Returns}_i = b_0 + b_1 \text{EXP}_i + b_2 \text{CASH}_i + b_3 \text{AGE}_i + b_4 \text{SIZE}_i + d_1 \text{BLEND}_i + d_2 \text{GROWTH}_i + d_3 \text{AGE_BLEND}_i + d_4 \text{AGE_GROWTH}_i + \varepsilon_i$$

Regression Statistics

 R^2 0.123

Adjusted R^2 0.123

Standard Error 4.224

Observations 23,025

ANOVA

df

SS

MS

F

Significance F

Regression

8

57,760.46

7,220

404.6

0

Residual

23,016

410,692.9

17.84

Total

23,024

468,453.3

Are the independent variables, as a group, explains the variation in the dependent variable at the significance level of 1%?

p-value (Significance F $\approx$ 0) < Significance level ($\alpha = 1\%$) $\rightarrow$ Reject H_0

The set of independent variables, as a group, **can** explain the variation in the dependent variable at the significant level of 1%.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

$$\text{Returns}_i = b_0 + b_1 \text{EXP}_i + b_2 \text{CASH}_i + b_3 \text{AGE}_i + b_4 \text{SIZE}_i + d_1 \text{BLEND}_i + d_2 \text{GROWTH}_i + d_3 \text{AGE_BLEND}_i + d_4 \text{AGE_GROWTH}_i + \varepsilon_i$$

Regression Statistics

 R^2 0.123

Adjusted R^2 0.123

Standard Error 4.224

Observations 23,025

ANOVA

df

SS

MS

F

Significance F

Regression

8

57,760.46

7,220

404.6

0

Residual

23,016

410,692.9

17.84

Total

23,024

468,453.3

Are the independent variables, as a group, explains the variation in the dependent variable at the significance level of 1%?

p-value (Significance F $\approx$ 0) < Significance level ($\alpha = 1\%$) $\rightarrow$ Reject H_0

The set of independent variables, as a group, **can** explain the variation in the dependent variable at the significant level of 1%.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

We will demonstrate the testing based on the following example:

$$\text{Returns}_i = b_0 + b_1 \text{EXP}_i + b_2 \text{CASH}_i + b_3 \text{AGE}_i + b_4 \text{SIZE}_i + d_1 \text{BLEND}_i + d_2 \text{GROWTH}_i + \varepsilon_i$$

	Coefficients	Standard Error	t-Statistic	P-value
Intercept	-2.810	0.3060	-9.183	0.00
EXP	-0.587	0.0496	-11.839	0.00
CASH	-0.032	0.0029	-11.211	0.00
AGE	0.065	0.0059	11.012	0.00
SIZE	0.267	0.0141	18.906	0.00
BLEND	0.603	0.1088	5.546	0.00
GROWTH	2.262	0.1204	18.779	0.00
AGE_BLEND	0.005	0.0077	0.627	0.53
AGE_GROWTH	0.020	0.0081	2.478	0.00

Are the independent variables statistically significant, and how do we interpret them?

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

	Coefficients	Standard Error	t-Statistic	P-value
Intercept	-2.810	0.3060	-9.183	0.00
EXP	-0.587	0.0496	-11.839	0.00
CASH	-0.032	0.0029	-11.211	0.00
AGE	0.065	0.0059	11.012	0.00
SIZE	0.267	0.0141	18.906	0.00
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AGE_BLEND	0.005	0.0077	0.627	0.53
AGE_GROWTH	0.020	0.0081	2.478	0.00

Are the independent variables statistically significant, and how do we interpret them?

The p-values and t-statistics suggest that most of the slope coefficients are statistically significant, except for AGE_BLEND (p-value > 0.01).

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables

3.

Testing for Statistical Significance of Dummy Variables

	Coefficients	Standard Error	t-Statistic	P-value
Intercept	-2.810	0.3060	-9.183	0.00
EXP	-0.587	0.0496	-11.839	0.00
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AGE_BLEND	0.005	0.0077	0.627	0.53
AGE_GROWTH	0.020	0.0081	2.478	0.00

For the base group (VALUE), the **AGE coefficient** suggest those funds earn extra return of 0.065 as time passes when BLEND = GROWTH = 0.

AGE_GROWTH is statistically significant → The extra annual return of Growth fund with each additional year of age is the sum of **AGE** and **AGE_GROWTH coefficients** = 0.065% + 0.020% = 0.085%.

With each constant year of age, Growth fund's return exceeds those of Value funds by: 0.085% + 2.262% = 2.347%.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables



Knowledge check

You are tasked with studying the relation between stock market returns and GDP growth for multiple countries and must use a binary variable in your regression model to categorize countries by stock market type, emerging (1) or developed (0) markets.

1. *Identify the new variable and its function.*

- A. Slope dummy: It allows for a change in slope to classify countries into weak stock performance countries and strong stock performance countries.
- B. Intercept dummy: It allows for a change in intercept to classify countries by their stock market development status.
- C. Interaction term: It allows for a change in intercept to classify countries into low-GDP growth and high-GDP growth countries.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables



Knowledge check

You are tasked with studying the relation between stock market returns and GDP growth for multiple countries and must use a binary variable in your regression model to categorize countries by stock market type, emerging (1) or developed (0) markets.

1. Identify the new variable and its function.

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- B. Intercept dummy: It allows for a change in intercept to classify countries by their stock market development status.
- C. Interaction term: It allows for a change in intercept to classify countries into low-GDP growth and high-GDP growth countries.

Answer:

To classify countries by emerging (1) versus developed (0) stock market status, the new variable, an **intercept dummy** should be used.

→ **B is correct.**

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables



Knowledge check

You are tasked with studying the relation between stock market returns and GDP growth for multiple countries and must use a binary variable in your regression model to categorize countries by stock market type, emerging (1) or developed (0) markets.

2. Refine the model to capture the effect on stock returns of the interaction of each country's GDP growth and its stock market development status. Identify the model you should use (noting these definitions).

GDPG: Country GDP growth

EM: Indicates emerging stock market country

DM: Indicates developed stock market country

- A. Stock return = $b_0 + b_1 \text{GDPG} + d_1 \text{EM} + d_2 \text{DM} + d_3 (\text{EM} \times \text{GDPG}) + \epsilon$.
- B. Stock return = $b_0 + b_1 \text{GDPG} + d_1 \text{EM} + d_2 \text{DM} + \epsilon$.
- C. Stock return = $b_0 + b_1 \text{GDPG} + d_1 \text{EM} + d_2 (\text{EM} \times \text{GDPG}) + \epsilon$.

[LOS 4.b] *Formulate and interpret* a multiple regression model that includes qualitative independent variables



Knowledge check

You are tasked with studying the relation between stock market returns and GDP growth for multiple countries and must use a binary variable in your regression model to categorize countries by stock market type, emerging (1) or developed (0) markets.

2. Refine the model to capture the effect on stock returns of the interaction of each country's GDP growth and its stock market development status. Identify the model you should use (noting these definitions).

GDPG: Country GDP growth

EM: Indicates emerging stock market country

DM: Indicates developed stock market country

- A. $\text{Stock return} = b_0 + b_1 \text{GDPG} + d_1 \text{EM} + d_2 \text{DM} + d_3 (\text{EM} \times \text{GDPG}) + \epsilon.$
- B. $\text{Stock return} = b_0 + b_1 \text{GDPG} + d_1 \text{EM} + d_2 \text{DM} + \epsilon.$
- C. $\text{Stock return} = b_0 + b_1 \text{GDPG} + d_1 \text{EM} + d_2 (\text{EM} \times \text{GDPG}) + \epsilon.$

Answer:

- This model must include a variable for country GDP growth, (GDPG)
- Only one dummy variable should be used because of two market status (EM or DM)

→ **C is correct.**

[LOS 4.c] *Formulate and interpret* a logistic regression model

1. Qualitative dependent variables

Qualitative dependent variables (categorical dependent variables) are outcome variables that describe data that fit into categories.

Example: to predict if a company will go bankrupt, we need to use a qualitative dependent variable $Y = \{\text{bankrupt} (= 1) \text{ or not bankrupt} (= 0)\}$.

Qualitative dependent variable Y

$Y = 0$ (not bankrupt)

$Y = 1$ (bankrupt)

 *Why is the linear regression not suitable for models with qualitative dependent variables?*

In linear regression, the predicted value could be $Y > 1$ or $Y < 0$, whereas the probability of something cannot be greater than 1 or less than 0.

→ The predicted value of the dependent variable could be invalid.

→ We need a transformation called **logistic transformation** (see next slide).

[LOS 4.c] *Formulate and interpret a logistic regression model*

2. Logistic transformation

Logistic transformation (logit transformation) is a mathematical function used to transform probabilities or proportions into values on a continuous scale, typically ranging from negative infinity to positive infinity.

The transformation is carried out by 3 steps as the following:

01

p : the probability that a condition is fulfilled, or an event happens

Problem: Because $0 \leq p \leq 1$, if p is used as dependent variable, the predicted value of the dependent variable could be invalid when its value is > 1 or < 0 .

02

$\frac{p}{1-p}$: the odds of an event happening

Problem: Because $\frac{p}{1-p} \geq 0$, if $\frac{p}{1-p}$ is used as dependent variable, the predicted value of the dependent variable could be invalid when its value is < 0 .

[LOS 4.c] *Formulate and interpret a logistic regression model*

2. Logistic transformation

03

$\text{Ln}(\frac{p}{1-p})$: *log odds or logit function*

Because $\text{Ln}(\frac{p}{1-p})$ is continuous from negative infinity to positive infinity, if $\text{Ln}(\frac{p}{1-p})$ is used as dependent variable, the predicted value of the dependent variable will satisfy the linear regression assumptions.

→ *All problems are solved*

**Logistic
transformation**

$$\text{Logit}(p) = \text{Ln}(\frac{p}{1-p})$$

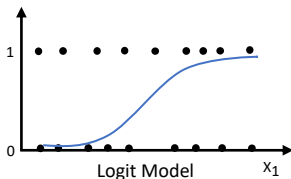
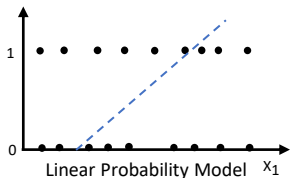
[LOS 4.c] *Formulate and interpret a logistic regression model*

3. Logistic regression

Logistic regression model (logit model) is the model uses the logistic transformation of the event probability (P) into the log odds, $\ln[P/(1 - P)]$, as the dependent variable to estimate the probability of a discrete outcome given the values of the independent variables.

$$\ln\left(\frac{p}{1-p}\right) = b_0 + b_1X_1 + b_2X_2 + \dots + \varepsilon$$

$$P = \frac{1}{1 + e^{-(b_0 + b_1X_1 + b_2X_2 + b_3X_3)}}$$



The logit model's nonlinear function takes on a sigmoidal shape and is approximately linear except when probability estimates are close to 0 or 1.

[LOS 4.c] *Formulate and interpret a logistic regression model*

4.

Compare logit model to linear model

	Linear model	Logit model
Method of estimating	Ordinary least squares (OLS)	Maximum likelihood estimation (MLE)
Interpretation of regression coefficients	The change in the dependent variable per unit change in the independent variable, holding all other independent variables constant	The change in the “log odds” that the event happens per unit change in the independent variable, holding all other independent variables constant
Test single independent variable	Similar to each other, using t-test	
Test overall model	Using F-test	Using likelihood ratio (LR) test follows chi-square distribution: <i>LR test statistics = $-2 \times (\text{Log likelihood restricted model} - \text{Log likelihood unrestricted model})$</i>
Coefficient of determination	R^2	pseudo- R^2 but can only be used to compare different specifications of the same model

[LOS 4.c] *Formulate and interpret a logistic regression model*

5.

Interpret a logistic regression

We will interpret the logistic regression based on the following example:

You are assigned to examine the propensity of companies to repurchase their shares, so for a sample of 500 companies, you have identified those that repurchased shares (Repurchase = 1) and those that did not (Repurchase = 0). You also collected company data for the year prior to the repurchase, including cash-to-total-assets ratio (CASH), debt-to-equity ratio (DE), and net profit margin (NPM), and estimated the following logistic regression:

$$\text{Repurchase}_i = b_0 + b_1 \text{CASH}_i + b_2 \text{DE}_i + b_3 \text{NPM}_i + \varepsilon_i$$

95% CI					
	Coefficient	Standard Error	z-Statistic	Lower	Upper
Intercept	-0.4738	0.196	-2.415	-0.858	-0.089
CASH	-0.9118	1.154	-0.790	-3.174	1.351
DE	-0.3186	0.133	-2.396	-0.579	-0.058
NPM	0.9407	0.417	2.255	0.123	1.758

[LOS 4.c] *Formulate and interpret a logistic regression model*

5. Interpret a logistic regression

5.1. Interpreting the intercept

Estimated logistic regression:

$$\text{Repurchase}_i = b_0 + b_1 \text{CASH}_i + b_2 \text{DE}_i + b_3 \text{NPM}_i + \varepsilon_i$$

	Coefficient	Standard Error	z-Statistic	Lower	Upper
Intercept	-0.4738	0.196	-2.415	-0.858	-0.089

The intercept of -0.4738 is the log odds of the probability of being a share repurchaser if CASH, DE, and NPM are all zero.

Let p be the probability of being a share repurchaser.

$$\ln\left(\frac{p}{1-p}\right) = -0.4738$$

$$\rightarrow \frac{p}{1-p} = e^{-0.4738} = 0.6226$$

$$\rightarrow p = (1-p) \times 0.6226$$

$$\begin{aligned} \rightarrow p &= 0.6226 / (1 + 0.6226) \\ &= 0.3837 \text{ or } 38.37\% \end{aligned}$$

[LOS 4.c] Formulate and interpret a logistic regression model

5.

Interpret a logistic regression

5.2. Interpreting the independent variables

Estimated logistic regression:

$$\text{Repurchase}_i = b_0 + b_1 \text{CASH}_i + b_2 \text{DE}_i + b_3 \text{NPM}_i + \varepsilon_i$$

	Coefficient	Standard Error	z-Statistic
CASH	-0.9118	1.154	-0.790
DE	-0.3186	0.133	-2.396
NPM	0.9407	0.417	2.255

A coefficient is interpreted as the expected change in the log odds for a one-unit change in the variable.

We can convert each coefficient from log odds into odds and then into probability:

Independent variable	Coefficient (log odds)	Odds ($e^{\text{log odds}}$)	Probability [Odds/(1 + Odds)]
CASH	-0.9118	0.4018	0.2866
DE	-0.3186	0.7272	0.4210
NPM	0.9407	2.5618	0.7192

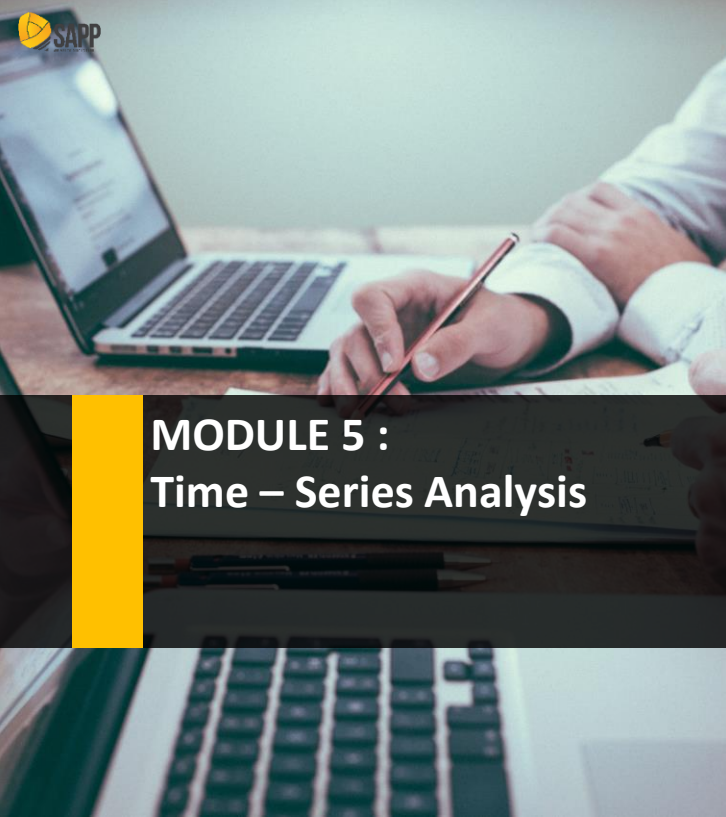
[LOS 4.c] *Formulate and interpret a logistic regression model*

5.

Interpret a logistic regression

5.2. Interpreting the independent variables

Independent variable	Probability	Interpretation
CASH	0.2866	One-unit change in the cash-to-total assets ratio results in a 28.7% increase in the probability of repurchasing shares.
DE	0.4210	One-unit change in the debt-to-equity ratio results in a 42.1% increase in the probability of repurchasing shares.
NPM	0.7192	One-unit change in net profit margin results in a 71.9% increase in the probability of repurchasing shares.



MODULE 5 : Time – Series Analysis

LEARNING OUTCOMES

- 5.a.** Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or a log-linear trend, given the estimated trend coefficients
- 5.b.** Describe factors that determine whether a linear or a log-linear trend should be used with a particular time series and evaluate limitations of trend models
- 5.c.** Explain the requirement for a time series to be covariance stationary and describe the significance of a series that is not stationary
- 5.d.** Describe the structure of an autoregressive (AR) model of order p and calculate one- and two-period-ahead forecasts given the estimated coefficients
- 5.e.** Explain how autocorrelations of the residuals can be used to test whether the autoregressive model fits the time series
- 5.f.** Explain mean reversion and calculate a mean-reverting level
- 5.g.** Contrast in-sample and out-of-sample forecasts and compare the forecasting accuracy of different time-series models based on the root mean squared error criterion
- 5.h.** Explain the instability of coefficients of time-series models

LEARNING OUTCOMES

- 5.i.** Describe characteristics of random walk processes and contrast them to covariance stationary processes
- 5.j.** Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and how to test for them, and demonstrate how a time series with a unit root can be transformed so it can be analyzed with an AR model
- 5.k.** Describe the steps of the unit root test for nonstationarity and explain the relation of the test to autoregressive time-series models
- 5.l.** Explain how to test and correct for seasonality in a time-series model and calculate and interpret a forecasted value using an AR model with a seasonal lag
- 5.m.** Explain autoregressive conditional heteroskedasticity (ARCH) and describe how ARCH models can be applied to predict the variance of a time series
- 5.n.** Explain how time-series variables should be analyzed for nonstationarity and/or cointegration before use in a linear regression
- 5.o.** Determine an appropriate time-series model to analyze a given investment problem and justify that choice

[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or a log-linear trend, given the estimated trend coefficients

1.

Introduction: Time series

A **time series** is a set of observations for a variable over successive periods of time.

The series has a **trend** if a consistent pattern can be seen by plotting the data on a graph.

Example: Stock market returns in Vietnam from 2011 - 2020



[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or...

2.

Linear Trend Model

A **linear trend** is a time series pattern that can be graphed using a straight line. A downward sloping line indicates a negative trend, while an upward-sloping line indicates a positive trend.

The **simplest form of a linear trend** is represented by the following linear trend model:

$$y_t = b_0 + b_1(t) + \epsilon_1$$

where:

y_t = the value of the time series (the dependent variable) at time t

b_0 = intercept at the vertical axis (y-axis)

b_1 = slope coefficient (or trend coefficient)

ϵ_1 = error term (or residual term or disturbance term)

t = time (the independent variable); $t = 1, 2, 3 \dots T$

[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or...

2.

Linear Trend Model

Ordinary least squares (OLS) regression is used to estimate the coefficient in the trend line, which provides the following prediction equation:

$$\hat{y}_t = \hat{b}_0 + \hat{b}_1(t)$$

where:

$\hat{y}_t$ = the predicted value of y (the dependent variable) at time t

$\hat{b}_0$ = the estimated value of the intercept term

$\hat{b}_1(t)$ = the estimated value of the slope coefficient

Example: Using a linear trend model

Suppose you are given a linear trend model with $\hat{b}_0 = 1.70$ and $\hat{b}_1 = 3.0$. Calculate $\hat{y}_t$ for $t = 1$ and $t = 2$.

Answer:

When $t = 1$, $\hat{y}_1 = 1.70 + 3.0(1) = 4.70$

When $t = 2$, $\hat{y}_2 = 1.70 + 3.0(2) = 7.70$

[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or...

3.

Log - Linear Trend Model

In fact, time series data, particularly financial time series, often display *exponential growth* (growth with continuous compounding), not linear growth. Thus, we can not use linear trend model in this situation.

When a series exhibits exponential growth, it can be modeled as:

$$y_t = e^{b_0 + b_1(t)}$$

where:

y_t = the value of the dependent variable at time t

b_0 = the intercept term

b_1 = the constant rate of growth

e = the base of the natural logarithm

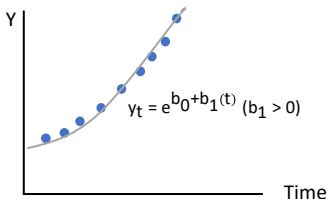
t = time = 1, 2, 3...T

[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or...

3.

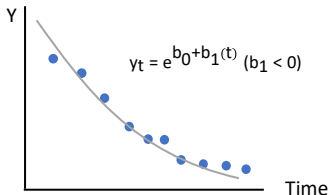
Log - Linear Trend Model

Positive exponential growth



The random variable (i.e., the time series) tends to increase at some constant rate of growth. If we plot the data, the observations will form a convex curve ($b_1 > 0$).

Negative exponential growth



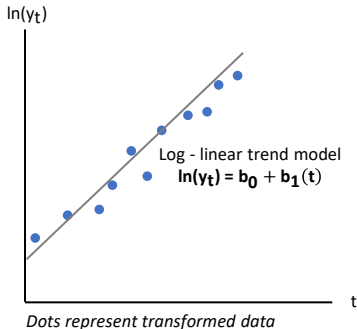
The data tends to decrease at some constant rate of decay, and the plotted time series will be a concave curve ($b_1 < 0$).

[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or...

3.

Log - Linear Trend Model

Hereafter we transform the exponential growth model into a log linear model by changing the form of the dependent variable:



$$y_t = e^{b_0 + b_1(t)}$$

Take the natural
logarithm of both sides

$$\ln(y_t) = \ln(e^{b_0 + b_1(t)})$$

$$\ln(y_t) = b_0 + b_1(t)$$

This is a **log - linear trend** – a time series pattern that can be graphed using a straight line of natural logarithm of dependent variable in term of independent variable (t).

[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or...

3.

Log - Linear Trend Model

Example: Log-linear trend model

An analyst estimates a log-linear trend model using quarterly revenue data (in millions of \$) from the first quarter of 2005 to the fourth quarter of 2016 for JP Northfield, Inc.:

$$\ln \text{revenue}_t = b_0 + b_1(t) + \varepsilon_t$$

with $t = 1, 2, \dots, 48$

The results are shown in the following table:

	Coefficient	Standard Error	t-statistic
Intercept	4.00	0.05	80.0
Trend	0.09	0.01	9.0

What is the revenue estimated in the first quarter of 2017?

[LOS 5.a] Calculate and evaluate the predicted trend value for a time series, modeled as either a linear trend or...

3.

Log - Linear Trend Model

Example: Log-linear trend model

	Coefficient	Standard Error	t-statistic
Intercept	4.00	0.05	80.0
Trend	0.09	0.01	9.0

What is the revenue estimated in the first quarter of 2017?

Answer:

We have:

$$\ln \text{revenue}_t = 4.00 + 0.09 (t)$$

We have 48 quarters in 12 full years from the first quarter of 2005 to the fourth quarter of 2016. Thus, for the first quarter of 2017, $t = 48 + 1 = 49$

$$\ln \text{revenue}_{49} = 4.00 + 0.09 (49) = 8.41$$

$$\rightarrow \text{Revenue}_{49} = e^{\ln \text{revenue}_{49}} = e^{8.41} = \$4,492 \text{ million.}$$

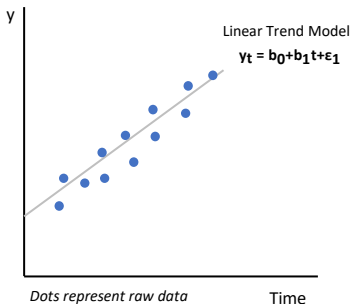
[LOS 5.b] Describe factors that determine whether a linear or a log-linear trend should be used with a particular time series and evaluate limitations of trend models

1.

Factors that determine which is the better model

To determine if a linear or log-linear trend model should be used, we should plot the data by linear trend model. Two cases that are likely to happen are presented below:

Case 1: Data points are equally distributed above and below the regression line



The data points appear to be equally distributed above and below the regression line.

We conclude that raw data grow over time by a constant amount.

The **linear trend model** is selected without considering the remaining model.

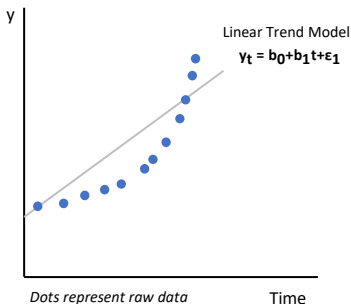
[LOS 5.b] Describe factors that determine whether a linear or a log-linear trend should be used with...

1.

Factors that determine which is the better model

To determine if a linear or log-linear trend model should be used, we should plot the data by linear trend model. Two cases that are likely to happen are presented below:

Case 2: Data points are persistently distributed above and below the regression line for a period of time



The residuals from a linear trend model is persistently positive or negative for a period of time.

We conclude that raw data grow over time by a constant rate.

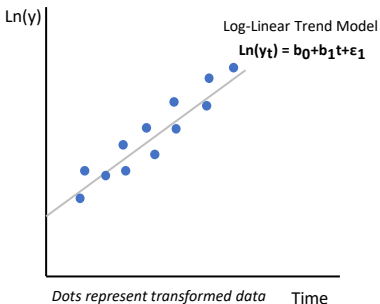
The **linear trend model** is not suitable and another model need to be considered.

[LOS 5.b] Describe factors that determine whether a linear or a log-linear trend should be used with...

1.

Factors that determine which is the better model

Case 2: Data points are persistently distributed above and below the regression line for a period of time



By transforming linear data to natural logarithm form, the data points now appear to be equally distributed above and below the regression line.

We conclude that transformed data grow over time by a constant amount.

The **Log-linear trend model** is selected without considering the remaining model.

[LOS 5.b] Describe factors that determine whether a linear or a log-linear trend should be used with...

2.

Limitations of Trend Models



Recall: One of the assumptions underlying linear regression is that the *residuals are uncorrelated with each other*.

In fact, time series data often shows interaction throughout multiple periods (the result of prior periods has impacts on it of current period) .

The trend models have the residuals that are persistently positive or negative for periods of time.

The model exhibits **serial correlation**

The trend models are **not appropriate** for the time series and we should not use them to predict future values.

Turn to other models, in which the first priority is **an autoregressive model**.

[LOS 5.c] Explain the requirement for a time series to be covariance stationary and describe the significance of a series that is not stationary

1.

Introduction to Autogressive model (AR)

Autoregressive model (AR) is the model that the dependent variable is regressed against one or more lagged values of itself.

$$x_t = b_0 + b_1 x_{t-1} + \varepsilon_t$$

where:

x_t = value of time series at time t

b_0 = intercept at the vertical axis (y-axis)

b_1 = slope coefficient

x_{t-1} = value of time series at time $t - 1$

ε_t = error term (or residual term or disturbance term)

t = time; $t = 1, 2, 3 \dots T$

As we have mentioned, current-period time-series values are related to previous-period values → The AR model effectively represent this relationship.

For example: the sales for a firm in one month could be regressed against its sale in the previous month using AR model.

[LOS 5.d] Describe the structure of an autoregressive (AR) model of order p and calculate one- and two-period-ahead forecasts given the estimated coefficients

1.

Introduction to Autoregressive model (AR)

We have seen the autoregressive model in prior slide expressed as:

$x_t = b_0 + b_1x_{t-1} + \varepsilon_t$, which is regarded as **AR(1) model**.

Second-order autoregressive model, or AR(2) model, which illustrates how variable x would be regressed on itself with a lag of one and two periods:

$$x_t = b_0 + b_1x_{t-1} + b_2x_{t-2} + \varepsilon_t$$

In general, an AR model of order p , **AR(p)**, is expressed as:

$$x_t = b_0 + b_1x_{t-1} + b_2x_{t-2} + \dots + b_px_{t-p} + \varepsilon_t$$

where p indicates the number of lagged values that the autoregressive model will include as independent variables.

[LOS 5.c] Explain the requirement for a time series to be covariance stationary and describe the significance of...

2.

Covariance stationary

The time series being modeled must be **covariance stationary** to ensure that statistical inferences based on ordinary least squares (OLS) estimates for an autoregressive (AR) time series model are valid.

A covariance stationary series must satisfy **three principal requirements**.

1. Constant and finite expected value: The expected value of the time series is constant over time: $E(y_t) = \mu$

2. Constant and finite variance: The time series' volatility around its mean does not change over time. $E[(y_t - \mu)^2] = \sigma^2$

3. Constant and finite covariance between values at any given lag: The covariance of the time series with leading or lagged values of itself is constant: $\text{cov}(y_t, y_{t-1}) = \text{cov}(y_t, y_{t+1}) = \lambda$

[LOS 5.c] Explain the requirement for a time series to be covariance stationary and describe the significance of...

2.

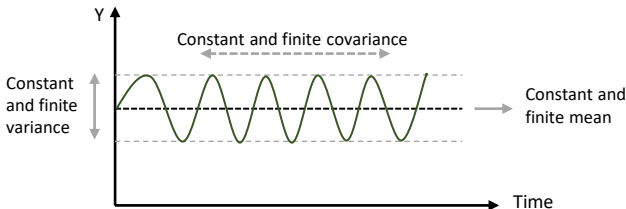
Covariance stationary

✓ A covariance stationary series must satisfy **three principal requirements**.

1. Constant and finite expected value: The expected value of the time series is constant over time: $E(y_t) = \mu$

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[LOS 5.c] Explain the requirement for a time series to be covariance stationary and describe the significance of...

2.

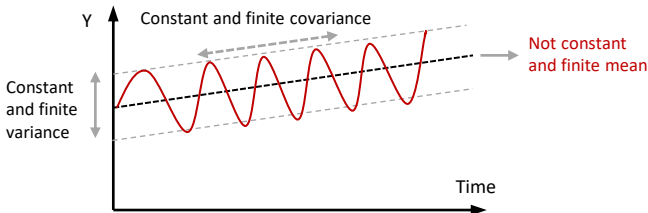
Covariance stationary

✗ The time series is not **covariance stationary**

VIOLATE: 1. Constant and finite expected value. The expected value of the time series is **not** constant over time: $E(y_t) \neq \mu$

2. Constant and finite variance. The time series' volatility around its mean does not change over time. $E[(y_t - \mu)^2] = \sigma^2$

3. Constant and finite covariance between values at any given lag. The covariance of the time series with leading or lagged values of itself is constant: $\text{cov}(y_t, y_{t-1}) = \text{cov}(y_t, y_{t+1}) = \lambda$



[LOS 5.c] Explain the requirement for a time series to be covariance stationary and describe the significance of...

2.

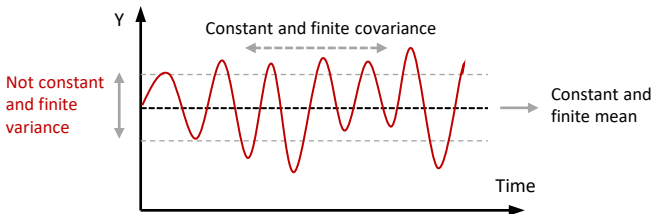
Covariance stationary

✗ The time series is not **covariance stationary**

1. Constant and finite expected value. The expected value of the time series is constant over time: $E(y_t) \neq \mu$

VIOLATE: 2. Constant and finite variance. The time series' volatility around its mean **does** change over time. $E[(y_t - \mu)^2] = \sigma^2$

3. Constant and finite covariance between values at any given lag. The covariance of the time series with leading or lagged values of itself is constant: $\text{cov}(y_t, y_{t-1}) = \text{cov}(y_t, y_{t+1}) = \lambda$



[LOS 5.c] Explain the requirement for a time series to be covariance stationary and describe the significance of...

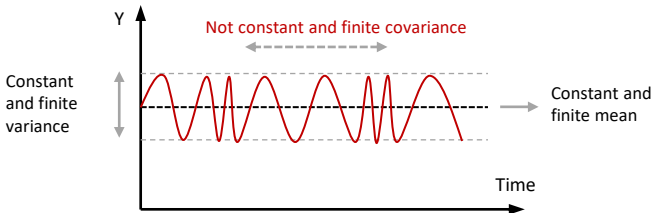
2. Covariance stationary

✗ The time series is not **covariance stationary**

1. Constant and finite expected value. The expected value of the time series is constant over time: $E(y_t) \neq \mu$

2. Constant and finite variance. The time series' volatility around its mean does not change over time. $E[(y_t - \mu)^2] = \sigma^2$

VIOLATE: 3. Constant and finite covariance between values at any given lag. The covariance of the time series with leading or lagged values of itself is **not** constant: $\text{cov}(y_t, y_{t-1}) = \text{cov}(y_t, y_{t+1}) = \lambda$



[LOS 5.d] Describe the structure of an autoregressive (AR) model of order p and calculate one- and...

2.

Forecasting With an Autoregressive Model

 Suppose we have an AR(1) model, the value at present (x_t) is given and we are required to forecast x_{t+2} , or two-step-ahead value. How can x_{t+2} be forecasted?

Step 1: Calculate one-period-ahead forecast for an AR(1) model:

$$\hat{x}_{t+1} = \widehat{b_0} + \widehat{b_1}x_t$$

Step 2: Use $\hat{x}_{t+1}$ at step 1 to calculate two-step-ahead forecast for an AR(1) model:

$$\hat{x}_{t+2} = \widehat{b_0} + \widehat{b_1}\hat{x}_{t+1}$$



The chain rule of forecasting

Due to several times of forecasting included in the procedure, multi-period forecasts are more uncertain than single-period forecasts.

[LOS 5.d] Describe the structure of an autoregressive (AR) model of order p and calculate one- and...

2.

Forecasting With an Autoregressive Model

Example: Forecasting two-step-ahead

Suppose that an AR(1) model has been estimated and has produced the following prediction equation: $x_t = 1.2 + 0.45x_{t-1}$. Calculate a two-step-ahead forecast if the current value of x is 5.0.

Answer:

One-step-ahead forecast:

If $x_t = 5$, then $\hat{x}_{t+1} = 1.2 + 0.45(5) = 3.45$.

Two-step-ahead forecast:

If $\hat{x}_{t+1} = 3.45$, then $\hat{x}_{t+2} = 1.2 + 0.45(3.45) = 2.75$.

[LOS 5.e] Explain how autocorrelations of the residuals can be used to test whether the autoregressive model fits the time series

1. Autocorrelation analysis

If the residuals have significant autocorrelation, the AR model that produced the residuals is not the best model for the time series being analyzed.

The procedure to test whether an AR time series model is correctly specified involves three steps:

01

Estimate the AR model being evaluated using linear regression: Start with a first-order AR model [i.e., AR(1)] using $\mathbf{x_t} = \mathbf{b_0} + \mathbf{b_1x_{t-1}} + \boldsymbol{\varepsilon_t}$

02

Calculate the autocorrelations of the model's residuals (i.e., the level of correlation between the forecast errors from one period to the next).

03

Test whether the autocorrelations are significantly different from zero.



Hypotheses

If the model is correctly specified, none of the autocorrelations will be statistically significant. We use two-tailed t-test and the hypotheses are structured as:

$$H_0: \rho_{\varepsilon_t, \varepsilon_{t-k}} = 0 \text{ (The residuals do not correlate)}$$

$$H_a: \rho_{\varepsilon_t, \varepsilon_{t-k}} \neq 0 \text{ (The residuals correlate)}$$

[LOS 5.e] Explain how autocorrelations of the residuals can be used to test whether the autoregressive model fits the time series

1.

Autocorrelation analysis

03

Test whether the autocorrelations are significantly different from zero.



Test statistic

t-statistic = estimated autocorrelation/standard error

$$= \frac{\rho_{\epsilon_t, \epsilon_{t-k}}}{1/\sqrt{T}}$$

where:

T is the number of observations

$1/\sqrt{T}$ = standard error of the correlation

$\rho_{\epsilon_t, \epsilon_{t-k}}$ = the correlation of error term t with the k^{th} lagged error term.

[LOS 5.e] Explain how autocorrelations of the residuals can be used to test whether the autoregressive model fits the time series

1.

Autocorrelation analysis

03

Test whether the autocorrelations are significantly different from zero.



Decision rule

Reject H_0 if $t\text{-test} > |T_c|$ (critical value) with $T_c = t_{\alpha}^{T-2}$

→ The time series have **autocorrelation** with the k^{th} lagged error term. Thus, the AR(1) model is not the best model for the data.

Now we add lag (x_{t-k}) in the model and test for AR(k) model:

$$x_t = b_0 + b_1 x_{t-1} + b_k x_{t-k} + \varepsilon_t.$$

This procedure will be followed until all autocorrelations are insignificant so that the AR model is correctly specified.

[LOS 5.f] Explain mean reversion and calculate a mean-reverting level

1.

Mean reversion

A time series exhibits **mean reversion** if it has a tendency to move toward its mean.

In other words, the time series has a tendency to decline when the current value is above the mean and rise when the current value is below the mean.

If a time series is at its mean-reverting level, the model predicts that the next value of the time series will be the same as its current value ($\hat{x}_t = x_{t-1}$).

$$\begin{aligned}\hat{x}_t &= x_{t-1} \\ x_t &= b_0 + b_1 x_{t-1}\end{aligned}$$



replace x_{t-1} by x_t

$$x_t = b_0 + b_1 x_t$$

$$\text{Mean-reverting level: } x_t = \frac{b_0}{1 - b_1}$$

[LOS 5.f] Explain mean reversion and calculate a mean-reverting level

2.

Mean - reverting level

$$\text{Mean-reverting level: } x_t = \frac{b_0}{1 - b_1}$$

$$\text{If } x_t > \frac{b_0}{1 - b_1}$$

The AR(1) model predicts that x_{t+1} will be lower than x_t

$$\text{If } x_t < \frac{b_0}{1 - b_1}$$

The AR(1) model predicts that x_{t+1} will be higher than x_t



Recall: All **covariance stationary time series** have a finite mean-reverting level (the first principal requirements).

An AR(1) time series will have a finite mean-reverting level when the absolute value of the lag coefficient is less than 1 (i.e., $|b_1| < 1$).

[Optional] Why do we need $|b_1| < 1$?

Indeed, the finite mean-reverting level (L) of AR(1) model is limit of the subsequence (x_t) with $x_t = b_0 + b_1 x_{t-1}$. We have:

$$|x_t - L| = |b_1| \cdot |x_{t-1} - L| = \dots = |b_1|^{t-1} \cdot |x_1 - L|$$

Thus, L is the limit of subsequence $(x_t) \leftrightarrow \lim_{t \rightarrow +\infty} |b_1|^{t-1} = 0$ or $|b_1| < 1$.

[LOS 5.f] Explain mean reversion and calculate a mean-reverting level

2.

Mean - reverting level

Example: Mean-reverting time series

Calculate the mean-reverting level for the manufacturing capacity utilization time series using the following regression results:

Time Series Regression Results for Manufacturing Capacity Utilization

Regression model: $x_t = b_0 + b_1 x_{t-1}$

R square 0.346508

Adjusted R square 0.29205

Standard error 1.333885

Observations 14

	Coefficients	Standard Error	t-Statistic
Intercept	82.137	0.753	109.080
Manufacturing utilization	-0.223	0.0884	-2.522

[LOS 5.f] Explain mean reversion and calculate a mean-reverting level

2.

Mean - reverting level

Example: Mean-reverting time series

Calculate the mean-reverting level for the manufacturing capacity utilization time series using the following regression results:

Answer:

Recall: Mean-reverting level is $x_t = \frac{b_0}{1 - b_1}$

With $b_0 = 82.137$ and $b_1 = -0.223$ as given, mean-reverting level is computed as:

$$x_t = \frac{b_0}{1 - b_1} = \frac{82.137}{1 - (-0.223)} = 67.16$$

- If the current level of manufacturing capacity utilization > 67.16
→ It is expected to fall in the next period.
- If the current level of manufacturing capacity utilization < 67.16
→ It is expected to rise in the next period.

[LOS 5.g] Contrast in-sample and out-of-sample forecasts and compare the forecasting accuracy of different time-series models based on the root mean squared error criterion

1.

Contrast in-sample and out-of-sample forecasts

In-sample forecasts

The forecasts are made within the range of data used in the estimation.

Example: We estimate AR models using raw inflation data from January 1995 to December 2018 and then forecast data within this sample's period.

Out-of-sample forecasts

The forecasts are made outside of the time period for the data used in the estimation.

Example: We estimate AR models using raw inflation data from January 1995 to December 2018 and then forecast data outside this sample's period, especially for the first 6-month of 2019.

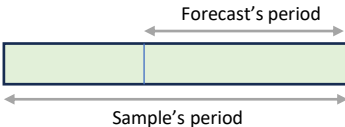
[LOS 5.g] Contrast in-sample and out-of-sample forecasts and compare the forecasting accuracy of...

1.

Contrast in-sample and out-of-sample forecasts

In-sample forecasts

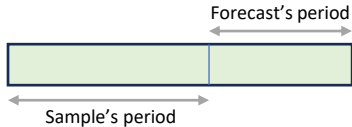
The forecasts are made within the range of data used in the estimation.



To estimate how accurate a model is in forecasting the actual data we used to develop the model, we use In-sample forecasts: **the lower in-sample forecast error variance, the more preferable.**

Out-of-sample forecasts

The forecasts are made outside of the time period for the data used in the estimation.



To estimate how accurate a model is in forecasting the y variable value for a time period outside the period used to develop the model, we use Out-of-sample forecasts: **the lower root mean squared error (RMSE), the more preferable.**

[LOS 5.g] Contrast in-sample and out-of-sample forecasts and compare the forecasting accuracy of...

2.

Gauge the forecasting accuracy

One of the most important role of modeling is forecasting the future, but the future is always out of sample, so we need to gauge the out-of-sample forecast error of the models.

The **root mean squared error (RMSE)** of the out-of-sample data is an indicator used to gauge the accuracy of autoregressive models in forecasting values.

The model has the **lower** RMSE for the out-of-sample data.



The model has **lower** forecast error.



The model is expected to have **better** predictive power in the future.

[LOS 5.g] Contrast in-sample and out-of-sample forecasts and compare the forecasting accuracy of...

2.

Compare forecast model performance

Example: Compare the forecasting accuracy

Which model is better in forecasting out-of-sample values?

Out-of-sample Forecast Error Comparisons: January 2019 – June 2019 US CPI Inflation (Annualized)

Date (2019)	Infl(t)	Infl(t-1)	Infl(t-2)	AR(1) Error	Squared Error	AR(2) Error	Squared Error
Jan	0.0000	0.0000	0.0000	0.1335	0.0178	-1.6000	2.5599
Feb	2.4266	0.0000	0.0000	-2.2931	5.2585	0.8266	0.6833
Mar	4.9070	2.4266	0.0000	-3.8068	14.4916	2.1522	4.6320
Apr	3.6600	4.9070	2.4266	-1.5716	2.4699	0.2014	0.0406
May	1.2066	3.6600	4.9070	0.3850	0.1482	-1.1714	1.3722
Jun	1.2066	1.2066	3.6600	-0.5924	0.3510	-0.2488	0.0619
				Average	3.7895	Average	1.5583
				RMSE	1.9467	RMSE	1.2483

[LOS 5.g] Contrast in-sample and out-of-sample forecasts and compare the forecasting accuracy of...

2.

Compare forecast model performance

Example: Compare the forecasting accuracy

Which model is better in forecasting out-of-sample values?

Answer:

Recall: The model that has the **lower** RMSE for the out-of-sample data is expected to have **better** predictive power in the future.

- The RMSE from the AR(1) model is 1.9467.
 - The RMSE from the AR(2) model is 1.2483.
- RMSE from the AR(2) model < RMSE from the AR(1) model.
- The AR(2) model was slightly more accurate than the AR(1) model in its out-of-sample forecasts of inflation from January 2019 to June 2019.

[LOS 5.h] Explain the instability of coefficients of time-series models

1.

Instability of regression coefficients

Financial and economic conditions are dynamic



Estimated regression coefficients varies in different periods

Financial and economic time series inherently exhibit some form of ***instability*** or ***nonstationarity***.

Models estimated with shorter time series are usually more stable than those with longer time series.

Tradeoff between the increased statistical reliability when using longer time periods and the increased stability of the estimates when using shorter periods.

Longer time periods
Increased statistical
reliability



Shorter time periods
Increased estimates'
stability

[LOS 5.h] Explain the instability of coefficients of time-series models

1.

Instability of regression coefficients

The primary concern when selecting a time series sample period is the underlying economic processes.

If there have been regulatory changes



If there have been dramatic changes in the underlying economic environment

The historical data may not provide a reliable model.

Whether the model is valid?

Examining the significance of the autocorrelation



Examining the covariance stationary

[LOS 5.i] Describe characteristics of random walk processes and contrast them to covariance stationary processes

1.

Random walk

A **simple random walk** is a time series in which the value of the series in one period is the value of the series in the previous period plus an unpredictable random error.

$$x_t = x_{t-1} + \varepsilon_t$$

where the best forecast of x_t is x_{t-1} and:

1. $E(\varepsilon_t) = 0$: The expected value of each error term is zero.
2. $E(\varepsilon_t^2) = \sigma^2$: The variance of the error terms is constant.
3. $\rho(\varepsilon_i \varepsilon_j) = 0$; if $i \neq j$: There is no serial correlation in the error terms.

A **random walk with drift** has $b_0 \neq 0$, compared to a simple random walk, which has $b_0 = 0$

$$x_t = b_0 + x_{t-1} + \varepsilon_t$$

where:

b_0 = the constant drift

[LOS 5.i] Describe characteristics of random walk processes and contrast them to covariance stationary processes

2.

Covariance stationary

Expressing a random walk:

$$x_t = b_0 + b_1 x_{t-1} + \varepsilon_t$$

where:

$b_0 = 0$ (for a random walk without a drift)

$b_0 \neq 0$ (for a random walk with a drift)

$b_1 = 1$ (for a random walk with or without a drift)

As we learnt about mean-reverting level, in either case (with or without a drift), the mean - reverting level is:

$$x_t = \frac{b_0}{1 - b_1} = \frac{b_0}{0} \text{ (undefined)}$$

Recall: We have learnt about Covariance stationary, which contains three characteristics: **Constant and finite expected value**, **Constant and finite variance** and **Constant and finite covariance between values at any given lag**.

[LOS 5.i] Describe characteristics of random walk processes and contrast them to covariance stationary processes

2.

Covariance stationary

In either case (with or without a drift), the mean - reverting level is:

$$x_t = \frac{b_0}{1 - b_1} = \frac{b_0}{0} \text{ (undefined)}$$

BUT

A time series must have a **finite mean-reverting level** (*Constant and finite expected value*) to be covariance stationary

A random walk, with or without a drift, is not covariance stationary, and exhibits what is known as a **unit root** ($b_1 = 1$) that we will discuss in the next slide.

CONCLUSION

- If the value of the lag coefficient is equal to one ($b_1 = 1$), the time series is said to have a **unit root** and will follow a random walk process.
- Since a time series that follows a random walk is not covariance stationary, modeling such a time series in an AR model can lead to incorrect inferences.

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and how to test for them, and demonstrate how a time series with a unit root can be transformed so it can be analyzed with an AR model. Describe the steps of the unit root test for nonstationarity and explain the relation of the test to autoregressive time-series models

1.

Unit Root Testing for Nonstationarity

To determine whether a time series is covariance stationary, we can ***run an AR model and examine autocorrelations***, or ***perform the Dickey Fuller test***.

1.1. Run an AR model and examine autocorrelations

The examination of the autocorrelations of a time series at various lags is a well-known prescription for inferring whether or not a time series is stationary.

- An AR model is estimated and the statistical significance of the autocorrelations at various lags is examined.
 - A stationary process will usually have residual autocorrelations insignificantly different from zero at all lags or residual autocorrelations that decay to zero as the number of lags increases.
- This approach is less definite than a currently more popular test for nonstationarity known as the Dickey Fuller test for a unit root.

Note: You can refer to previous slides in this module to recall the detail test of autocorrelations.

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and...

1.

Unit Root Testing for Nonstationarity

Example: Run an AR model and examine autocorrelations

The correlations of the error terms from the estimation of an AR(1) model using a sample with 102 observations are presented in the following figure. Determine whether the time series is covariance stationary.

Model: $x_t = b_0 + b_1 x_{t-1} + \varepsilon_t$.

Lag	Autocorrelation	T-test	Lag	Autocorrelation	T-test
1	0.0616	0.6222	7	-0.0101	-0.1024
2	0.0843	0.8517	8	0.0212	0.2138
3	0.0259	0.2613	9	-0.0959	-0.9690
4	0.0189	0.1908	10	0.0389	0.3936
5	0.1001	1.0113	11	-0.0677	-0.6838
6	-0.0638	-0.6445	12	-0.0122	-0.1240

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and...

1.

Unit Root Testing for Nonstationarity

Example: Run an AR model and examine autocorrelations

Determine whether the time series is covariance stationary.

Model: $x_t = b_0 + b_1 x_{t-1} + \epsilon_t$.



Hypotheses

We use two-tailed t-test and the hypotheses are structured as:

$$H_0: \rho_{\epsilon_t, \epsilon_{t-k}} = 0 \text{ (Time series is covariance stationary)}$$

$$H_a: \rho_{\epsilon_t, \epsilon_{t-k}} \neq 0 \text{ (Time series is covariance nonstationary)}$$



Test statistic

t-statistic = estimated autocorrelation/standard error = $\frac{\rho_{\epsilon_t, \epsilon_{t-k}}}{1/\sqrt{T}}$ with $T = 102$

Look at the figure above, for example, we can calculate the lag 1's t-statistic

$$= \frac{\rho_{\epsilon_t, \epsilon_{t-1}}}{1/\sqrt{T}} = \frac{0.0616}{1/\sqrt{102}} = 0.6222.$$

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and...

1.

Unit Root Testing for Nonstationarity

Example: Run an AR model and examine autocorrelations

Determine whether the time series is covariance stationary.

Model: $x_t = b_0 + b_1 x_{t-1} + \epsilon_t$.



Decision rule

Do not reject H_0 because with any lag, the t-test $< |T_c|$ (critical value) with

$$T_c = t_{0.05}^{100} = 1.98$$

→ The time series is covariance stationary and the AR(1) model is correctly specified.

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and...

1.

Unit Root Testing for Nonstationarity

To determine whether a time series is covariance stationary, we can *run an AR model and examine autocorrelations*, or *perform the Dickey Fuller test*.

1.2. Perform the Dickey Fuller test

Dickey and Fuller created a rather creative test for a unit root. It transforms the AR(1) model to run a simple regression.

Basic form of the AR model

$$x_t = b_0 + b_1 x_{t-1} + \varepsilon_t$$

Subtract x_{t-1} from both sides:

$$x_t - x_{t-1} = b_0 + (b_1 - 1)x_{t-1} + \varepsilon_t$$

Now testing whether the new coefficient $g_1 = (b_1 - 1)$ is different from zero using a **modified t-test**

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and...

1.

Unit Root Testing for Nonstationarity



Hypotheses

As discussed, an AR(1) time series will have a finite mean-reverting level when the absolute value of the lag coefficient is less than 1. Thus, Dickey Fuller test is a **one-tailed t-test** and the hypotheses are structured as:

$H_0: g_1 = 0$ (The series have a unit root)

$H_a: g_1 < 0$ (The series have no unit root)



Test statistic

$$\text{DF t-test} = \frac{g_1}{s_{g_1}}$$

where:

s_{g_1} = standard error of coefficient g_1

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and...

1.

Unit Root Testing for Nonstationarity



Decision rule

Reject H_0 if DF t-test $< -T_c$ (critical value) with $T_c = t_\alpha$

→ The series have **no unit root**.

→ The covariance stationary is present.

Otherwise, if H_0 is not rejected:

→ The series must have a unit root.

→ The nonstationarity is present.

[LOS 5.j,k] Describe implications of unit roots for time-series analysis, explain when unit roots are likely to occur and...

2.

First differencing

*If we believe a time series is a random walk (i.e., has a unit root), we can transform the data to a covariance stationary time series using a procedure called **first differencing**.*

Subtracting the value of the time series in the immediately preceding period from the current value to define a new dependent variable, y_t .

If time series of x has a unit root:

$$\begin{aligned}x_t &= x_{t-1} + \varepsilon_t \\ \rightarrow y_t &= x_t - x_{t-1} = \varepsilon_t\end{aligned}$$



Stating y in the form of an AR(1) model:

$$\begin{aligned}y_t &= b_0 + b_1 y_{t-1} + \varepsilon_t \\ \text{where: } b_0 &= b_1 = 0\end{aligned}$$

Mean-reverting level:

$$y_t = \frac{b_0}{1 - b_1} = \frac{0}{1 - 0} = 0 \text{ (finite value)}$$

This transformed time series is covariance stationary.

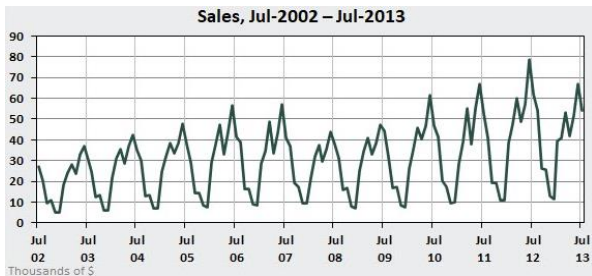
[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and interpret a forecasted value using an AR model with a seasonal lag

1.

Seasonality

Seasonality in a time series is a time-series is a pattern that **tends to repeat** from year to year.

Example: Given that sales data normally vary according to the time of year, we might expect this month's sales (x_t) to be related to sales for the same month last year ($x_t - 12$).



When seasonality is present, modeling the associated time series data would be misspecified unless the AR model incorporates the effects of the seasonality.

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

1.

Seasonality

Example: Detecting seasonality

You are interested in predicting occupancy levels for a resort hotel chain and have obtained the chain's quarterly occupancy levels for the most recent 40 quarters (10 years). You decide to model the quarterly occupancy time-series using the AR(1) model: $\ln x_t = b_0 + b_1 \ln x_{t-1} + \varepsilon_t$

Determine whether seasonality exists using the results presented in the following example.

Autoregression Output for Log-Quarterly Hotel Occupancy Resort

AR(1) Model: $\ln x_t = b_0 + b_1 \ln x_{t-1} + \varepsilon_t$

R² 0.7929

Standard error 0.1952

Observations 39

	<i>Coefficients</i>	<i>Standard</i>	<i>Error t-Statistic</i>
Intercept	0.0375	0.0274	1.369
Lag 1	0.5318	0.1635	3.2526

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

1.

Seasonality

Example: Detecting seasonality

Residual autocorrelations for the first four lags of the time series are shown as below:

Autocorrelation of Residuals

<i>Residual Lag</i>	<i>Autocorrelation</i>	<i>Standard Error</i>	<i>t-statistic</i>
1	-0.0615	0.1601	-0.3841
2	-0.0121	0.1601	-0.0756
3	-0.0212	0.1601	-0.1324
4	0.8719	0.1601	5.4460

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

1.

Seasonality

Answer:

We have $df = 37$, $\alpha = 5\% \rightarrow$ critical t -value = 2.026

T-statistic $> 2.026 \rightarrow$ Reject the null hypothesis that the Lag 4 autocorrelation is 0

T-statistic $< 2.026 \rightarrow$
Do not reject that autocorrelation is 0

Seasonality is present in the time-series

This model is misspecified and will be unreliable for forecasting purposes

A seasonality term needs to be included to make the model more correctly specified

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

2.

Correcting for Seasonality

To adjust for seasonality in an AR model, an **additional lag** of the dependent variable (corresponding to the same period in the previous year) is **added** to the original model as another independent variable.

Continue with the example above:

Significant residual correlation at Lag 4 indicates **seasonality** in the quarterly time series.

Add a lagged value of the dependent variable to the original AR(1) model that corresponds to the seasonal pattern:

$$\ln x_t = b_0 + b_1 \ln x_{t-1} + b_2 \ln x_{t-4} + \varepsilon_t$$

Note: The inclusion of a seasonal lag, does not result in an AR(2) model. It results in an AR(1) model incorporating a seasonal lag term.

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

2.

Correcting for Seasonality

Resort Occupancy Levels

AR(1) Model with a Seasonal Lag:

$$\ln x_t = b_0 + b_1 \ln x_{t-1} + b_2 \ln x_{t-4} + \varepsilon_t$$

R²

0.948983874

Standard error

0.3754

Observations

36

**R-square has
been improved,
from 79.3%**

	<i>Coefficients</i>	<i>Standard</i>	<i>Error t-Statistic</i>
Intercept	0.0085	0.0049	1.7347
Lag 1	0.2598	0.0527	4.9298
Lag 4	0.7921	0.2166	3.6570

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

2.

Correcting for Seasonality

Autocorrelation of Residuals

<i>Residual Lag</i>	<i>Autocorrelation</i>	<i>Standard Error</i>	<i>t-statistic</i>
1	-0.0526	0.1667	-0.3156
2	0.0715	0.1667	0.4290
3	-0.0241	0.1667	-0.1446
4	-0.0435	0.1667	-0.2610

Fourth-lag residual autocorrelation has dropped substantially, no longer statistically significant.

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

3.

Forecasting with an AR model with a seasonal lag

Example: Forecasting with an autoregressive model

Based on the regression results from the previous example and the occupancy levels over the past year (presented below).

Forecast the level of hotel occupancy for the first quarter of 2023.

Quarterly Hotel Occupancy Levels

Quarter	2022.1	2022.2	2022.3	2022.4
Occupancy Level	250,000	750,000	450,000	600,000

[LOS 5.I] Explain how to test and correct for seasonality in a time-series model and calculate and...

3.

Forecasting with an AR model with a seasonal lag
Answer:

Based on information from previous slide, Seasonally adjusted forecasting equation:

$$\ln x_t = 0.0085 + 0.2598 \ln x_{t-1} + 0.7921 \ln x_{t-4}$$

where x_t : the occupancy level for the t^{th} quarter.

To forecast the occupancy level for the hotel chain for the first quarter of 2023 (i.e., 2023.(1)), the following computation is made:

$$\ln x_{2023.(1)} = 0.0085 + 0.2598(\ln x_{2022.4}) + 0.7921(\ln x_{2022.1})$$

$$\rightarrow \ln x_{2023.(1)} = 0.0085 + 0.2598(\ln 600,000) + 0.7921(\ln 250,000)$$

$$\rightarrow \ln x_{2023.(1)} = 0.0085 + 0.2598(13.3047) + 0.7921(12.4292)$$

$$\rightarrow \ln x_{2023.(1)} = 13.3103$$

$$\text{Since } x = e^{\ln x}, x_{2023.(1)} = e^{13.3103} = 603,378.52$$

Thus, the forecasted level of hotel occupancy for the first quarter of 2023 is 603,379 (a significant increase over the same quarter the previous year).

[LOS 5.m] Explain autoregressive conditional heteroskedasticity (ARCH) and describe how ARCH models can be applied to predict the variance of a time series

1.

Autoregressive conditional heteroskedasticity

Examining a single time series, such as an AR model

Variance of the residuals
in period (t)

depends on

Variance of the residuals
in previous period (t-1)

Autoregressive conditional heteroskedasticity (ARCH) exists.

The **standard errors** of the regression coefficients in AR models and the **hypothesis tests** of these coefficients are **invalid**

[LOS 5.m] Explain autoregressive conditional heteroskedasticity (ARCH) and describe how ARCH models can be applied...

2.

Testing for autoregressive conditional heteroskedasticity

An **ARCH model** is used to express the autoregressive conditional heteroskedasticity, for which the variance of the residuals in one period is dependent on the variance of the residuals in the previous period.

The ARCH(1) regression model is expressed as:

$$\hat{\epsilon}_t^2 = a_0 + a_1 \hat{\epsilon}_{t-1}^2 + \mu_t$$

where: a_0 is the constant and μ_t is an error term

We use t-test to test ARCH. If the coefficient, a_1 , is statistically different from 0, the time series is ARCH(1), and vice versa.



Hypotheses

$H_0: a_1 = 0$ (the time series is not ARCH(1))

$H_a: a_1 \neq 0$ (the time series is ARCH(1))

[LOS 5.m] Explain autoregressive conditional heteroskedasticity (ARCH) and describe how ARCH models can be applied...

2.

Testing for autoregressive conditional heteroskedasticity



Test statistic

$$t\text{-statistics} = \frac{\hat{a}_1}{s_{\hat{a}_1}}$$

where: $s_{\hat{a}_1}$ is standard error of $\hat{a}_1$



Decision rule

Reject H_0 if $|t\text{-test}| > t_c$ (critical value) with $t_c = t_{\alpha/2}^{(n-2)}$

→ The time series is ARCH(1).

Correcting ARCH

Use regression procedures that correct for heteroskedasticity (such as **generalized least squares**) to estimate the standard error of the parameters.

[LOS 5.m] Explain autoregressive conditional heteroskedasticity (ARCH) and describe how ARCH models can be applied...

3.

Predicting the Variance of a Time Series

If a time series has ARCH errors, we use that ARCH model to predict the variance of the residuals in future periods.

$$\hat{\epsilon}_{t+1}^2 = a_0 + a_1 \hat{\epsilon}_t^2 + \mu_t$$

Example: Predicting the Variance of a Time Series

The next figure contains the results from the regression of an ARCH(1) model. Determine whether the results indicate autoregressive conditional heteroskedasticity (ARCH), and if so, calculate the predicted variance of the error terms in the next period if the current period squared error is 0.5625.

Model: ARCH (1) Regression Results

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t-statistic</i>	<i>P-value</i>
Constant	5.9068	1.08631	5.4375	< 0.001
Lag 1	0.4515	0.09558	4.7238	< 0.001

[LOS 5.m] Explain autoregressive conditional heteroskedasticity (ARCH) and describe how ARCH models can be applied...

3.

Predicting the Variance of a Time Series

Example: Predicting the Variance of a Time Series

Determine whether the results indicate autoregressive conditional heteroskedasticity (ARCH), and if so, calculate the predicted variance of the error terms in the next period if the current period squared error is 0.5625.

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t-statistic</i>	<i>P-value</i>
Constant	5.9068	1.08631	5.4375	< 0.001
Lag 1	0.4515	0.09558	4.7238	< 0.001

The time-series is ARCH (1)

Small p-value indicates statistical significance

Answer:

The variance of the error term in the next period can be computed as:

$$\hat{\epsilon}_{t+1}^2 = a_0 + a_1 \hat{\epsilon}_t^2 = 5.9068 + 0.4515(0.5625) = 6.1608$$

[LOS 5.n] Explain how time-series variables should be analyzed for nonstationarity and/or cointegration before use in a linear regression

1.

Cointegration

Cointegration means that two time series are **economically linked** (related to the same macro variables) or follow the **same trend** and that relationship is not expected to change.

To test whether two time series are cointegrated, we regress one variable on the other using the following model:

$$y_t = b_0 + b_1 x_t + \varepsilon (*)$$

where:

y_t = value of time series y at time t

x_t = value of time series x at time t

We use Dickey Fuller t-test to test for a unit root of the residuals from model (*).

If the test rejects the null hypothesis of a unit root:

→ Error terms generated by 2 time series are covariance stationary and 2 series are cointegrated.

→ The regression can be used to model their relationship, and vice versa.

[LOS 5.n] Explain how time-series variables should be analyzed for nonstationarity and/or cointegration...

2.

Multiple time series analysis

Occasionally, regression using two time series will be run (*e.g.: time series utilizing two different variables*)

Example: Using the market model to estimate the equity beta for a stock, an analyst regresses a time series of the stock's returns (y_t) on a time series of returns for the market (x_t).

$$y_t = b_0 + b_1 x_t + \varepsilon$$

Either one or both of these time series could be susceptible to nonstationarity

Runs separate DF t-tests to test whether the two-time series **have unit roots**

5 possible outcomes

[LOS 5.n] Explain how time-series variables should be analyzed for nonstationarity and/or cointegration...

2.

Multiple time series analysis

5 possible outcomes

Able to use linear regression?

01

Both time series are covariance stationary

Yes, the coefficients should be statistically reliable

02

Only the dependent variable time series is covariance stationary

No, the coefficients would not be reliable

03

Only the independent variable time series is covariance stationary

Yes or no, depends upon whether the two time series are **cointegrated**

04

Neither time series is covariance stationary and the two series are not cointegrated

05

Neither time series is covariance stationary and the two series are cointegrated

[LOS 5.n] Explain how time-series variables should be analyzed for nonstationarity and/or cointegration...

2.

Multiple time series analysis

Can Linear Regression Be Used to Model the Relationship Between Two Time Series?

		Independent variable time series	
		Is Covariance Stationary	Is not Covariance Stationary
Dependent Variable Time Series	Is Covariance Stationary	Yes	No
	Is not Covariance Stationary	No	Yes, if the two times series are cointegrated

[LOS 5.o] Determine an appropriate time-series model to analyze a given investment problem and justify that choice

1.

Guidelines for determining best suited model

To determine what type of model is best suited to meet your needs, follow these guidelines to make sure that your model is correctly specified:

01

Determine the goal.

Is the relationship of a variable to other variables (e.g., cointegrated time series, cross-sectional multiple regression) or the variable over time (e.g., trend model) being attempted to be modeled?

[LOS 5.o] Determine an appropriate time-series model to analyze a given investment problem and justify that choice

1.

Guidelines for determining best suited model

To determine what type of model is best suited to meet your needs, follow these guidelines to make sure that your model is correctly specified:

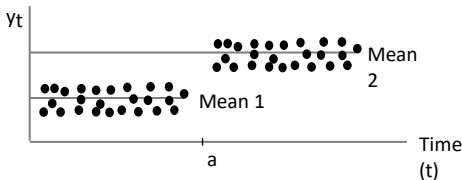
02

If an individual variable is modeled, plot the values of the variable over time and look for characteristics that would indicate non-stationarity.

A **structural change** is indicated by a significant *shift* in the plotted data at a point in time that seems to divide the data into two or more distinct patterns.

→ Two different models should be used, to test whether the time series has actually shifted.

→ If the time series has shifted significantly, a single time series encompassing the entire period will likely produce unreliable results.



[LOS 5.o] Determine an appropriate time-series model to analyze a given investment problem and justify that choice

1.

Guidelines for determining best suited model

To determine what type of model is best suited to meet your needs, follow these guidelines to make sure that your model is correctly specified:

03

If there is no seasonality or structural shift, use a trend model.

- If the data plot on a straight line with an upward or downward slope, use a linear trend model.
- If the data plot in a curve, use a log-linear trend model.

04

Run the trend analysis, compute the residuals, and test for serial correlation using the Durbin Watson test.

- If there is no serial correlation, *the model can be used.*
- If there is serial correlation, *another model should be used (e.g., AR model).*

[LOS 5.o] Determine an appropriate time-series model to analyze a given investment problem and justify that choice

1.

Guidelines for determining best suited model

To determine what type of model is best suited to meet your needs, follow these guidelines to make sure that your model is correctly specified:

05

If the data has serial correlation, reexamine the data for stationarity before running an AR model.

If it is not stationary, treat the data for use in an AR model as follows:

- If the data has a linear trend, first-difference the data.
- If the data has an exponential trend, first-difference the natural log of the data.
- If there is a structural shift in the data, run two separate models as discussed above.
- If the data has a seasonal component, incorporate the seasonality in the AR model as discussed in the following.

[LOS 5.o] Determine an appropriate time-series model to analyze a given investment problem and justify that choice

1.

Guidelines for determining best suited model

To determine what type of model is best suited to meet your needs, follow these guidelines to make sure that your model is correctly specified:

06

After first-differencing, if the series is covariance stationary, run an AR(1) model and test for serial correlation and seasonality.

- If there is no remaining serial correlation, *the model can be used.*
- If there is still serial correlation, incorporate lagged values of the variable into the AR model until any serial correlation have been removed.

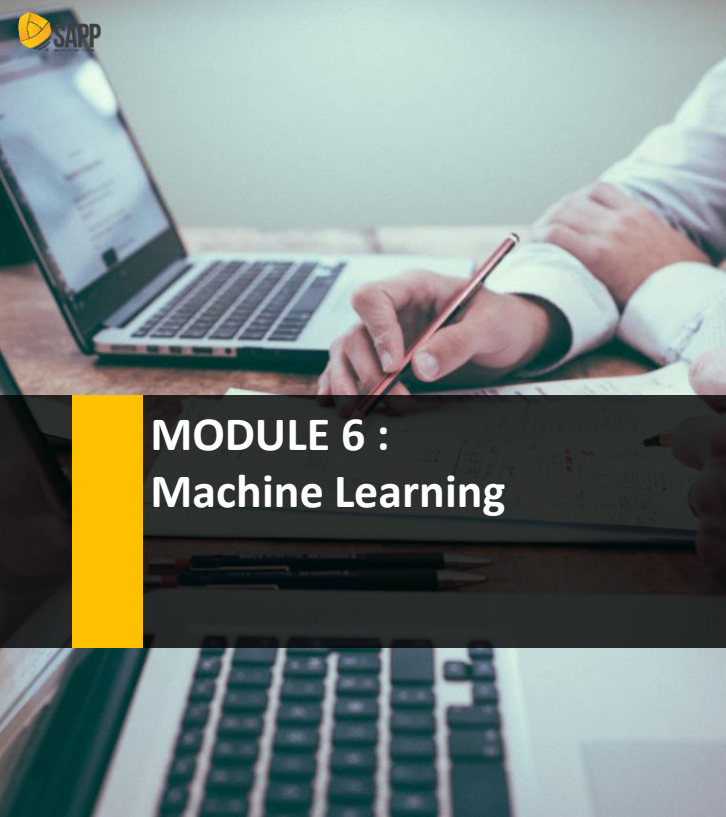
07

Test for ARCH: Regress the square of the residuals on squares of lagged values of the residuals.

- If the coefficient is not significantly different from zero, *the model can be used.*
- If the coefficient is significantly different from zero, ARCH is present. Correct using generalized least squares.

08

If two statistically reliable models have been developed, and it needs to be determined which one is better at forecasting, calculate their out-of-sample RMSE.



MODULE 6 : Machine Learning

LEARNING OUTCOMES

- 6.a.** Describe supervised machine learning, unsupervised machine learning, and deep learning
- 6.b.** Describe overfitting and identify methods of addressing it
- 6.c.** Describe supervised machine learning algorithms - including penalized regression, support vector machine, k-nearest neighbor, classification and regression tree, ensemble learning, and random forest - and determine the problems for which they are best suited
- 6.d.** Describe unsupervised machine learning algorithms - including principal components analysis, k-means clustering, and hierarchical clustering - and determine the problems for which they are best suited
- 6.e.** Describe neural networks, deep learning nets, and reinforcement learning

[LOS 6.a] Describe supervised machine learning, unsupervised machine learning, and deep learning

1.

Introduction: Machine learning

Machine learning (ML) is defined as the use of algorithms to make decisions by generalizing (or finding patterns) in a given data set.

Data set

Algorithms

Patterns

Decisions

ML performs better than standard statistical approaches because of:

- The ability to deal with a **large number of variables** (high dimension) and when the relationships are **nonlinear**.
- **No assumptions** requirement about the distribution of the underlying data as the statistical models.

ML terms

Target variable: This is the dependent variable. It can be continuous, categorical, or ordinal.

Features: These are the independent variables.

Training data set: This is the sample used to fit the model.

Hyperparameter: This is a model input specified by the researcher.

[LOS 6.a] Describe supervised machine learning, unsupervised machine learning, and deep learning

1.

Introduction: Machine learning

We can have equivalent terms between ML and statistical model:

ML	Statistical model
Target variable	Dependent variable
Features	Independent variables
Training data set	Data sample
Hyperparameter	Coefficients

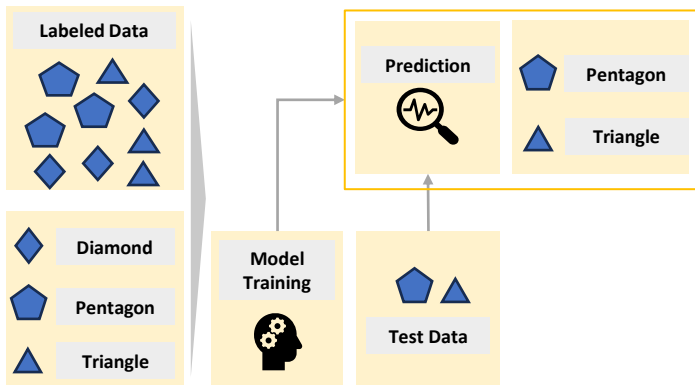
[LOS 6.a] Describe supervised machine learning, unsupervised machine learning, and deep learning

2.

Supervised machine learning

Supervised learning uses labeled training data that contains matched sets of observed inputs and the associated output to guide the ML program toward superior forecasting accuracy.

Working of Supervised Learning



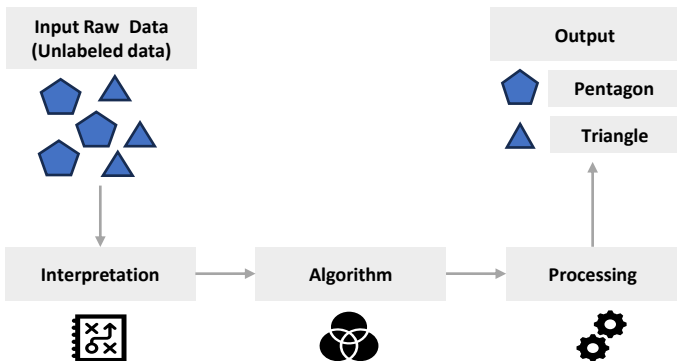
[LOS 6.a] Describe supervised machine learning, unsupervised machine learning, and deep learning

3.

Unsupervised machine learning

Unsupervised learning is machine learning that does not make use of labeled data. In the absence of any target variable, the program seeks out structure or interrelationships in the data.

Working of Unsupervised Learning



[LOS 6.a] Describe supervised machine learning, unsupervised machine learning, and deep learning

4.

Compare supervised and unsupervised machine learning

	Supervised machine learning	Unsupervised machine learning
Definition	The algorithm learns from a labeled dataset	The algorithm is provided with unlabeled data
Process	Learn a pattern from input data to output data, to predict on new, unseen data	Extract meaningful information from the data, often by grouping similar data points together

[LOS 6.a] Describe supervised machine learning, unsupervised machine learning, and deep learning

5.

Deep learning

Deep learning is a type of machine learning based on artificial neural networks in which multiple layers of processing are used to extract progressively higher level features from data and complex tasks such as image classification, face recognition, speech recognition, and natural language processing.

Deep learning algorithms

Are used for complex tasks such as image recognition, natural language processing

Reinforced learning algorithms

Programs that learn from their own prediction errors (*discussed later*)

Neural networks

A group of ML algorithms applied to problems with significant nonlinearities (*discussed later*)

[LOS 6.a] Describe supervised machine learning, unsupervised machine learning, and deep learning

6.

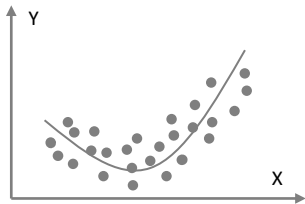
Compare Deep learning and Reinforcement learning

	Deep learning	Reinforcement learning
Definition	A subset of machine learning that focuses on neural networks with many layers (deep neural networks).	A type of machine learning where an agent interacts with an environment to achieve a goal.
Process	Automatically learn hierarchical representations of data, extract features, and recognize patterns within large and complex datasets.	Training agents to make sequences of decisions in an environment to maximize a cumulative reward signal.

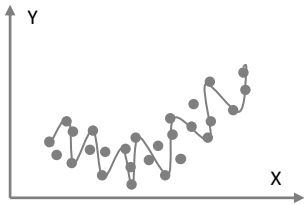
[LOS 6.b] Describe overfitting and identify methods of addressing it**1. Overfitting and generalization**

Overfitting is a modeling error which occurs when a machine learning model learns the training data too well, including its noise and random fluctuations, to the point that it performs poorly on new, unseen data.

The figures below will illustrate the good fitting and overfitting:



Just right fitting



Overfitting

[LOS 6.b] Describe overfitting and identify methods of addressing it

1.

Overfitting and generalization

When?

Occurs when the noise in the target variables seems to improve the model fit (i.e., randomness is misperceived to be a pattern, resulting in high in-sample R-squared)

Consequence?

Will decrease the accuracy of model forecasts on other (out-of-sample) data - overfit models do not generalize well to new data (i.e., out-of-sample R-squared will be low).

There are **three nonoverlapping data sets** to measure how well a model generalizes

Sample type	Use cases	Which errors occur?
Training sample	Used to develop the model	In-sample prediction errors
Validation sample	Used for tuning the model	Out-of-sample prediction errors
Test sample	Used for evaluating the model using new data	Out-of-sample prediction errors

[LOS 6.b] Describe overfitting and identify methods of addressing it

2. Errors and overfitting

These errors (in-sample and out-of-sample) are decomposed into the following:

01

Bias error: This is the in-sample error representing the difference between the average value that the model predicts and the actual value of the data resulting from models with a poor fit.

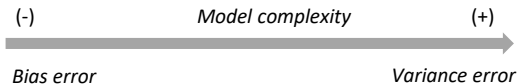
02

Variance error: This is the out-of-sample error representing the degree of dispersion of the values that the model predicts compared to the actual values resulting from overfitted models that do not generalize well.

03

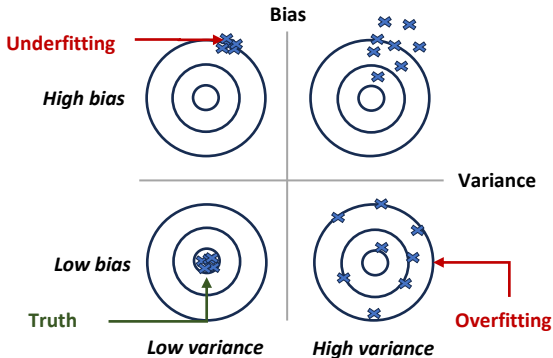
Base error: These are residual errors due to random noise

As a model becomes more complex, bias error decreases, and variance error increases.



[LOS 6.b] Describe overfitting and identify methods of addressing it**2. Errors and overfitting**

The figures following will illustrate the matrix of bias and variance error



The true value is at the center of the circles. The **X** marks represent the predicted values.

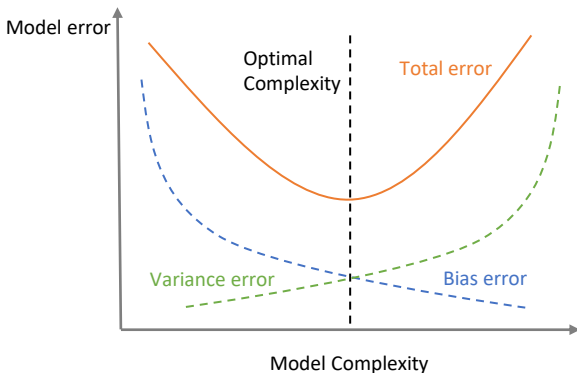
- With high bias, the predicted values are far from the center.
- With high variance, the predicted values are widely scattered, leading to deviations from the actual values.

→ We aim for **low bias** and **low variance**.

[LOS 6.b] Describe overfitting and identify methods of addressing it

2. Errors and overfitting

A **fitting curve**, which shows in- and out-of-sample error rates (E_{in} and E_{out}) on the y-axis plotted against model complexity on the x-axis, is presented here and illustrates the tradeoff between cost and complexity.

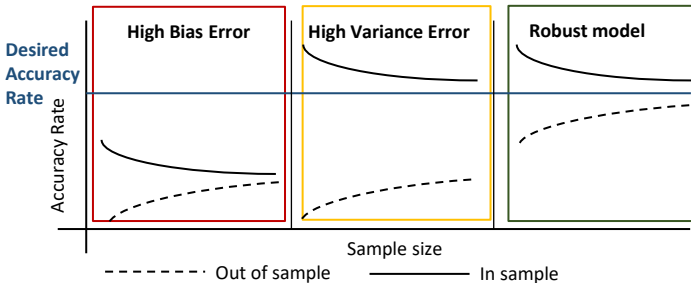


An optimal level of complexity minimizes the total error and is a key part of successful model generalization.

[LOS 6.b] Describe overfitting and identify methods of addressing it

3. Learning curve

A **learning curve** plots the accuracy rate (i.e., $1 - \text{error rate}$) in the validation or test sample versus the size of the training sample.



This model will see the error rates converge, but far below the desired level.

This model will see only the in-sample accuracy rate converge toward the desired level, while the out-of-sample accuracy rate lags far behind.

This model will show an improving accuracy rate as the sample size is increased, and the in-sample and out-of-sample error rates will converge toward a desired accuracy level.

[LOS 6.b] Describe overfitting and identify methods of addressing it

4. Methods to reduce the problem of overfitting

There are two methods to reduce the problem of overfitting

Complexity reduction

Involves streamlining or minimizing the number of components, factors, or elements within the system to make it more manageable, understandable, and efficient.

- A penalty is imposed to exclude features that are not meaningfully contributing to out-of-sample prediction accuracy.
- This penalty value increases with the number of independent variables (features) used by the model.

Cross validation

Involves partitioning the available dataset into subsets, training the model on some of these subsets, and then evaluating the model's performance on the remaining data.

- Researchers must ensure that the training data set is both large and representative of the population.
- The validation sample, similarly, should be large and representative to properly test the model.

[LOS 6.b] Describe overfitting and identify methods of addressing it

4.

Methods to reduce the problem of overfitting

Cross validation

K - fold cross validation is the technique used to mitigate the problem of such **holdout samples** (i.e., data samples not used to train the model) reducing the training set size too much.

The sample is randomly divided equally into k parts

The training sample comprises $(k - 1)$ parts, with one part left for validation

Error is then measured for the model in each of the parts.

This process is repeated k times, and the average in-sample and out-of-sample error rates are compiled.

[LOS 6.b] Describe overfitting and identify methods of addressing it

4. Methods to reduce the problem of overfitting

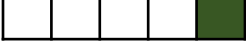
This figure below will illustrate the **5 – fold cross validation**:

1st  → Performance 1

2nd  → Performance 2

3rd  → Performance 3

4th  → Performance 4

5th  → Performance 5

Validation fold



Training folds



$$\text{Performance} = \frac{1}{5} \cdot \sum_{i=1}^5 \text{Performance } i$$

[LOS 6.b] Describe overfitting and identify methods of addressing it

4.

Methods to reduce the problem of overfitting

Example: Evaluating ML Algorithm Performance

Shreya Anand is a portfolio manager based in the Mumbai headquarters office of an investment firm, where she runs a high-dividend-yield fund for wealthy clients. Anand has some knowledge of data science from her university studies.

She is interested in classifying companies in the NIFTY 200 Index—an index of large- and mid-cap companies listed on the National Stock Exchange of India—into two categories: dividend increase and no dividend increase. She assembles data for training, validating, and testing an ML-based model that consists of 1,000 observations of NIFTY 200 companies, each consisting of 25 features (fundamental and technical) and the labeled target (dividend increase or no dividend increase).

After training her model, Anand discovers that while it is good at correctly classifying using the training sample, it does not perform well on new data. In consulting her colleagues about this issue, Anand hears conflicting explanations about what constitutes good generalization in an ML model:

Statement 1: The model retains its explanatory power when predicting using new data (i.e., out-of-sample).

Statement 2: The model shows low explanatory power after training using in-sample data (i.e., training data).

Statement 3: The model loses its explanatory power when predicting using new data (i.e., out-of-sample).

[LOS 6.b] Describe overfitting and identify methods of addressing it**4.****Methods to reduce the problem of overfitting****Example: Evaluating ML Algorithm Performance**

1. *Which statement made to Anand is most accurate?*
2. *Anand's model is most likely being impaired by which of the following?*
 - A. Underfitting and bias error
 - B. Overfitting and variance error
 - C. Overfitting and bias error
3. *By implementing which one of the following actions can Anand address the problem?*
 - A. Estimate and incorporate into the model a penalty that decreases in size with the number of included features.
 - B. Use the k-fold cross-validation technique to estimate the model's out-of-sample error, and then adjust the model accordingly.
 - C. Use an unsupervised learning model.

[LOS 6.b] Describe overfitting and identify methods of addressing it

4.

Methods to reduce the problem of overfitting

Answer:

1. Which statement made to Anand is most accurate?

Statement 1, is correct. Statement 2, is incorrect because it describes a poorly fitting model with high bias. Statement 3, is incorrect because it describes an overfitted model with poor generalization.

2. Anand's model is most likely being impaired by which of the following?

B is correct. Anand's model is good at correctly classifying using the training sample, but it does not perform well using new data. The model is overfitted, so it has high variance error.

3. By implementing which one of the following actions can Anand address the problem?

B is correct. A is incorrect because the penalty should increase in size with the number of included features. C is incorrect because Anand is using labeled data for classification, and unsupervised learning models do not use labeled data.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine, k-nearest neighbor, classification and regression tree, ensemble learning, and random forest - and determine the problems for which they are best suited

Supervised machine learning

**Penalized
regression**

**Support vector
machine**

**K – nearest
neighbor**

**Classification and
regression tree**

Ensemble learning

Random forest

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

1.

Penalized regression

*Recall from previous slides about **Overfitting**:* it will decrease the accuracy of model forecasts on other (out-of-sample) data when a large number of features (i.e., independent variables) are included in the data sample.



We need a method to minimize the number of components, factors, or elements within the system to make it more manageable, understandable, and efficient.



Penalized regression is a method that has been useful for reducing the problem of overfitting by imposing a **penalty** based on the number of features used by the model. It seeks to **minimize** the sum of square errors (**SSE**) as well as a **penalty value** (see next slide).

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

1. Penalized regression

Least absolute shrinkage and selection operator (LASSO): one popular type of **penalized regression**

- In addition to minimizing SSE, LASSO minimizes the sum of the absolute values of the slope coefficients.
- Through optimization, LASSO automatically eliminates the least predictive features. The formula below performs a type of feature selection of LASSO:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{k=1}^K |\hat{b}_k|$$

where:

- $\lambda \sum_{k=1}^K |\hat{b}_k|$ is penalty term
- λ is the hyperparameter that determines the balance between overfitting the model and keeping it parsimonious.

With $\lambda = 0$, the LASSO penalized regression is equivalent to an OLS regression.

The greater the number of included features (i.e., variables with non-zero coefficients), the larger the penalty term.

→ A feature is included only if the SSE declines by more than the penalty term increases.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

1.

Penalized regression

Application:

Nowadays, investment analysts are increasingly using LASSO and other regularization techniques to remove less pertinent features and build parsimonious model. Some of Penalized regression's application are as the following:

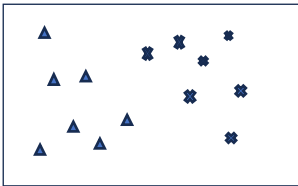
- **Financial forecasting:** forecasting default probabilities in industrial sectors where scores of potential features have been reduced to fewer than 10 variables.
- **Portfolio management:** estimation of a stable covariance matrix that can be used for mean-variance optimization.
- **Credit scoring:** determining which borrower characteristics and financial variables are most predictive of creditworthiness.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

2. Support vector machine (SVM)

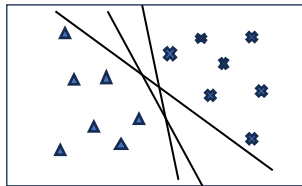
Support vector machine (SVM) is a linear classification algorithm that separates the data into one of two possible classifiers.

Data Labelled in Two Groups



A simple dataset labeled in two groups (triangles and crosses).

Data is Linearly Separable



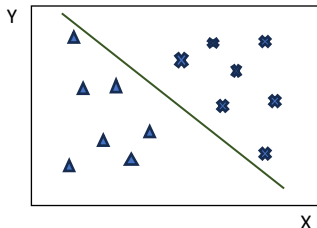
Two regions can be easily separated by an infinite number of straight lines called **linear classifiers**.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

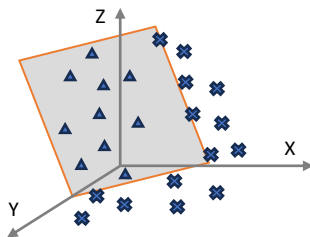
2. Support vector machine (SVM)

Observations with n features can be represented in an n - dimension space, and the dataset would be linearly separable if the observations can be separated into two distinct regions by a **linear space boundary (($n-1$)-dimensional hyperplane.)**

Example: Linear Support Vector Machine Classifier with 2 and 3 dimensions of features



With 2 dimensions of features, the linear space boundary is a **line (1 dimensional hyperplane)**.



With 3 dimensions of features, the linear space boundary is a **plane (2 dimensional hyperplane)**.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

2. Support vector machine (SVM)

Problem: In fact, most real-world datasets are not linearly separable. Some observations may fall on the wrong side of the boundary and be misclassified by the SVM algorithm.

We often use 2 methods to handle this problem

Soft margin classification

Adding a **penalty** to the objective function for observations in the training set that are misclassified. In essence, the SVM algorithm will choose a **discriminant boundary** that optimizes the trade-off between a wider margin and a lower total error penalty.

Non-linear SVM algorithm

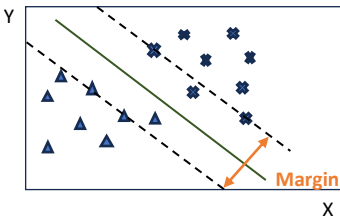
Finding a **hyperplane** (decision boundary) that best separates data points of different classes while maximizing the margin between these classes. These algorithms may reduce the number of misclassified instances in the training datasets but are more complex.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

2. Support vector machine (SVM)

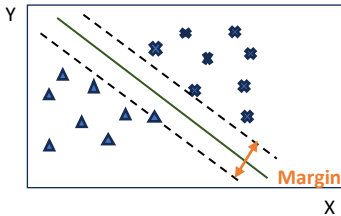
To illustrate in detail of Soft margin classification, we have the comparison with Hard margin classification as the following

Soft margin classification



Soft margin SVM allows some misclassification by introducing slack variables.

Hard margin classification



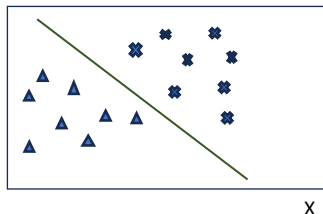
Hard margin SVM aims to find a hyperplane that perfectly separates the data into two classes without any misclassification.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

2. Support vector machine (SVM)

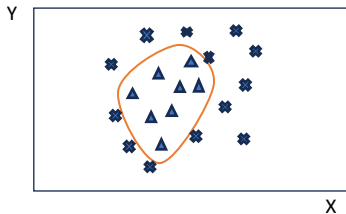
To illustrate in detail of Non-linear SVM algorithm, we have the comparison with Linear SVM algorithm as the following

Linear SVM algorithm



The data points are linearly separable into two classes.

Non-linear SVM algorithm



The data points are not linearly separable into two classes.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

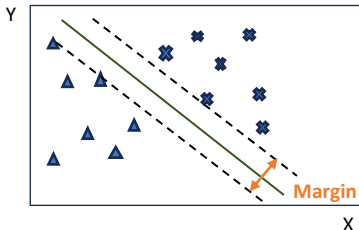
2.

Support vector machine (SVM)

We can see from below figure about 2 dimensions of features that SVM is classifying all observations perfectly.

Features of SVM:

- SVM maximizes the probability of making a correct prediction by determining the boundary that is farthest away from all the observations.
- This boundary comprises a discriminant boundary as well as margins on the side of the boundary.
- The margins are determined by the support vectors, observations that are closest to the boundary.



[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

2. Support vector machine (SVM)

Application:

Support Vector Machine (SVM) find several applications in the field of finance due to their ability to handle classification and regression tasks effectively. Here are some notable applications of SVM in finance:

- **Bankruptcy Prediction:** SVM is employed to predict the likelihood of a company facing financial distress or bankruptcy by analyzing financial ratios, credit ratings, and other relevant factors.
- **Classification:** Classify text from documents (e.g., news articles, company announcements, and company annual reports) into useful categories for investors (e.g., positive sentiment and negative sentiment)
- **Portfolio Management:** SVM can assist in optimizing investment portfolios by selecting a combination of assets that maximize returns while managing risk. It can take into account factors such as historical returns, volatility, and correlations between assets.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

3. K-nearest neighbor (KNN)

K-nearest neighbor (KNN) is the technique used to classify an observation based on nearness to the observations in the training sample.

*KNN consists of **3 steps** as the following:*

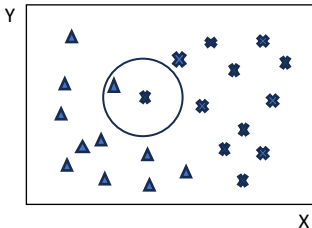
- 01 Specifying the value of k (the hyperparameter)
- 02 Looking for the k observations in the sample that are closest to the new observation that is being classified
- 03 Choosing the classification with the largest number out of k observations being considered

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

3.

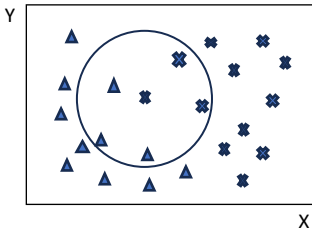
K-nearest neighbor (KNN)

KNN with new observation, $k = 1$



With $k = 1$, the cross will be classified into the same category as its nearest neighbor (i.e., the triangle category).

KNN with new observation, $k = 5$



With $k = 5$, there are three triangles and two crosses, so the cross again classified as belonging to the triangle category.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

3. K-nearest neighbor (KNN)

In practice, several different techniques can be used to determine an optimal value for k , taking into account the number of categories and their partitioning of the feature space.

If k is too small, it will result in a high error rate.

If k is too large, it will dilute the result by averaging across too many outcomes.

If k is even, there may be ties, with no clear winner.

Application:

In the investment industry, application of KNN include bankruptcy prediction, stock price prediction, corporate bond credit rating assignment, and customized equity and bond index creation.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

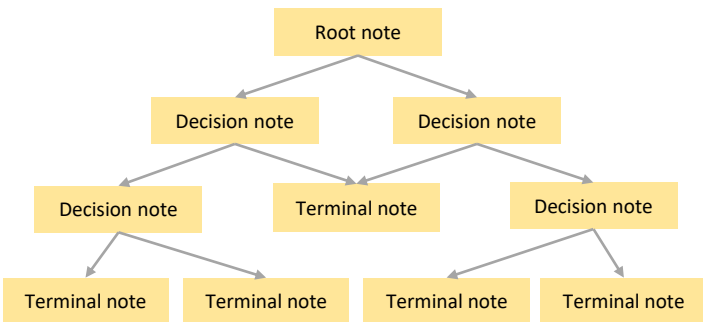
4.

Classification and regression trees (CART)

Classification and regression tree (CART) is a common supervised machine learning technique that predict target variable of either:

- Categorical → Producing a binary classification tree
- Continuous → Producing a regression tree

Illustration of CART

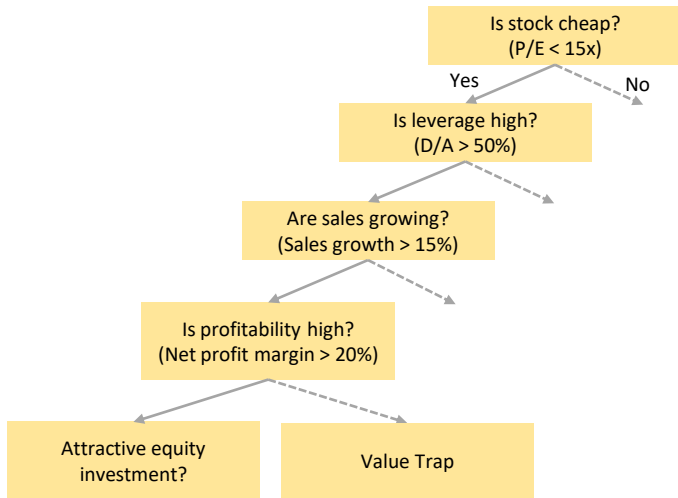


[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

4.

Classification and regression trees (CART)

Example: Stylized Decision Tree—Attractive Investment or Value Trap?



[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

4.

Classification and regression trees (CART)

To avoid overfitting:

- Regularization criteria such as maximum tree depth, maximum number of decision nodes, and so on are specified by the researcher.
- Sections of tree with minimal explanatory power are pruned.

What are the advantages of CART models over other algorithms?

Inducing robust rules despite noisy data and complex relationships between high numbers of features. → A powerful tool to build expert systems for decision-making processes.

Providing a visual explanation for the prediction that contrasts favorably with other algorithms considered to be “black boxes” because it may be difficult to understand the reasoning behind their outcomes.

Application:

In the investment industry, application of CART include, among others, enhancing detection of fraud in financial statements, generating consistent decision processes in equity and fixed-income selection, and simplifying communication of investment strategies to clients.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

5. Ensemble Learning

Ensemble learning is the technique of combining predictions from multiple models rather than a single model, and the combination of multiple learning algorithms is known as the **ensemble method**.

Ensemble learning can be divided into ***two main categories***

Aggregation of heterogeneous learners

- Different algorithms are combined with a **voting classifier**.
- The different algorithms each get a vote, and then we go with whichever answer gets the most votes.
- The models selected will have sufficient diversity in approach, resulting in a greater level of confidence in the predictions.

Aggregation of homogenous learners

- The same algorithm is used, but on different training data.
- The different training data samples (used by the same model) can be derived by **bootstrap aggregating** or **bagging**.
- The process relies on generating random samples (bags) with replacement from the initial training sample.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

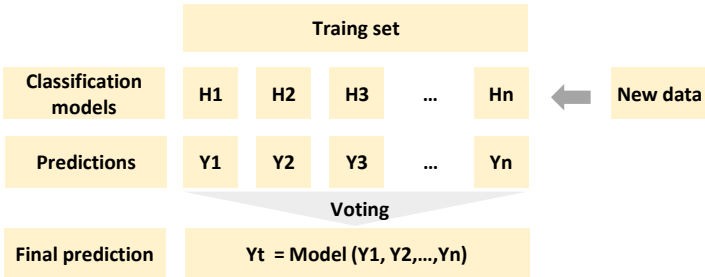
5.

Ensemble Learning

Example: Voting classifier

Suppose we have trained and compared the results of several algorithms, such as SVM, KNN, and CART. A **majority-vote classifier** will assign to a new data point the predicted label with the most votes.

For example, if the SVM and KNN models are both predicting the category “stock outperformance” and the CART model is predicting the category “stock underperformance,” then the majority-vote classifier will choose stock outperformance.”



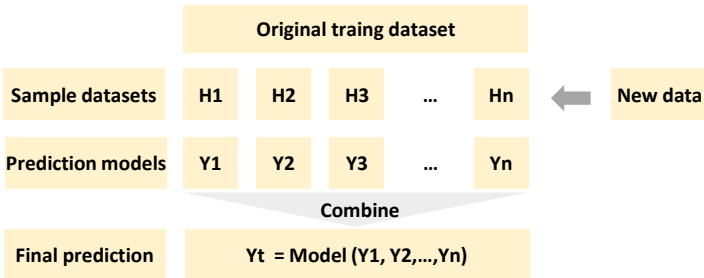
[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

5. Ensemble Learning

Example: Bootstrap aggregating (Bagging)

Bootstrap aggregating (or bagging) is a technique whereby the original training dataset is used to generate n new training datasets or bags of data by random sampling with replacement from the initial training set.

The algorithm can now be trained on n independent datasets that will generate n new models. Then, for each new observation, we can aggregate the n predictions using a majority-vote classifier for a classification or an average for a regression.



[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

5.

Ensemble Learning

Application:

Ensemble learning is widely used in finance for a variety of applications. Here are some key applications of ensemble learning in finance:

- **Credit Scoring:** improve the accuracy of credit scoring models.
- **Asset Allocation:** optimize asset allocation by combining the forecasts of multiple models or strategies.
- **Algorithmic Trading:** use in algorithmic trading systems to make trading decisions.
- **Portfolio Hedging:** develop effective portfolio hedging strategies by combining multiple models that predict asset price movements and market conditions.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

6.

Random Forest

Random forest is a variant of classification trees whereby a large number of classification trees are trained using data bagged from the same data set.

*We have **4 steps** involved in Random Forest Algorithm:*

01

A subset of data points and a subset of features is selected for constructing each decision tree. Simply put, n random records and m features are taken from the data set having k number of records.

02

Individual decision trees are constructed for each sample.

03

Each decision tree will generate an output.

04

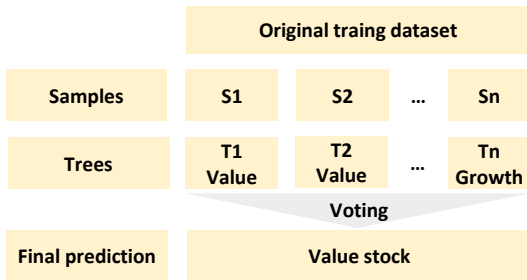
Final output is considered based on Majority Voting or Averaging for Classification and regression, respectively.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

6. Random Forest

Example:

Consider the stock basket as the data as shown in the figure below. Now n number of samples are taken from the stock basket, and an individual decision tree is constructed for each sample. Each decision tree will generate an output, as shown in the figure. The final output is considered based on majority voting. In the below figure, we can see that the majority decision tree gives output as a value stock when compared to a growth stock, so the final output is taken as a value stock.



[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

6.

Random Forest

Compare Decision Tree and Random Forest:

Random forest is a collection of decision trees; still, there are a lot of differences in their behavior. Random forests are much more successful than decision trees only if the trees are diverse and acceptable.

Decision Tree

Suffer from the problem of overfitting if it's allowed to grow without any control.

Faster in computation.

When a data set with features is taken as input, it will formulate some rules to make predictions.

Random Forest

Created from subsets of data, and the final output is based on average or majority ranking; hence the problem of overfitting is taken care of.

Comparatively slower.

Randomly selects observations, builds a decision tree, and takes the average result. It doesn't use any set of formulas.

[LOS 6.c] Describe supervised machine learning algorithms - including penalized regression, support vector machine,...

6.

Random Forest

Application:

Random forest is a powerful algorithm with many investment applications such as:

- **Asset allocation:** use factor-based investment strategies for asset allocation and investment selection;
- **Market prediction:** Predicting whether an IPO will be successful (e.g., percent oversubscribed, first trading day close/IPO price) given the attributes of the IPO offering and the corporate issuer.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis, k-means clustering, and hierarchical clustering - and determine the problems for which they are best suited

Unsupervised machine learning

**Principal component analysis
(PCA)**

Clustering

**K-means
Clustering**

**Hierarchical
Clustering**

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

1.

Principal component analysis (PCA)

Problem: too much noise often arise when the number of features in a data set (i.e., its dimension) is excessive. Dimension reduction seeks to reduce this noise by discarding those attributes that contain little information.

Principal component analysis (PCA) is an unsupervised method which summarizes the information in a large number of correlated factors into a much smaller set of uncorrelated factors (principal components) specified by the researcher called **eigenvectors**, which are linear combinations of the original features.

Each **eigenvector** has an **eigenvalue**—the proportion of total variance in the data set explained by the eigenvector.

The first factor



The second factor


The k^{th} factor

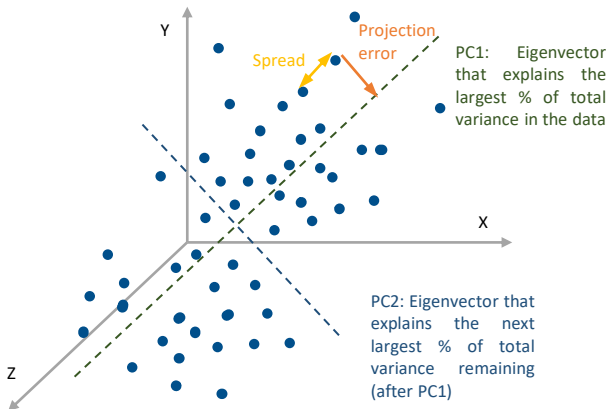
The eigenvalue and the importance of factor is smaller throughout this follow

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

1.

Principal component analysis (PCA)

First and Second Factors (Principal Components) of a Hypothetical Three-Dimensional Dataset



[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

1.

Principal component analysis (PCA)

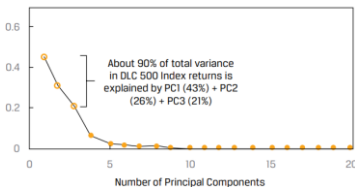
Understanding how many principal components to retain is crucial because it involves finding a **balance** between having a simplified, easier-to-handle representation of a complex dataset when **selecting a few components** and **accepting a certain amount of information loss**.

Scree plots show the proportion of total variance explained by each of the principal components. In practice, the smallest number of principal components that collectively capture 85%–95% of the total variance are retained.

Example: Scree Plots of Percent of Total Variance Explained by Each Principal Component for Hypothetical DLC 500 and VLC 30 Equity Indexes

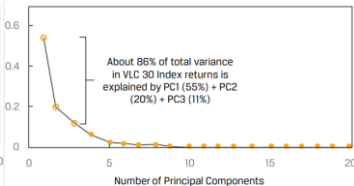
A. Diversified Large Cap 500 Index (DLC 500)

Percent of Variance Explained



B. Very Large Cap 30 Index (VLC 30)

Percent of Variance Explained



[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

1.

Principal component analysis (PCA)

Disadvantages of PCA:

- Since the principal components are combinations of the dataset's initial features, they typically cannot be easily labeled or directly interpreted by the analyst.
- Compared to modeling data with variables that represent well-defined concepts, the end user of PCA may perceive PCA as something of a "black box."

Application:

Principal Component Analysis (PCA) has several applications due to its ability to reduce dimensionality and extract relevant information from large datasets as the following:

- **Risk Management:** PCA can help identify and quantify sources of risk in a portfolio.
- **Asset Pricing Models:** In asset pricing models like the Capital Asset Pricing Model (CAPM), PCA can be employed to estimate the betas (sensitivities) of assets to common risk factors.
- **Portfolio Optimization:** PCA can be used to select a subset of assets or factors that explain the most significant portion of portfolio variance.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

2.

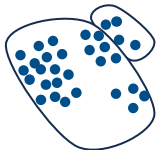
Clustering

Clustering is another type of unsupervised machine learning, which aim is to find a good clustering of the data:

- The observations inside each cluster are similar or close to each other (a property known as **cohesion**).
- The observations in two different clusters are as far away from one another or are as dissimilar as possible (a property known as **separation**).

Evaluating Clustering—Intra-Cluster Cohesion and Inter-Cluster Separation

Bad Clustering



Good Clustering



(Maybe) Better Clustering



We now introduce two of the more popular clustering approaches: **k-means** and **hierarchical clustering**.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

2.

Clustering

2.1. K-means Clustering:

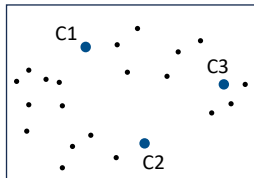
K-means is an algorithm that repeatedly partitions observations into a fixed number, k , a model hyperparameter of non-overlapping clusters.

Each cluster has a **centroid** (the center of the cluster), and each new observation is assigned to a cluster based on its proximity to the centroid.

The k-means algorithm follows an iterative process that groups the observations in the following steps:

01

K-means starts by determining the position of the k initial random centroids.



We assume that there is 3 centroids randomly chosen as figure above.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

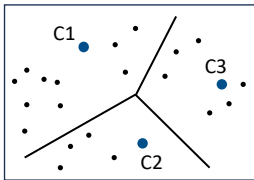
2.

Clustering

2.1. K-means Clustering:

02

The algorithm then analyzes the features for each observation. Based on the distance measure that is used, k-means assigns each observation to its closest centroid, which defines a cluster.



We assign each observation to the nearest centroid (defining initial 3 clusters).

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

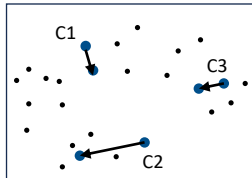
2.

Clustering

2.1. K-means Clustering:

03

Using the observations within each cluster, k-means then calculates the new (k) centroids for each cluster, where the centroid is the average value of their assigned observations.



We calculate new centroid as the average values of observations in a cluster.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

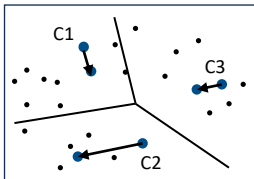
2.

Clustering

2.1. K-means Clustering:

04

K-means then reassigns the observations to the new centroids, redefining the clusters in terms of included and excluded observations.



We reassign each observation to the nearest centroid (from step 03).

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

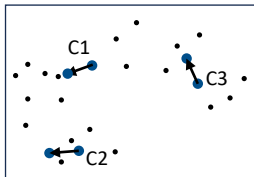
2.

Clustering

2.1. K-means Clustering:

05

The process of recalculating the new (k) centroids for each cluster is reiterated.



We reiterate the process of recalculating new centroid.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

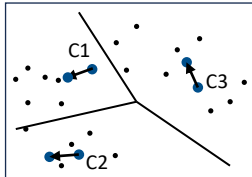
2.

Clustering

2.1. K-means Clustering:

06

K-means then reassigns the observations to the revised centroids, again redefining the clusters in terms of observations that are included and excluded.



We reassign each observation to the nearest centroid from step 05, completing second iteration.

The k-means algorithm will continue to iterate until no observation is reassigned to a new cluster (i.e., no need to recalculate new centroids).

Disadvantage of K-means Clustering: the hyperparameter k is chosen before clustering starts, meaning that one has to have some idea about the nature of the data set.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

2.

Clustering

2.2. Hierarchical Clustering:

Hierarchical clustering is a procedure used to build a hierarchy of cluster without any predefined number of clusters that create intermediate rounds of clusters of increasing (in “agglomerative”) or decreasing (in “divisive”) size until a final clustering is reached.

Hierarchical clustering

Agglomerative clustering
(bottom-up)

Divisive clustering
(top-down)

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

2.

Clustering

2.2. Hierarchical Clustering:

Agglomerative clustering (bottom-up)

- **Step 1:** It begins with each observation being treated as its own cluster.
 - **Step 2:** The algorithm finds the two closest clusters, defined by some measure of distance (similarity), and combines them into one new larger cluster.
- This process is repeated iteratively until all observations are clumped into a single cluster.

Divisive clustering (top-down)

- **Step 1:** It starts with all the observations belonging to a single cluster.
 - **Step 2:** The observations are then divided into two clusters based on some measure of distance (similarity).
- The algorithm progressively partitions the intermediate clusters into smaller clusters until each cluster contains only one observation.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

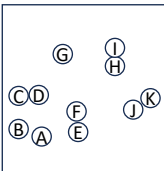
2.

Clustering

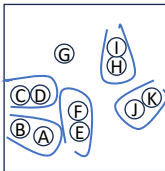
2.2. Hierarchical Clustering:

Example: Agglomerative Hierarchical Clustering

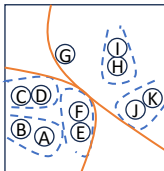
11 data points



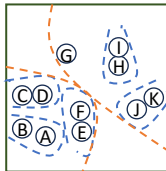
6 clusters



2 bigger clusters



1 big cluster



The process begins with 11 individual clusters and then generates a sequence of groupings.

There are five clusters with two observations each and one cluster with a single observation for a total of six clusters.

It then generates two clusters—one cluster with six observations and the other with five observations.

This final large cluster includes the two main sub-clusters, with each containing three smaller sub-clusters, and total 11 observations.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

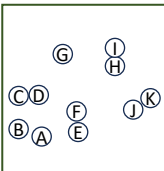
2.

Clustering

2.2. Hierarchical Clustering:

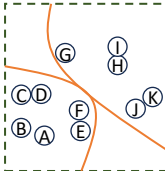
Example: Divisive Hierarchical Clustering

1 big cluster



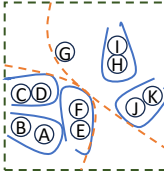
The process begins with 11 observations in one large cluster.

2 smaller clusters



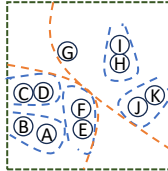
The algorithm generates two smaller clusters, one with six observations and the other with five observations.

6 clusters



There are now six clusters, with two observations each except for observation G, which is its own cluster.

11 data points



11 clusters are generated, with each cluster containing only one observation.

[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

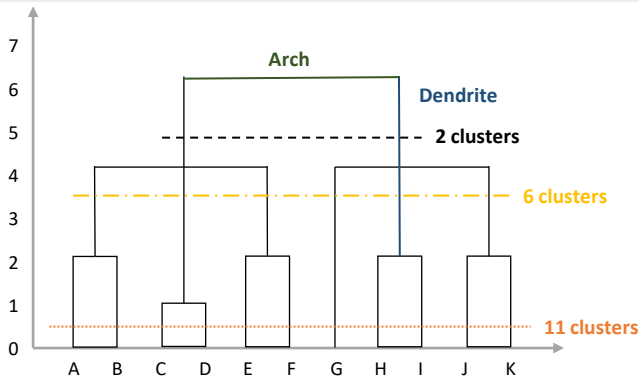
2.

Clustering

2.2. Hierarchical Clustering:

A type of tree diagram for visualizing a hierarchical cluster analysis is known as a **dendrogram**, which highlights the hierarchical relationships among the clusters. The x-axis shows the clusters, and the y-axis indicates some distance measure.

Dendrogram of Agglomerative Hierarchical Clustering



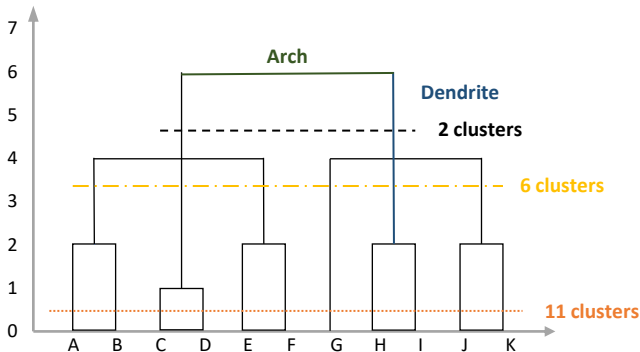
[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

2.

Clustering

2.2. Hierarchical Clustering:

- Clusters are represented by a horizontal line, the **arch**, which connects two vertical lines, called **dendrites**.
- Shorter dendrites represent a shorter distance (and greater similarity) between clusters.
- The horizontal dashed lines cutting across the dendrites show the number of clusters into which the data are split at each stage.



[LOS 6.d] Describe unsupervised machine learning algorithms - including principal components analysis,...

2.

Clustering

Application:

Clustering techniques have several important applications that help financial professionals and organizations make informed decisions, manage risk, and optimize their operations as the following:

- **Portfolio Construction:** group assets with similar risk and return profiles into portfolios. By creating clusters of assets with similar characteristics, portfolio managers can diversify their investments effectively and manage risk.
- **Risk Management:** identify correlations and dependencies among financial assets.
- **Data Analysis:** While clusters are themselves not clearly labeled or defined, clustering can be valuable in uncovering hidden structures or similarities between observations in complex data sets.

[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

1.

Neural Networks

Neural networks (also called **artificial neural networks**, or **ANNs**) are a highly flexible type of ML algorithm applied to a variety of tasks characterized by ***non-linearities*** and ***complex interactions*** such as image recognition, speech recognition, natural language processing (NLP), etc., among features that constructed as ***nodes connected by links***.

Neural networks have **three** types of layers

Input layer with a node for each of the features

Hidden layers, where learning occurs in training and inputs are processed on trained nets

Output layer, which passes information outside the network

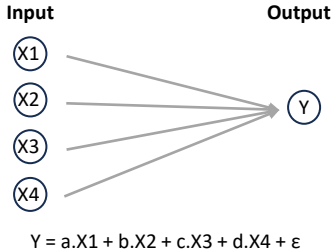
The figure in the next slide shows the connection between multiple regression and neural networks.

[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

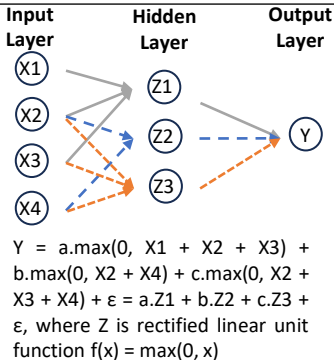
1.

Neural Networks

Conceptual Illustration of Regression



Conceptual Illustration of Hypothetical Neural Network



The nodes (Z_i) in the neural network's hidden layer transform the inputs in a **non-linear** fashion into new values that are then combined into the target value.

For neural networks, the feature inputs would be scaled (i.e., standardized) to account for differences in the units of the data.

[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

1.

Neural Networks

The figure below shows a more complex neural network, including:

- One input layer consisting of four nodes (X_i) (i.e., four features),
- One hidden layer consisting of five hidden nodes (Z_i) (five neurons),
- One output node (Y).

These three numbers: 4, 5, and 1—for the neural network are **hyperparameters** that determine the structure of the neural network.

**Input
Layer**

(X1)

(X2)

(X3)

(X4)

Hidden Layer

(Z1)

(Z2)

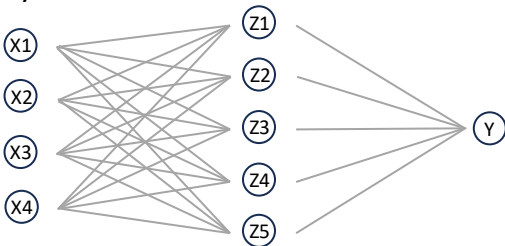
(Z3)

(Z4)

(Z5)

**Output
Layer**

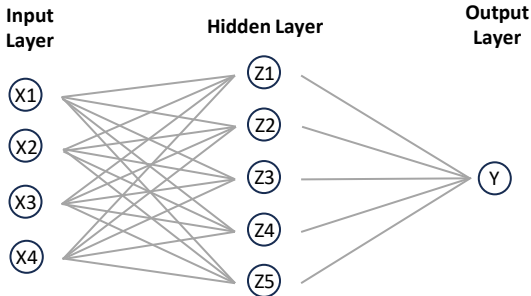
(Y)



[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

1.

Neural Networks



- Take the topmost neuron (Z1), four links connect to that node from the inputs (X1, X2, X3, X4), so the node gets four values transmitted by the links.
- Each link has a weight meant to represent its importance, and each node has, conceptually, two functional parts: a **summation operator** and an **activation function**.

[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

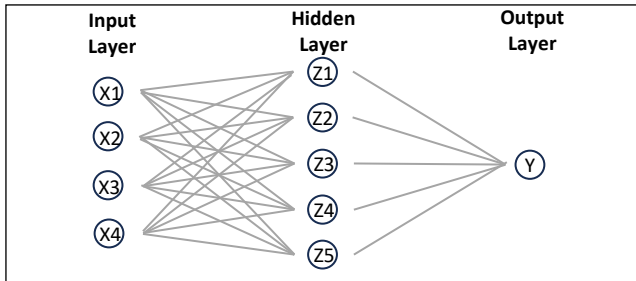
1.

Neural Networks

Hereafter is process of Neural Network that contain **three** main steps:

01

These neurons comprise a **summation operator** that collates the information and passes it on to a (typically nonlinear) **activation function**.



Summation operator



Activation function

Total net input $\sum_{i=1}^4 X_i.w_i$

Final output (Y)
(decrease or increase the strength of X_i)

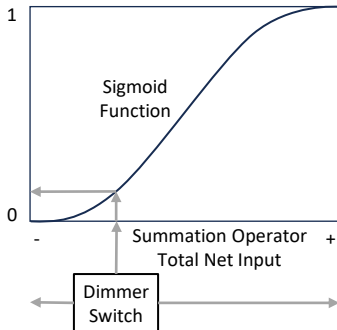
[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

1.

Neural Networks

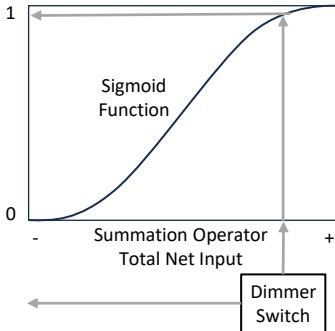
Activation Function as “Light Dimmer Switch” at Each Node in a Neural Network

Activation Function Output



A **negative** total net input is transformed via the S-shaped function into an output close to **0**, so the node **does not** trigger.

Activation Function Output



A **positive** total net input is transformed via the S-shaped function into an output close to **1**, so the node **does** trigger.

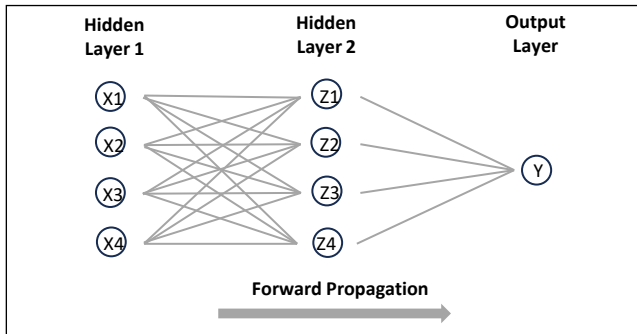
[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

1.

Neural Networks

02

The output of the activation function is then transmitted to the next set of nodes if there is a second hidden layer or to the output layer node as the predicted value (a process called **forward propagation**).



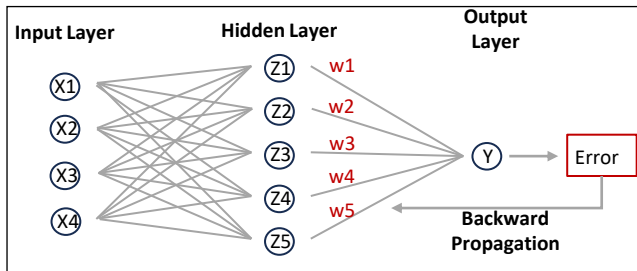
[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

1.

Neural Networks

03

A related process, **backward propagation**, is employed to revise the weights used in the summation operator as the network learns from its errors.



Learning occurs as the network adjusts its weights in order to **minimize the overall error**:

New weight = (Old weight) – (Learning rate) × (Partial derivative of the total error with respect to the old weight)

where: **Partial derivative** is a gradient or rate of change of the total error with respect to the change in the old weight and **Learning rate** is a hyperparameter that affects the magnitude of adjustments.

[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

1.

Neural Networks

Application:

Neural networks have various applications due to their ability to model complex patterns and make predictions based on data. Here are some notable applications:

- **Algorithmic Trading:** develop predictive models for trading strategies and analyze historical price data, market news, and other relevant information to make buy or sell decisions.
- **High-Frequency Trading:** make rapid trading decisions based on real-time market data.
- **Customer Support and Chatbots:** used in chatbots and virtual assistants to provide customer support, answer queries, and assist with account management.

[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

2. Deep Learning Networks

Deep Learning Networks (Deep Neural Networks, DLNs, DNNs) are unsupervised neural networks with many hidden layers (often more than 20).

Operation of DLNs:

- The input layer pass a set of inputs to a layer of non-linear mathematical functions (neurons) with weights w_{ij} (for neuron i and input j), each of which usually produces a scaled number in the range $(0, 1)$ or $(-1, 1)$.
- These numbers are then passed to another layer of functions and into another and so on until the final layer produces a set of probabilities of the observation being in any of the target categories (nodes in the output layer).
- The last layer in a DLN calculates the expected probability of an observation belonging to a category, and the observation is assigned to the category with the highest probability.

Application:

- Image, pattern, and character recognition.
- Credit card fraud detection, autonomous cars, natural language processing, and investment decision-making.
- Advances in analytical methods, increases in computing speed, and availability of large quantities of machine-readable data.

[LOS 6.e] Describe neural networks, deep learning nets, and reinforcement learning

3.

Reinforcement Learning (RL)

Reinforcement Learning (RL) is a type of machine learning paradigm where an agent must make a sequence of decisions to achieve a goal that seeks to maximize a defined reward given defined constraints.

Operation of RL:

- Reinforcement learning has neither direct labeled data for each observation nor instantaneous feedback.
- The algorithm needs to observe its environment, learn by testing new actions (some of which may not be immediately optimal), and reuse its previous experiences.
- The learning subsequently occurs through millions of trials and errors.

Application:

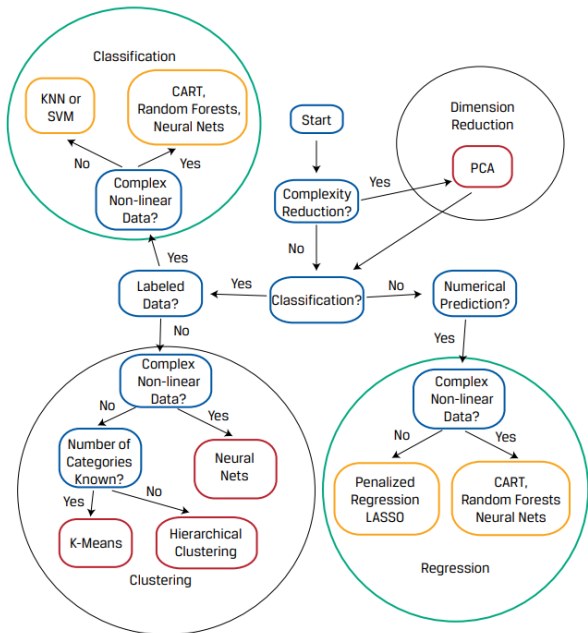
- Valuable in the investment field due to its ability to adapt to changing market conditions, learn from data, and optimize decisions based on real-time feedback.
- It is essential to consider the challenges and risks associated with using RL in finance, including the potential for overfitting, model interpretability, and the need for careful consideration of reward functions to ensure desired investment objectives are met.

Summarizes the suitability of various ML algorithms

Variables	ML Algorithm Type	
	Supervised (Target Variable)	Unsupervised (No Target Variable)
Continuous	Regression - Linear, Penalized Regression/ LASSO - Logistic - Classification and Regression Tree (CART) - Random Forest	Dimensionality Reduction - Principal Components Analysis (PCA) Clustering - K – Means - Hierarchical
Categorical	Classification - Logit - Support Vector Machine (SVM) - K – Nearest Neighbor (KNN) - Classification and Regression Tree (CART)	Dimensionality Reduction - Principal Components Analysis (PCA) Clustering - K – Means - Hierarchical
Continuous or Categorical	Neural Networks Deep learning Reinforcement Learning	Neural Networks Deep Learning Reinforcement Learning

Choice of Appropriate ML Algorithm

Stylized Decision Flowchart for Choosing ML Algorithms



Choice of Appropriate ML Algorithm

We have **4 steps** involved in selecting the appropriate ML algorithm to use, based on the problem to be solved and the characteristics of the data.

01

Decide if the data set is complex (too many features). If so, apply a dimension reduction algorithm before proceeding to Step 2.

02

Decide if the problem is that of classification. If yes, go to Step 3. If no, (i.e., it is a numerical prediction problem):

- Use penalized regression if the data is linear.
- Or, for nonlinear and complex data, use CART, random forests, or neural networks.

03

Is it unsupervised classification? If no, go to Step 4. For supervised classification:

- For linear data, use KNN or SVM.
- For complex nonlinear data, use CART, random forests, or neural networks.

04

For unsupervised classification:

- For linear data, use k-means if the number of categories is known. If the number of categories is not known, use hierarchical clustering.
- For complex nonlinear data, use neural networks.



MODULE 7 : Big Data Projects

LEARNING OUTCOMES

- 7.a.** Identify and explain steps in a data analysis project
- 7.b.** Describe objectives, steps, and examples of preparing and wrangling data
- 7.c.** Evaluate the fit of a machine learning algorithm
- 7.d.** Describe objectives, methods, and examples of data exploration
- 7.e.** Describe methods for extracting, selecting and engineering features from textual data
- 7.f.** Describe objectives, steps, and techniques in model training
- 7.g.** Describe preparing, wrangling, and exploring text-based data for financial forecasting

Introduction to big data

Big data is characterized by the four Vs

Volume

Big data refers to a huge volume of data.

Variety

Big data is collected from various sources: user-generated, traditional transactional, emails, images, clickstreams, and so on.

Velocity

The speed with which the data is created and collected (e.g., all the social media postings created during a specified time interval).

Veracity

Not all data sources are reliable, and the researcher has to separate quality from quantity to generate robust forecasts.

Big data includes two main categories of data

Structured data

It refers to the data that is organized and formatted in a way that is easily recognizable (rows and columns).

Unstructured data

It refers to the data that is unorganized, and the ML algorithm has to sift through the noise to pick out information.

[LOS 7.a] *Identify and explain* steps in a data analysis project

1.

Structured data analysis steps

Example of Structured data

Id	Name	Age
1	Jim	28
2	Pam	26
3	Michael	42

Id	Subject	Teacher
1	Chinese	John
2	Maths	Wally
3	Music	Arthur

Student_Id	Subject_Id	Grade
2	1	98
1	2	100
1	3	75
3	3	60
2	3	76
3	2	88

→ The tables above contain information including id, name, age, subject, teacher and grade. Because this information is organized and formatted in rows and columns, it is structured data.

[LOS 7.a] *Identify and explain* steps in a data analysis project

1.

Structured data analysis steps

We will illustrate **5 steps** involved in analyzing data for financial forecasting as the following:

01

Conceptualization of the modeling task

Determine:

- The problem at hand
- The output of the model
- How the model will be used and for whom
- Whether the model will be embedded in existing (or new) business processes.

02

Data collection

- For financial forecasting, usually numeric data (tabular format structured data) is collected from internal and external sources.
- The researcher has to determine which sources (internal or external) to use to collect this data.

03

Data preparation and wrangling

Cleansing the data:

- Addressing any missing values
- Verification of any out-of-range values

Preprocessing data: Aggregating, filtering, or extracting relevant variables.

[LOS 7.a] *Identify and explain* steps in a data analysis project

1.

Structured data analysis steps

04

Data exploration

- Feature selection and engineering
- Initial data analysis

04

Model training

- Determining the appropriate ML algorithm to use
- Evaluating the algorithm using a training data set
- Tuning the model

[LOS 7.a] *Identify and explain* steps in a data analysis project

2.

Unstructured data analysis steps

Example of Unstructured data

"John Smith
123 Main Street
City, State ZIP Code
Phone: (123) 456-7890
Email: john.smith@email.com"

→ The text above contains contact information, including a name, address, phone number, and email address. While this information is recognizable, it's not structured in a standardized way like a database entry. Thus, it is unstructured data.

*The steps used for structured data need to be modified in order to analyze unstructured, text-based data. The first **4 steps** would then be modified as the following:*

01

Text problem formulation

Determine:

- The problem
- The exact inputs and output of the model
- How the output will be used

[LOS 7.a] *Identify and explain* steps in a data analysis project

2.

Unstructured data analysis steps

02

Data collection (curation)

- Determine the data source (relevant data)
- If using supervised learning, annotating a reliable target variable is also necessary.

03

Text preparation and wrangling

Cleanse and preprocess data to convert streams of unstructured data into format that is usable by traditional modeling methods designed for structured inputs.

04

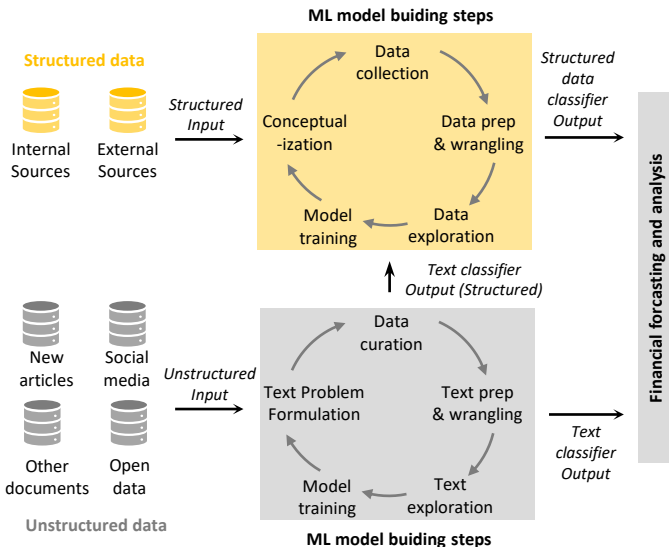
Text exploration

- Test visualization
- Text feature selection
- Engineering

The output of a model using unstructured data may then be used in isolation, or combined with other structured variables as an input into another model.

[LOS 7.a] *Identify and explain* steps in a data analysis project

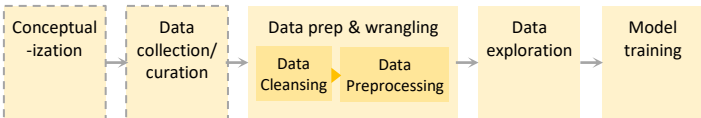
Model Building for Financial Forecasting Using Big Data: Structured (Traditional) vs. Unstructured (Text)



[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

1.

Conceptualization and Data collection/ curation

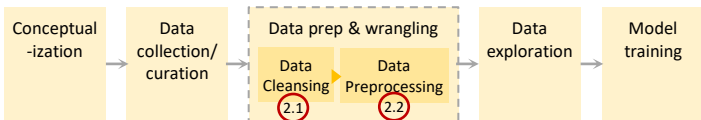


- **Conceptualization** is the step like a blueprint on a drawing board, a modifiable plan that is necessary to initiate the model building process.
- It include stating the problem, defining objectives, identifying useful data points, and conceptualizing the model.
- **Data collection** involves downloading data from internal and external sources.
- External data comes with a price, but saves time and money in the data wrangling step.

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

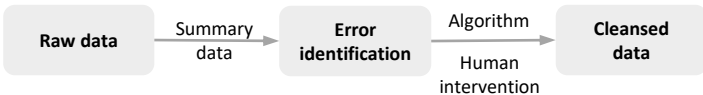
2.

Data preparation and wrangling for structured data



2.1. Data preparation (Data Cleansing) for structured data

This is the initial and most common task in data preparation that is performed on raw data. Data cleansing is the process of examining, identifying, and mitigating errors in raw data.



Data cleansing deals with reducing errors in the raw data.

Invalidity error	Non uniformity error	Incompleteness error
Duplication error	Inaccuracy error	Inconsistency error

[LOS 7.b-g] Describe objectives, steps, and examples of preparing and wrangling data

2. Data preparation and wrangling

2.1. Data preparation (Data Cleansing) for structured data

Example: Raw Data Before Cleansing

1	ID	Name	Gender	Date of Birth	Salary	Other Income	State	Credit Card
2	1	Mr. ABC	M	12/5/1970	\$50,200	\$5,000	VA	Y
3	2	Ms. XYZ	M	15 Jan, 1975	\$60,500	\$0	NY	Y
4	3	EFG		1/13/1979	\$65,000	\$1,000	CA	N
5	4	Ms. MNO	F	1/1/1900	—	—	FL	Don't Know
6	5	Ms. XYZ	F	15/1/1975	\$60,500	\$0		Y
7	6	Mr. GHI	M	9/10/1942	NA	\$55,000	TX	N
8	7	Mr. TUV	M	2/27/1956	\$300,000	\$50,000	CT	Y
9	8	Ms. DEF	F	4/4/1980	\$55,000	\$0	British Columbia	N

Data After Cleansing

1	ID	Name	Gender	Date of Birth	Salary	Other Income	State	Credit Card
2	1	Mr. ABC	M	12/5/1970	USD 50200	USD 5000	VA	Y
3	2	Ms. XYZ	F	1/15/1975	USD 60500	USD 0	NY	Y
4	3	Mr. EFG	M	1/13/1979	USD 65000	USD 1000	CA	N
5	6	Mr. GHI	M	9/10/1942	USD 0	USD 55000	TX	N
6	7	Mr. TUV	M	2/27/1956	USD 300000	USD 50000	CT	Y
7	8	Ms. DEF	F	4/4/1980	CAD 55000	CAD 0	British Columbia	N

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

2.

Data preparation and wrangling for structured data

2.1. Data wrangling for structured data

This task performs transformations and critical processing steps on the cleansed data to make the data ready for ML model training.

DATA WRANGLING (PREPROCESSING)

Data transformation

Extraction

Aggregation

Filtration

Selection

Conversion

Removal of outlier

Trimming

Winsorization

Scaling

Normalization

Standardization

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

2. Data preparation and wrangling for structured data

2.1. Data wrangling for structured data

The following **transformations** are common in practice:

- 01 **Extraction:** A new variable can be extracted from the current variable for ease of analyzing and using for training the ML model.
- 02 **Aggregation:** Two or more variables can be aggregated into one variable to consolidate similar variables.
- 03 **Filtration:** The data that are not needed for the project must be identified and filtered
- 04 **Selection:** The data that are intuitively not needed for the project can be removed.
- 05 **Conversion:** The variables can be of different types: nominal, ordinal, continuous, categorical → must be converted into appropriate types to further process and analyze them correctly.

[LOS 7.b-g] Describe objectives, steps, and examples of preparing and wrangling data

2. Data preparation and wrangling for structured data

2.1. Data wrangling for structured data

There are several practical methods for handling outliers:

01

Trimming: extreme values and outliers are simply removed from the dataset.

Example: a 10% trimmed dataset is one for which the 10% highest and the 10% lowest values have been removed.



02

Winsorization: extreme values may be replaced by the maximum value allowable for that variable.



[LOS 7.b-g] Describe objectives, steps, and examples of preparing and wrangling data

2. Data preparation and wrangling for structured data

2.1. Data wrangling for structured data

Example: Data After Cleansing

1	ID	Name	Gender	Date of Birth	Salary	Other Income	State	Credit Card
2	1	Mr. ABC	M	12/5/1970	USD 50200	USD 5000	VA	Y
3	2	Ms. XYZ	F	1/15/1975	USD 60500	USD 0	NY	Y
4	3	Mr. EFG	M	1/13/1979	USD 65000	USD 1000	CA	N
5	6	Mr. GHI	M	9/10/1942	USD 0	USD 55000	TX	N
6	7	Mr. TUV	M	2/27/1956	USD 300000	USD 50000	CT	Y
7	8	Ms. DEF	F	4/4/1980	CAD 55000	CAD 0	British Columbia	N

Data After Applying Transformations (and removing outliers)

1	ID	Gender	Age	Total Income	State	Credit Card
2	1	M	48	55200	VA	Y
3	2	F	43	60500	NY	Y
4	3	M	39	66000	CA	N
5	6	M	76	55000	TX	N

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

2. Data preparation and wrangling for structured data

2.1. Data wrangling for structured data

Some ML algorithms (e.g., neural networks, SVM) require features to be homogenous (i.e., feature values in the same range).

→ **Scaling** is the conversion of data features to a common unit of measurement.

*Two common methods of scaling are **normalization** and **standardization**:*

Normalization

The process of rescaling numeric variables in the range of [0, 1].

$$X_{i(\text{normalized})} = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}}$$

Sensitive to outliers, so treatment of outliers is necessary before normalization is performed.

Standardization

The process of both centering and scaling the variables.

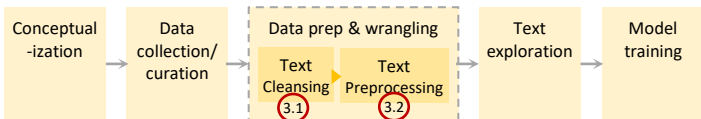
$$X_{i(\text{standardized})} = \frac{X_i - \mu}{\sigma}$$

Less sensitive to outliers as it depends on the mean and standard deviation of the data but assumes that the variable is normally distributed.

[LOS 7.b-g] Describe objectives, steps, and examples of preparing and wrangling data

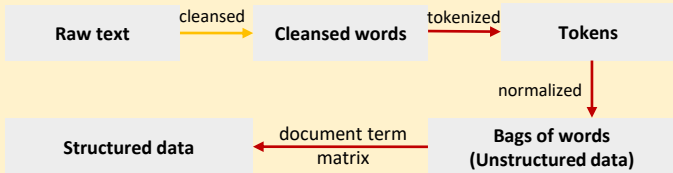
3.

Data preparation and wrangling for unstructured data



Process Overview

Computers can't read normal texts
→ They need to be cleansed, tokenized



- Text cleansing
- Text wrangling

[LOS 7.b-g] Describe objectives, steps, and examples of preparing and wrangling data

3. Data preparation and wrangling for unstructured data

3.1. Text cleansing

Raw text data are a sequence of characters and contain other non-useful elements, including html tags, punctuations, and white spaces.

Example: Raw Text from the Source

```
<h1 class="text-left mb-3">Robots Are Us</h1>
```

```
<h2> Every home and business should have a robot </h2>
```

→ We need to clean the text by **removing unnecessary elements** from the raw text. Text cleansing (Text preparing) involves 4 steps as the following:

01

Remove html tags: Text collected from web pages has embedded HTML tags, which may need to be removed before processing.

02

Remove punctuations: Some punctuations (e.g., %, \$, ?) may be needed for analysis, and if so, they are replaced with annotations (i.e., textual expressions) for model training.

03

Remove number: Digits are removed or replaced with annotations (/number/). If the value of a number is important for analysis, such values are first **extracted via text applications**.

04

Remove white spaces: Extra formatting-related white spaces (e.g., tabs, indents) do not serve any purpose in text processing and are removed.

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

3.

Data preparation and wrangling for unstructured data

3.1. Text cleansing

Example: Text Cleansing Process

Original text from a financial statement as shown on a webpage

CapEx on the normal operations remained stable on historically low levels, \$800,000 compared to \$1.2 million last year.

Quarter 3, so far, is 5% sales growth quarter-to-date, and year-to-date, we have a 4% local currency sales development.

Raw text after scraping from the source

```
<p><font size = "4"> CapEx on the normal operations remained stable on
historically low levels, $800,000 compared to $1.2 million last year. <b/><b/>
Quarter 3, so far, is 5% sales growth quarter-to-date, and year-to-date, we have a
4% local currency sales development.</font></p>
```

Text after removing html tags

CapEx on the normal operations remained stable on historically low levels, \$800,000 compared to \$1.2 million last year.

Quarter 3, so far, is 5% sales growth quarter-to-date, and year-to-date, we have a 4% local currency sales development.

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

3.

Data preparation and wrangling for unstructured data

3.1. Text cleansing

Example: Text Cleansing Process

Text after removing and replacing punctuations

CapEx on the normal operations remained stable on historically low levels /dollarSign/800000 compared to /dollarSign/12 million last year /endSentence/ Quarter 3 so far is 5 /percentSign/ sales growth quarter-to-date and year-to-date we have a 4 /percentSign/ local currency sales development /endSentence/

Text after replacing numbers

CapEx on the normal operations remained stable on historically low levels /dollarSign//number/ compared to/dollarSign//number/ million last year /endSentence/ Quarter/number/ so far is /number/ /percentSign/sales growth quarter-to-date and year-to-date we have a /number/ /percentSign/ local currency sales development /endSentence/

Text after removing extra white spaces

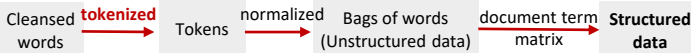
CapEx on the normal operations remained stable on historically low levels/dollarSign//number/compared to/dollarSign//number/million last year/endSentence/Quarter/number/so far is/number//percentSign/sales growth quarter-to-date and year-to-date we have a/number//percentSign/local currency sales development/endSentence/

[LOS 7.b-g] Describe objectives, steps, and examples of preparing and wrangling data

3.

Data preparation and wrangling for unstructured data

3.2. Text wrangling



To further understand text processing, **tokens** and **tokenization** need to be defined.

- A **token** is equivalent to a word, and **tokenization** is the process of splitting a given text into separate tokens.
- In other words, a **text** is considered to be a **collection of tokens**.

Example: Tokenization of Four Texts

Cleaned Texts

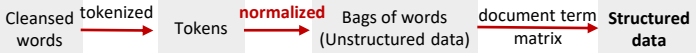
Tokens

Text 1	The man went to the market	The_man_went_to_the_market
Text 2	Market values are increasing	Market_values_are_increasing
Text 3	Increased market is needed	Increased_market_is_needed
Text 4	There is no market for products	There_is_no_market_for_products

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

3. Data preparation and wrangling for unstructured data

3.2. Text wrangling



Normalization Process involves 4 steps as the following:

01

Lowercasing the alphabet removes distinctions among the same words due to upper and lower cases.

This action helps the computers to process the same words appropriately.

Example: “The” → “the”

02

Removal of stop words: Stop words are such commonly used words as “the,” “is,” and “a.” They are removed to reduce the number of tokens in the training data.

03

Stemming: is a rules-based algorithm that converts all variations of a word into a common value. While stemming makes the text confusing for human processing, it is ideally suited for machines.

04

Lemmatization: is the process of converting inflected forms of a word into its **morphological root** (known as lemma). It is similar to stemming, but is more computationally advanced and resource intensive.

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

3. Data preparation and wrangling for unstructured data

3.2. Text wrangling

After the data is cleansed and normalized, a **bag-of-words (BOW)** procedure is applied, which simply collects all the words or tokens without regard to the sequence of occurrence.

Example: Bag-of-Words Representation of Four Texts Before and After Normalization Process

BOW before normalizing

"The" "today" "marketing" "product" "man" "Market" "is" "went" "values"
"needed" "to" "are" "There" "the" "increasing" "no" "market" "Increased" "for"

BOW after removing uppercase letters

"the" "values" "needed" "man" "are" "there" "went" "increasing" "no" "to"
"increased" "for" "market" "marketing" "product" "today" "is"

BOW after removing stop words

"man" "increased" "went" "marketing" "market" "needed" "today" "product"
"values" "increasing"

BOW after stemming

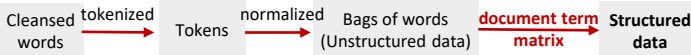
"man" "went" "market" "today" "valu" "increas" "need" "product"

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

3.

Data preparation and wrangling for unstructured data

3.2. Text wrangling



The last step of text preprocessing is using the final BOW after normalizing to build a **document term matrix (DTM)**.

Document term matrix: is a matrix that is similar to a data table for structured data and is widely used for text data. Each row of the matrix belongs to a document (or text file), and each column represents a token (or term).

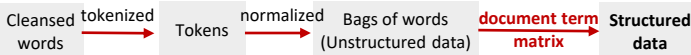
Feature of DTM:

- The number of rows = the number of documents.
- The number of columns = the number of token in the normalized BOW.

[LOS 7.b-g] Describe objectives, steps, and examples of preparing and wrangling data

3. Data preparation and wrangling for unstructured data

3.2. Text wrangling



Example: DTM of four Texts and using Normalized BOW filled with counts of occurrence

	man	went	market	today	valu	increas	need	product
Text 1	1	1	1	1	0	0	0	0
Text 2	0	0	1	0	1	1	0	0
Text 3	0	0	1	0	0	1	1	0
Text 4	0	0	1	0	0	0	0	1

In the example, the word **“no”** is treated as a single token and has been removed during the normalization because it is a stop word.

→ Consequently, this fails to signify the negative meaning (“no market”) of the text (i.e., Text 4).

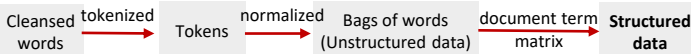
→ To overcome such problems, a technique called **N-grams** can be employed.

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

3.

Data preparation and wrangling for unstructured data

3.2. Text wrangling



If the sequence of text is important, **N-grams** can be used to represent word sequences.

A **two-word** sequence is a **bigram**, a **three-word** sequence is **trigram**, and so forth.

For example: Consider the sentence, “The market is up today.”

→ **Bigrams** of this sentence include “the_market”, “market_is”, “is_up”, and “up_today”.

→ BOW is then applied to the bigrams instead of the original words.

N-gram implementation will affect the normalization of the BOW because *stop words will not be removed*.

[LOS 7.b-g] *Describe* objectives, steps, and examples of preparing and wrangling data

3.

Data preparation and wrangling for unstructured data

3.2. Text wrangling

Example: N-Grams and N-Grams BOW

Clean text

The man went to the market today

Unigrams

"The" "man" "went" "to" "the" "market" "today"

Bigrams

"The_man" "man_went" "went_to" "to_the" "the_market" "market_today"

Trigrams

"The_man_went" "man_went_to" "went_to_the" "to_the_market"
"the_market_today"

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

Data exploration is a crucial part of big data projects to examine and analyze a dataset to understand its structure, patterns, relationships, and potential insights.



Exploratory data analysis (EDA): involves looking at data descriptors such as summary statistics, heat maps, word clouds, and so on.

The objectives of EDA include:

- Understanding data properties, distributions, and other characteristics.
- Finding patterns or relationships, and evaluating basic questions and hypotheses.
- Planning modeling in future steps.

Feature selection: is a process to select only the needed attributes of the data for ML model training.

The higher the number of features selected, the higher the model complexity and training time.

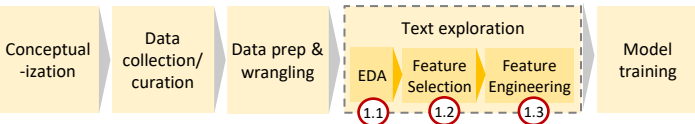
Feature engineering: is the process of creating new features by transforming, decomposing, or combining multiple features.

A related term is **feature extraction**, whereby a feature is created from the data set (e.g., creating a value for age using date-of-birth data).

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

1.

Data exploration for structured data



1.1. Exploratory data analysis (EDA)

EDA can be performed for a **single feature** (one dimension) or **multiple features** (multidimension).

For a single feature, summary statistics include the mean, standard deviation, skewness, and kurtosis. → EDA visualizations include **box plots, histograms, density plots, and bar charts**.

Box plots

Used for continuous variables and highlight the median, quartiles, and outliers of a normally distributed feature.

Histograms

Capture the frequency of observations in a series of equal-width bins.

Density plots

Smoothed histograms for continuous data overlaid on top of histograms.

Bar charts

Show frequency distributions of categorical variables.

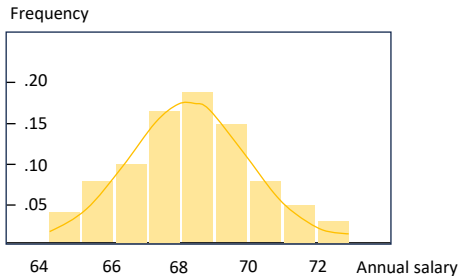
[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

1.

Data exploration for structured data

1.1. Exploratory data analysis (EDA)

Example: Histogram with Superimposed Density Plot



This histogram shows a hypothetical annual salary distribution (in £) of entry-level analyst positions at UK banks. The data represent a normal distribution with an approximate mean of £68,500.

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

1.

Data exploration for structured data

1.1. Exploratory data analysis (EDA)

When the **number of features is large**, dimension reduction models such as principal component analysis (PCA) can facilitate data exploration.

For multiple features, the summary statistic can be a correlation matrix. Data can be visually graphed using a **scatterplot, box plots, stacked bar or a line graph**.

Multiple box plots can be arranged in a single chart, where each individual box plot represents a feature (*see next slide*).

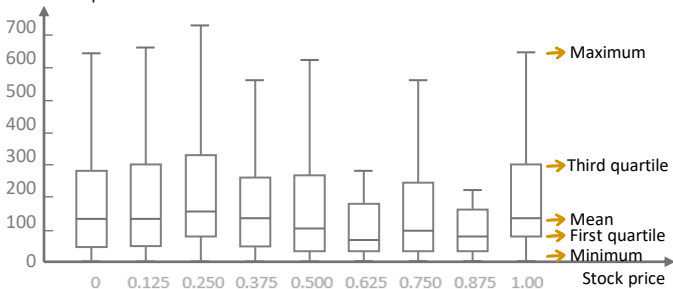
[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

1.1. Exploratory data analysis (EDA)

Multiple box plots can be arranged in a single chart, where each individual box plot represents a feature.

Example: Multiple Box Plots in One Chart

Unit of shares purchased



The individual box plots indicate the distribution of shares purchased at the different stock prices.

When the stock price is \$0.25, the median number of shares purchased is the highest.

When the stock price is \$0.625, the median number of shares purchased is the lowest.

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

1.

Data exploration for structured data

1.2. Feature Selection

- We try to select only the features that contribute to the out-of-sample predictive power of the model → it is essential to identify and remove unneeded, irrelevant, and redundant features.

Two main methods of feature selection is **statistical measures** and **dimensionality reduction**.

Statistical measures

Assign a score gauging the importance of each feature. The features can then be ranked using this score and either retained or eliminated from the dataset.

Dimensionality reduction

Identify the features in the data that account for the greatest variance between observations and allows for the processing of a reduced volume of data.

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

1.

Data exploration for structured data

1.3. Feature Engineering

- **Feature Engineering (FE)** involves optimizing and improving the selected features to produce good features that describe the structures inherent in the dataset.
- Feature engineering involves either **decomposing** a feature into multiple features or **converting** an existing feature into a new feature.

Example of feature engineering

For continuous data

New feature may be created by taking the logarithm of the product of two or more features

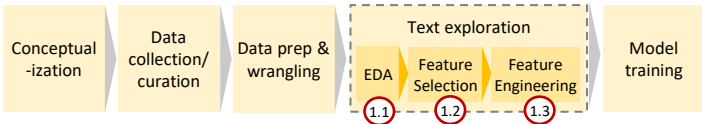
For categorical data

New feature can be created by using “one hot encoding” to covert categorical features in to dummy variables.

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

2.

Data exploration for unstructured data



2.1. Exploratory data analysis (EDA)

Just like with structured data, it is important to gain insight into existing patterns in the unstructured data for further analysis

EDA can be performed by:

- Computing **term frequency** (TF): number of times the word appears in the text.
- Analyzing **co-occurrence**: looking at where two or more words appear together.

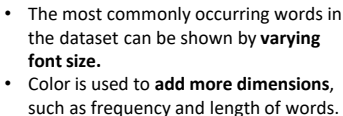
EDA can be presented by:

- Using the same methods as explained in the structured data section.
- Words clouds are common visualizations when working with text data as they can be made to visualize the most informative words and their TF values.

2. Data exploration for unstructured data

2.1. Exploratory data analysis (EDA)

Example: Idea Lightbulb Word Cloud Design



[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

2.

Data exploration for unstructured data

2.2. Feature Selection

- **Feature selection** involves selecting a subset of tokens in the BOW and effectively eliminating noisy features (features that do not contribute to ML model training).
- Typically noisy features are both the most frequent and most sparse (or rare) tokens in the dataset.

The general feature selection methods in text data are as follows:

01

Frequency – Document frequency (DF)

- Document frequency accounts for the number of document that contains the token.
- Document frequency =
$$\frac{\text{The number of document containing the token}}{\text{Total number of documents}}$$
- *Very high and low term frequency (TF) or document frequency (DF) indicates noisy features.*

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

2.

Data exploration for unstructured data

2.2. Feature Selection

02

Chi-square

- **Chi-square** test the independence of two events: occurrence of the token and occurrence of the class.
- It is used to identify tokens that appear to be associated with a specific class.
- *Tokens with the highest chi-square test statistic values can be selected as features due to higher discriminatory potential.*

03

Mutual information (MI)

- **Mutual information (MI)** measure how much information is contributed by a token to a class of texts. (value ranges from 0 to 1).
- If the token is equally distributed among all text classes (no information) → $MI = 0$
- If the token mostly appears in just one class of text (much information) → $MI = 1$

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

2.

Data exploration for unstructured data

2.2. Feature Selection

Example: Tokens with MI Values for Two Given Text Classes

Text Classes: Sports or Politics

Sports

Politics

Token	MI Value	Token	MI Value
soccer	0.0781	election	0.0612
cup	0.0525	president	0.0511
match	0.0456	polls	0.0341
play	0.0387	vote	0.0288
game	0.0299	party	0.0202
team	0.0265	candidate	0.0201
win	0.0189	campaign	0.0201

[LOS 7.d-e] *Describe* objectives, methods, and example of data explorations

2.

Data exploration for unstructured data

2.3. Feature Engineering

As with structured data, feature engineering can greatly improve ML model training. The following are some techniques for feature engineering:

Numbers

Mechanism: Numbers are converted into a token, to differentiate between different types of number (years, tax, ID, ...)

N-grams

Mechanism: Multi-word patterns that are particularly discriminative can be identified and their connection kept intact.

Ex.: expansionary monetary policy → expansionary_monetary_policy

Name entity recognition (NER)

Mechanism: NER algorithm analyzes the individual tokens and their surrounding semantics while referring to its dictionary to **tag an object class** to the token.

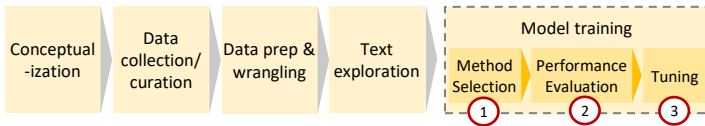
Ex.: CFA = organization.

Part of speech (POS)

Mechanism: uses language structure and dictionaries to tag every token in the text with a corresponding part of speech.

Ex.: Market = noun.

[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training



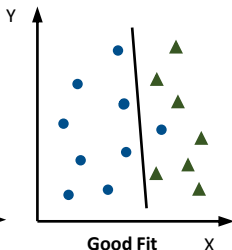
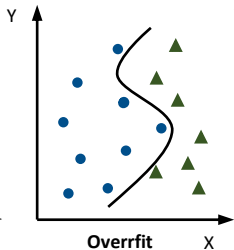
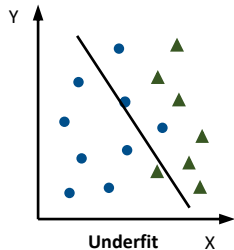
- The ML model training process for structured and unstructured data is typically the same.
- The fundamental idea of ML model training is **fitting a system of rules on a training dataset** to reveal a pattern in the data.
- In other words, **fitting** describes the degree to which (or how well) an ML model can be generalized to new data.
- A good model fit results in good model performance and can be validated using new data outside of the training dataset (i.e., out-of-sample).

Model fitting errors are caused by several factors - the main ones being **dataset size** and **number of features** in the dataset.

- **Dataset Size:** Small datasets can lead to underfitting of the model so the model may not recognize important patterns.
- **Number of Features:** A dataset with a **small number** of features can lead to **underfitting**, and a dataset with a **large number** of features can lead to **overfitting**.

[LOS 7.f-c] Describe objectives, steps, and techniques in model training

Model Fitting Scenarios: Underfit, Overfit, and Good Fit



This model is **underfit** because it does not fit the training data well enough.

Although the center model that generates the "S"-shaped line has the best accuracy (no errors) on the training data, it is **overfit** and thus unlikely to perform well on future test cases.

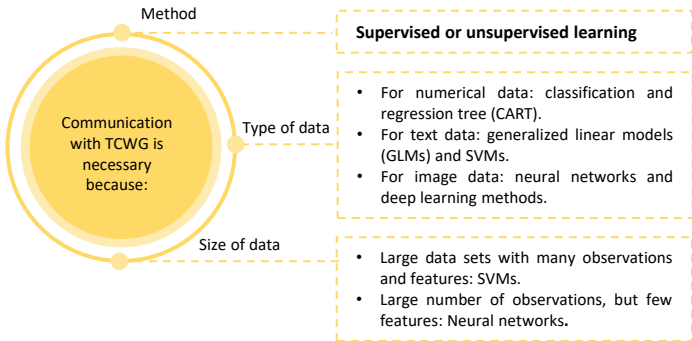
This model is **good fit** because it fits the training data well but not so well that it cannot be generalized to out-of-sample data.

[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

1.

Method selection

Method selection is the art and science of choosing the appropriate ML method (i.e., algorithm) given the objectives and data characteristics. It is based on three factors as the following



[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

1.

Method selection

Once a method is selected, appropriate **hyperparameters** need to be specified:

- The regularization term (λ) in supervised models
- Activation function and number of hidden layers in NN
- Number of trees and tree depth in ensemble methods
- k in k -nearest neighbor classification and k -means clustering
- p -threshold in logistic regression.

For mixed data sets (containing numerical and textual data), **multiple methods** are often used.

For a model to be able to discriminate well, it should be provided with a wide variety of training data.

→ There is a problem called **Class imbalance**, which occurs when one class has a large number of observations relative to other classes.

Example: in a model for predicting bond default, if the data set has a large number of high-grade bonds, then the model would be more likely to predict nondefault for a new observation.

One way to overcome class imbalance is to **undersample** the overrepresented class and **oversample** the underrepresented class (*see example in the next slide*).

[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

2.

Performance Evaluation

In order to validate a model, we must measure its training performance or goodness of fit. There are three most popular methods to measure this performance.

Error analysis

(2.1)

Receiver operating
characteristic (ROC)

(2.2)

Root mean square
error (RMSE) (2.3)

2.1. Error analysis

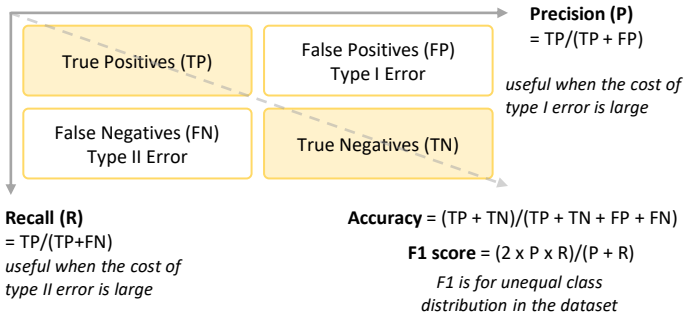
- For classification problems, error analysis involves **computing four basic evaluation metrics**: true positive (TP), false positive (FP), true negative (TN), and false negative (FN) metrics.
- FP is also called a Type I error, and FN is also called a Type II error.

[LOS 7.f-c] Describe objectives, steps, and techniques in model training

2.1. Error analysis

Actual Training Labels

		Class "1"	Class "0"
Predicted Results	Class "1"	True Positives (TP)	False Positives (FP) Type I Error
	Class "0"	False Negatives (FN) Type II Error	True Negatives (TN)



[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

2.

Performance Evaluation

2.1. Error analysis

Example: Performance Metrics and Scores Computation

		Actual Training Labels	
		Class "1"	Class "0"
Predicted Results	Class "1"	3 (TP)	1 (FP)
	Class "0"	2 (FN)	4 (TN)

Performance Metrics

$$TP = 3, FP = 1, FN = 2, TN = 4$$

$$P = 3 / (3+1) = 0.75$$

$$R = 3 / (3+2) = 0.60$$

$$F1 \text{ Score} = (2 \times 0.75 \times 0.60) / (0.75 + 0.60) = 0.67$$

$$\text{Accuracy} = (3 + 4) / (3 + 1 + 4 + 2) = 0.70$$

[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

2.

Performance Evaluation

2.2. Receiver operating characteristic (ROC)

ROC definition: a curve showing the tradeoff between FPs and TPs (This trade off represent business decisions)

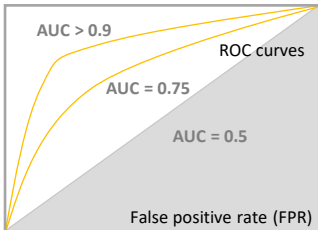
- Y axis: True positive rate (TPR) – or “Recall” / “Sensitivity” = $TP / (TP + FN)$;
- X axis: False positive rate (FPR) = $FP / (FP + TN)$

A **more convex curve** indicates better model performance.

The area under the curve (AUC): (is a value from 0 to 1) the closer the value of AUC is to 1, the higher the predictive accuracy of the model.

ROC Curves and AUC

True positive rate (TPR)



A **more convex curve** – and higher AUC indicates better model performance.

[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

2.

Performance Evaluation

2.3. Root mean square error (RMSE)

- The RMSE is computed by finding the square root of the mean of the squared differences between the actual values and the model's predicted values (error).
- It is useful for data predictions that are continuous, such as regression models.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (\text{predicted}_i - \text{actual}_i)^2}{n}}$$

- A small RMSE indicates potentially better model performance.

[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

3. Model tuning

After model evaluation, the model needs to be revised until it reaches an acceptable performance level.

Model tuning is process of adjusting the hyperparameters of a machine learning algorithm or model in order to find the best configuration for improving its performance on a specific task or dataset.

2 types of errors in model fitting

Bias error is the prediction error in the training data resulting from **underfit models**. It occurs from **oversimplified** models, which don't learn adequately from the training sample.

Variance error is the prediction error in the validation sample resulting from **overfitting models** that do not generalize well. It occurs when too many features are included in the training sample (i.e., the model is too complicated).

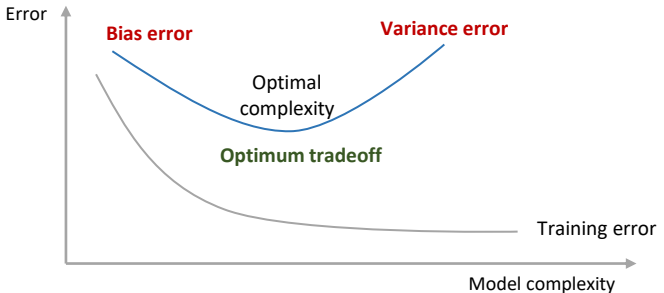
It is necessary to find an optimum **tradeoff** between bias and variance errors, such that the model is neither underfitting nor overfitting.

[LOS 7.f-c] Describe objectives, steps, and techniques in model training

3. Model tuning

- A **fitting curve** is a plot of training error and cross-validation prediction error with varying model complexity (more complex = more features).
- The trade-off can be accessed by looking at the fitting curve.

Example: Fitting curve



- As a model's complexity increases, it starts overfitting the training sample, and training error (i.e., bias error) declines. However, this decrease in bias error comes at the cost of increasing variance error.
- The optimal complexity lies somewhere in the middle of the fitting curve.

[LOS 7.f-c] *Describe* objectives, steps, and techniques in model training

3.

Model tuning

- Model tuning can be conducted by modifying hyperparameters, or increase the size of training data set,....
- Tuning techniques include *grid search* and *ceiling analysis*:
 - **A grid search** is an automated process of selecting the best combination of hyperparameters.
 - **Ceiling analysis** is an evaluation and tuning of each of the components in the entire model-building pipeline.